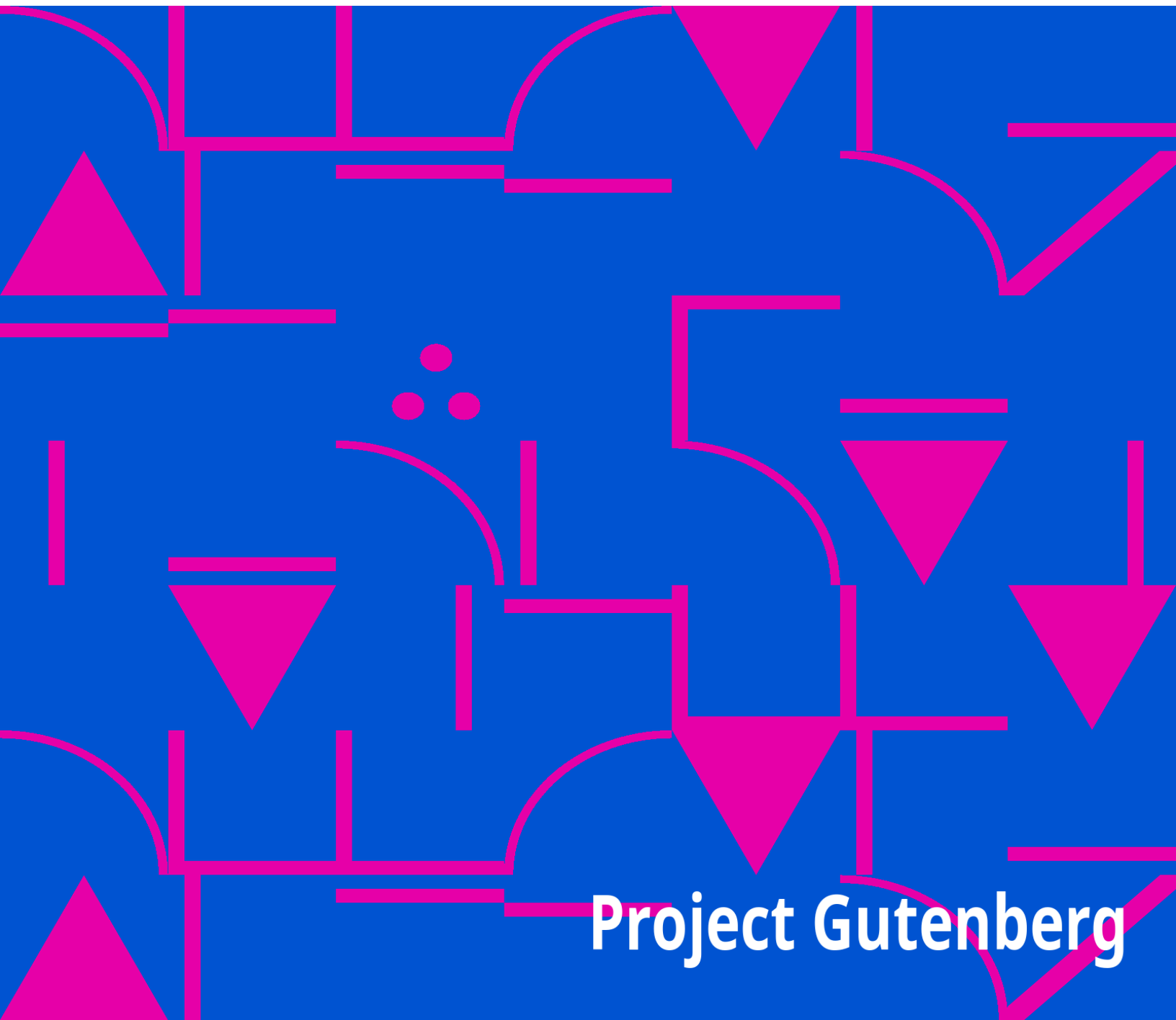


Illustrated Index of British Shells

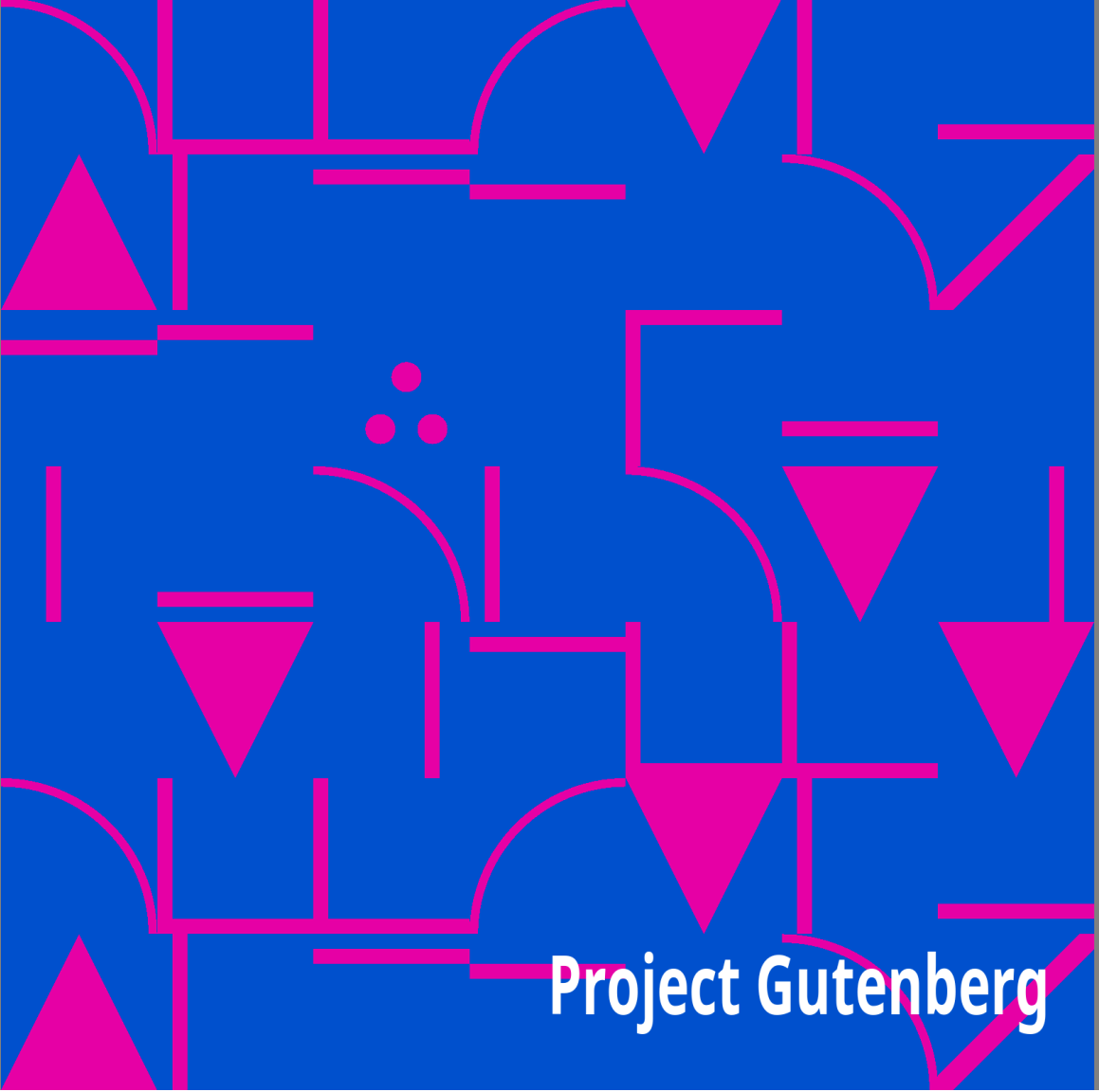
G. B. Sowerby

The background of the lower half of the image is a solid blue field. Overlaid on this is a complex, repeating pattern of magenta geometric shapes. These shapes include various sizes of triangles (some pointing up, some down), squares, rectangles, and semi-circles. Some shapes are solid magenta, while others are defined by magenta outlines. The arrangement is non-representational and serves as a decorative border or background for the text.

Project Gutenberg

Illustrated Index of British Shells

G. B. Sowerby

An abstract geometric pattern on a blue background, featuring various magenta shapes including triangles, circles, and lines. The pattern is dense and covers the lower two-thirds of the page.

Project Gutenberg

The Project Gutenberg eBook of Illustrated Index of British Shells

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg License included with this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

Title: Illustrated Index of British Shells

Author: G. B. Sowerby

Release date: July 26, 2013 [eBook #43305]

Most recently updated: October 23, 2024

Language: English

Other information and formats: www.gutenberg.org/ebooks/43305

Credits: Produced by Chris Curnow, Keith Edkins and the Online Distributed Proofreading Team at <http://www.pgdp.net> (This file was produced from images generously made available by The Internet Archive)

*** START OF THE PROJECT GUTENBERG EBOOK ILLUSTRATED
INDEX OF BRITISH SHELLS ***

ILLUSTRATED INDEX

OF
BRITISH SHELLS.
CONTAINING
FIGURES OF ALL THE RECENT SPECIES,
With Names and other Information.

BY
G. B. SOWERBY, F.L.S.
LONDON:
SIMPKIN, MARSHALL, AND CO., PATERNOSTER ROW
AND
THE AUTHOR, 9, PEMBROKE SQUARE, KENSINGTON.
1859.

PREFACE.

The first idea of this Work arose from the occasional use of Mr. Damon's 'Catalogue of the Shells of Great Britain and Ireland,' named after the 'History of British Mollusca,' etc. Here was, on one side, a very useful list of names, without figures or references to guide an inquirer to the species; and on the other, a very large and splendid Work, placed by its heavy price beyond the reach of Naturalists in general. There was nothing between the two; nothing within reach that would assist collectors of British Shells to name their acquisitions from ocean, beach, river, pond, or hedgerow. A public want seemed here to be discovered. It was thought that a list like the above, with a coloured drawing of every species, if produced at an available price, would supply that want. At first no more than this was thought of, but

afterwards, as the preparations advanced, various suggestions were made tending to some extension of the plan. The results are here presented.

About 700 Figures in the Plates represent all the recognized species. They amount to 600, being 60 more than are admitted in the latest monograph. The greater part of these are either new or newly introduced as British. The rest are raised from varieties to species.

As the nomenclature used by Messrs. Forbes and Hanley in their great Work will be most generally adopted in this country, it is followed here with very few alterations; and the synonyms given (in italics) are selected from those which are most familiar and likely to be met with.

Some information is added respecting habits and localities, range of depth of marine species, and comparative abundance. The letters C., Cc., Mc., R., Rr., and Mr., signifying 'common,' 'extremely common,' 'moderately common,' 'rare,' etc., must be taken with great caution, especially the latter, as subject to continual change.

A list is given of British Fossil Shells identical with now living species.

A familiar description of British Mollusca and some peculiarities in their families and genera will, it is hoped, form a useful and suitable introduction to the names and figures. The Author's best thanks are due to his friends, Madame De Burgh, who has collected many beautiful shells; R. M'Andrew, Esq., F.L.S., whose dredging experiences are so extensive; J. G. Jeffreys, Esq., F.R.S., etc., whose researches have resulted in so many additions to our Fauna; Mr. Brice Wright, the Naturalist, of Great Russell Street, and others, who have in the most liberal manner supplied him with specimens from which his drawings are taken. The loan of delicate and minute shells to authors and artists has so often proved disastrous, that nothing but a generous friendship, and a zealous love of Natural History, could have induced proprietors so freely to incur the risk.

INTRODUCTION

TO

SHELL-BEARING BRITISH MOLLUSCA.

A common garden Snail, observed crawling and feeding, is easily seen to possess a distinct head, upon which are four feelers or horns, with eyes at the end, and underneath which is a mouth; a broad disc, extending the whole length of the body, is the organ of locomotion; a spiral or coiled shell surmounts the whole.

If, on the other hand, an Oyster be opened, no distinct head, or place for eyes or mouth, can be seen, but only a seeming mass of pulp, with a round gristle in the centre (the muscle of attachment), the former enclosed in a soft skin with a double fringe round the edge (mantle), placed within a shell composed of two pieces.

All Mollusca are more or less completely represented by these two. The Snail and its class are named *Gasteropoda* (abdomen-creeping), and their shells *Univalves*; the Oyster and its class, *Acephala* (headless), and their shells *Bivalves*. We begin with the latter class at—

PLATE I.

Pholadidæ, or "Borers," 1-13, have the power of piercing and inhabiting holes in submarine wood, rocks, and stones. The hinder part of the body is lengthened out in the form of a double tube, with openings for the ingress and egress of fluids. *Teredo*, 1-6, has short valves, two pallets fixed at the sides of the siphons, and lines its hole with a shelly tube. *Pholas* and *Pholadidea*, 8-13, have long valves and sometimes accessory plates. The shell of *Xylophaga*, 7, is like that of *Teredo*, but the animal has neither pallets nor tube.

Gastrochæna, 14, has a gaping shell, and encloses itself in a bottle-shaped covering of cemented stones and sand. *Saxicava*, 15, 16, *Venerupis*, 18, and *Petricola*, 17, burrow in sand, or live in holes of rocks.

Myadæ, or "Gapers," *Mya*, 19, 20, and *Panopæa*, 21, 21*, burrow downwards in sand and mud, their siphons protected by a coriaceous covering. Their shells are not closed all round, and those of *Mya* have a spoon-shaped process in the hinge of one valve.

Corbulidæ, 22-28, have short siphons with fringed edges, and the foot protrudes through a hole in the otherwise closed mantle.

PLATE II.

Poromya, 1, 1*, and the beautiful pearly *Pandora*, 2, 3, might be included in the family of *Corbulidæ*. *Lyonsia*, 4, and *Thracia*, 5-10, represent the *Anatinidæ*. 11-18 are *Solenidæ*, or "Razor-shells." Solens burrow in deep holes in the sand, where they remain in a perpendicular position at a great depth below the surface, where yet their presence may be traced by the opening left. Their large muscular foot enables them to take great leaps when out of the holes.

PLATE III.

Psammobia, 1-4, and *Tellina*, 5-16, include some of our most beautiful shells. The animals have a large, fleshy, curved foot, and the mantle open and fringed. Their shells are provided with an external ligament, and teeth on the hinge. *Donax*, or "Wedge," 19, 20, has long, separated, fringed siphons. Besides the small external ligament, properly so called, the *Mactridæ* have a triangular pit in the hinge of their shell, containing an elastic substance or spring, designed to resist the action of the adductor muscles. *Macra*, 21-26, ends the Plate.

PLATE IV.

Lutraria, 1-3, is another genus of *Mactridæ*. The *Veneridæ*, 4-16 and 23, or "Venus" tribe, differ from it in not having the spring-holding pit. *Artemis*, 10, 11, is known by a large angular bend in the pallial impression; and *Cytherea*, 23, is distinguished from Venus by one tooth on the hinge standing out a little way from the central ones. *Astarte*, 17-22, belongs to the *Cyprinidæ*.

PLATE V.

Fig. 1-3 also represent genera of *Cyprinidæ*. They have no bend in the pallial impression. *Cardium*, or "Cockle," 4-13, presents a peculiarity in the foot, which is large and bent, and enables the animal to leap in a very lively

manner. The family of *Lucinidæ* occupy the remaining figures. *Lucina* has a curious tongue-shaped doubling of the pallial impression.

PLATE VI.

The *Kelliadæ*, 1-15, are marine, while the *Cycladidæ*, completing the Plate, are fresh-water bivalves, and may be found in many ponds and rivers.

PLATE VII.

Of the *Unionidæ*, or "Fresh-water Mussels," *Unio*, 1-3, has teeth on the hinge, and *Anodon*, 4, has not: it is a thin shell. The foot of these molluscs is very large, and is used in cutting the animal's way through the mud in which it burrows. Of the marine, or true Mussels, *Modiola*, 6-9, and *Crenella*, 12-17, have a portion of the shell reaching beyond the point or apex, which in *Mytilus* is terminal, 18-21. *Dreissina*, 5, is shaped like *Mytilus*, but lives in fresh-water docks. The *Mytilidæ* attach themselves by means of horny threads spun from the foot, and called a "byssus."

PLATE VIII.

Shells of *Arcadæ*, 1-14, have the hinge characterized by a series of teeth on each side of the apex. *Nucula*, *Arca*, and *Pectunculus* are easily distinguished. *Limopsis*, 14, has a spring pit between the two rows. Our figure of *Pinna*, 16, is from a young specimen: it grows to great size, and spins a very silky byssus. The shell of *Anomia*, 18, is fixed to rocks, etc., by means of a bony button passing through a hole or sinus in the lower valve. *Lima*, 22-24, has a light, thin shell, and its mantle is adorned with beautiful fringes. It swims rapidly through the water by the opening and shutting of its valves, and also has the habit, in seasons of rest, of protecting itself by a network or vest of marine fragments strung together by its byssal threads.

PLATE IX.

Contains the shells of our Pectens, or Scallops, 1-16. The animals swim like *Lima*, as above, and also spin a byssus.

The class *Brachiopoda*, 17-24, so named because what seem to be organs of locomotion consist of a pair of coiled, ciliated arms, ends the list of Headless Molluscs and their bivalve shells.

PLATE X.

Class *Pteropoda*, 1-4, contains minute Mollusca, with glassy shells variously formed, and wing-like expansions for swimming.

Chiton, 5-17, or "Coat of Mail," which may often be seen incrusting rocks, commences the *Gasteropoda*; although seemingly fixed, they are capable of locomotion, using the whole of the oval disc, which may be seen on turning them over, for the purpose. The same may be said of the "Limpet" tribe, *Patella*, 1-22.

PLATE XI.

This Plate commences with other forms of the Limpet tribe, *Fissurella*, 1, 2, with a hole at the top; *Puncturella*, 3, with a fissure near the top; and *Emarginula*, 4-6, with a slit at the margin. *Haliotis*, 7, or pearly "Ear-shell," leads to the *Trochidæ*, or "Tops," 8-27.

PLATE XII.

Ianthina, 1-4, is genus of oceanic molluscs, which are provided with a beautiful floating apparatus, on which the female carries her egg-bags.

Neritina, 5, and the *Paludinidæ*, 8-11, live in fresh-water. The animal of *Paludina* is sprinkled all over with bright, golden specks. The *Littorinæ*, or "Winkles," 12-24, are marine, and frequent the shore among seaweeds. *Lacuna*, 25-32, belongs to the same family.

PLATE XIII., XIV.

Are small genera, various in their characters and habits, but reasonably included in the last family, *Littorinidæ*.

PLATE XV.

The animals of *Turritella*, 1-3, and *Cæcum*, 6, 7, are not unlike, although the shells are so different; eyes at the base of the tentacles, a short foot, and horny operculum. The somewhat similar animal of *Aporrhais*, 4, 5, is brilliantly coloured with gold and red. *Cerithium*, 8-10, differs from *Cerithiopsis*, 11-15, more in regard to the animal than the shell; the latter animal has a retractile proboscis, and its operculum is not spiral.

PLATE XVI.

Chemnitzia, 1-11, and *Truncatella*, 12, together with *Stylifer*, *Eulima*, and *Odostomia*, form the family of *Pyramidellidæ*. Notwithstanding the difference in the shells, the animals are very similar, having a retractile proboscis, and eyes immersed at the base of their tentacles.

The animal of *Natica*, 13-19, has a lobe on the upper part of the foot, reflected over the shell in front, and another lobe behind. The shells of *Laminaria*, 23, 24, are completely enveloped in similar lobes.

PLATE XVII.

The shells of *Odostomia* are known by a fold in the inner lip of the aperture.

PLATE XVIII.

Two *Tritons*, 1, 2, are admitted for the first time as British, on what we consider fair evidence, although only two or three specimens have been taken off Guernsey. One of *T. nodiferus* was incrustated by a truly British *Lepralia*. *Murex*, 3, is now familiar as an aquarian, as well as *Purpura*, 5. A milky secretion found in the head turns purple when exposed, and gives the celebrated purple dye. *Buccinum*, 7-13, includes the common "Whelk," 8. *Fusus Berniciensis*, 14, is among the rarest and most beautiful of our British shells.

PLATE XIX.

Nassa reticulata, 3, is a favourite inhabitant of the tank; it burrows in search of food among the pebbles, elevating its siphon above the surface. *Mangelia*, 4-26, has a similar long siphon. The shell of *Erato*, 27, is

wrapped in the lobes of the animal's mantle, as is also that of *Cyprea*, 28, the latter presenting a beautiful object.

PLATE XX.

After *Tornatella*, 1, and *Ovula*, 2, 3, come *Bullidæ*, 4-27, including several genera differing remarkably in the shape and disposition of the lobes of their mantles, which in some instances, as in *Philine*, 20-25, cover the shell. The stomach of *Scapander*, 26, 27, is a remarkable kind of mill, composed of two bones, between which the food is ground. The shells of *Aplysia*, 28, and *Pleurotranchus*, 29, 30, are quite internal. *Spirula*, 31, belonging to the Cephalopodous or Cuttlefish tribe, is only introduced doubtfully, as there is no proof of the species living in our seas.

PLATE XXI., XXII., XXIII., XXIV.

These are occupied by shells of *Gasteropoda pulmonifera*, so named because they breathe air by a pulmonary cavity, instead of water by gills. The *Limnæadæ*, or fresh-water Snails, contained in Plate I., live in water, but breathe by exposing the pulmonary cavity to air at the surface. *Succinea*, Plate XXII., 1-3, lives by the sides of streams, sometimes immersed. *Conovulus*, 4-7, affects brackish marshes. Our common Slugs yield, on dissection, thin shells, which are concealed beneath the mantle, *Limax*, 9-12. The shells of *Testacellus*, 13, 14, are fixed near the end of the foot, outside. The glassy shells of *Vitrina*, 15, 16, do not enclose the whole Slug-like animal. The other genera of Land Snails have shells large enough to receive the animal when retracted and withdrawn for repose.

BRITISH FOSSIL SHELLS.

The following living species are also found in a fossil state in drift and strata of the British Isles. They are all enumerated in Mr. Searles Wood's monograph of the CRAG Mollusca.

The numbers refer to the species as figured in our Plates, Mr. Wood's nomenclature being inserted in italics when differing from ours.

PL. I. Teredo 2. Pholas 11. Pholadidea 12. Gastrochæna 14, as *dubia*. Saxicava 15, 16. Venerupis 18. Mya 19, 20. Panopea 21, 21*, as *Faujasii*. Corbula 22, 23? Sphænia 25. Neæra 27.

PL. II. Poromya 1. Pandora 2, as *inæquivalvis*, 3, as *pinna*. Thracia 7, 8. Solen, 13, 15. Solecurtus 18, as *Mactra strigilata*. Syndosmya, as *Abra*, 19, 22.

PL. III. Psammobia 1, 3, 4. Tellina 5, 6, 7, 9, as *ovata*, 15, as *Balthica*, 16. Scrobicularia 18, as *Trigonella plana*. Donax 19, as *vittatus*. Mactra 20, 21, 22, as *ovalis*, 23, 25, 26.

PL. IV. Lutraria 2. Tapes 7, 8. Artemis 11. Venus 12, 14, 15, 17. Astarte 18, 19, 20, 22, as *borealis*.

PL. V. Cyprina 1. Circe 2. Isocardia 3. Cardium 6, 8, 9, 12, 13. Lucina 14, as *Loripes*, 15, as *Cryptodon flexuosum*, 16. Diplodonta 19. Clausina 20, as *Cryptodon f.*

PL. VI. Montacuta 1, 2, 3. Kellia 5, 6. Poronia 7, as *Kellia r.* Cyclas 16, 18. Pisidium 23, 24, 25, 26.

PL. VII. Unio 2, 3. Anodon 4. Modiola 6, 9, 10. Crenella 13, 14, 15, 16. Mytilus 18, 19, 20.

PL. VIII. Nucula 1, 5. Leda 6, 7. Area 9, 10, 11, as *pectunculoides*. Pectunculus 13. Avicula 15. Pinna 16. Ostrea 17. Anomia 18, 19, 20, 21. Lima 22, 23, 24.

PL. IX. Pecten 1, 2, 5, 8, 9, 10, 11, 12, 13, 14.

PL. X. Chiton 7, as *fascicularis*. Patella 18. Acme 23. Propilidium, as *Tectura*, 24. Dentalium 26, 27. Capulus 28. Calyptræa 29.

PL. XI. Fissurella 1, as *græca*. Puncturella 3. Emarginula 5, 6. Trochus 8, 10, 11, 12, as *papillosus*, 15, 16, 17, 19. Margarita 21. Adeorbis 25. Scissurella 26.

PL. XII. Paludina 9. Valvata 10. Littorina 14. Lacuna 27.

PL. XIII. Rissoa 3*, as *Paludestrina subumbilicata*, 7, 11?, 13?, 15, 23, 27.

PL. XIV. Rissoa 2, 9, as *Chemnitzia*. Aclis 24, as *Alvania ascaris*, 25, as *Chemnitzia n.*

PL. XV. Turritella 2. Aporrhais 4. Cæcum 6, 7. Cerithium 9, 10. Cerithiopsis, as *Cerithium*, 11. Scalaria 17, 19, 20. Eulima 22, 25.

PL. XVI. Chemnitzia 1, 2, as *similis*, 4, 8. Natica 13, 17, as *Catena*, 19, as *Grænlandica*. Velutina 21. Trichotropis 26.

PL. XVII. Odostomia 1, 8, as *plicata*, 9, as *plicata*, 10?, 12.

PL. XVIII. Purpura 5. Buccinum 10? Fusus 15, as *Trophon*, 16, as *Tr.*, 17, as *Tr. gracile*, 18, as *Tr. T.*, 19, as *Tr. p.* Trophon 22, 24.

PL. XIX. Nassa 1, 2. Mangelia, as *Clavatula*, 4, 5, 8, as *Philberti*, 9, 11, as *Boothii*, 12, 14, 15, 17?, 21, 22. Erato 27. Cypræa 28.

PL. XX. Tornatella 1, as *Actæon tornatilis*. Ovula 3, as *Bulla*. Cylichna, as *Bulla*, 4, 5, as *regulbiensis*, 7, 9, 12. Bulla 19. Philine as *Bullæa*, 21, 22. Scaphander, as *Bulla*, 26.

PL. XXI. Limnæa 1, 4, 7, 11. Planorbis 18, 24, as *complanatus*, and in the upper fresh-water formation, all except 9, 11.

PL. XXII. Succinea 1, 3. Conovulus 5, and in the more recent fresh-water deposits, all except 7 and 13 to 16.

PL. XXIII. Helix 19, 22, and upper fresh-water, all except 1, 12, 14, 15, 24, 25, 26.

PL. XXIV. Upper fresh-water deposits, all except 16, 24.

INDEX TO THE GENERA.

Referring to Plates and Figures.

Achatina, [xxiv](#). 24.

Aclis, [xiv](#). 23, 25.

Acmaea, [x](#). 22, 23.

Acme, [XXIV](#). 26.
Adeorbis, [XI](#). 25.
Akera, [XX](#). 16.
Amphispira, [XX](#). 14, 15.
Ancylus, [XXI](#). 14-17.
Anodonta, [VII](#). 4.
Anomia, [VIII](#). 18-21.
Aplysia, [XX](#). 28.
Aporrhais, [XV](#). 4, 5.
Arca, [VIII](#). 18-12.
Argiope, [IX](#). 20, 23.
Artemis, [IV](#). 10, 11.
Assimineia, [XIII](#). 1, 2.
Astarte, [IV](#). 17, 22.
Avicula, [VIII](#). 15.
Azeca, [XXIV](#). 23.

Balea, [XXIV](#). 17.
Barleeia, [XIV](#). 12.
Bithinea, [XII](#). 6, 7.
Buccinum, [XVIII](#). 8-13.
Bulla, [XX](#). 17-19.
Bulimus, [XXIV](#). 1-4.

Calyptræa, [X](#). 29.
Capulus, Pileopsis.
Cardium, [V](#). 4-13.
Carychium, [XXII](#). 8.
Cæcum, [XV](#). 6-7.
Cemoria, Puncturella.
Ceratisolen, [II](#). 11.
Cerithiopsis, [XV](#). 11-15.
Cerithium, [XV](#). 8-10.
Chemnitzia, [XVI](#). 1-11.
Chiton, [X](#). 5-17.
Circe, [V](#). 2.
Clavatula, Mangelia.

Clausilia, [XXIV](#). 18-21.
Clausina, [V](#). 20-22.
Cochlodesma, [II](#). 10.
Conovulus, [XXII](#). 4-7.
Corbula, [I](#). 22-24.
Crania, [IX](#). 24.
Crenella, [VII](#). 12-17.
Cyclostoma, [XXIV](#). 25.
Cyclas, [VI](#). 16-20.
Cylichna, [XX](#). 4-13.
Cypræa, [XIX](#). 28.
Cyprina, [V](#). 1.
Cytherea, [IV](#). 23.

Dentalium, [X](#). 26, 27.
Diodonta, [II](#). 16, 16*.
Diplodonta, [V](#). 19.
Donax, [III](#). 19, 20.
Dreissina, [VII](#). 5.

Emarginula, [XI](#). 4-6.
Euomphalus, [XIV](#). 17, 18.
Ervillia, [III](#). 17.
Eulima, [XV](#). 22-26.
Eulimella, [XIV](#). 26-30.

Fissurella, [XI](#). 1, 2.
Fusus, [XVIII](#). 14-19.

Galeomma, [VI](#). 14, 15.
Gastrochæna, [I](#). 14.

Haliotis, [XI](#). 7.
Helix (Zonites), [XXII](#). 17-29.
Helix, [XXIII](#). 1-29.
Hinnites, Pecten, [IX](#). 1.
Hippothyris, [IX](#). 17.
Hyalæa, [X](#). 1.

Ianthina, [XII](#). 1-4.

Isocardia, [V](#). 3.

Jeffreysia, [XIV](#). 13-16.

Kellia, [VI](#). 5-8.

Lachesis, [XVIII](#). 6.

Lacuna, [XII](#). 25-32.

Lamellaria, [XVI](#). 23, 24.

Leda, [VIII](#). 6, 7.

Lepton, [VI](#). 9-13.

Lima, [VIII](#). 22-24.

Limax, [XXII](#). 9-12.

Limnæa, [XXI](#). 1-11.

Limopsis, [VIII](#). 14.

Littorina, [XII](#). 12-24.

Lucina, [V](#). 14-18.

Lucinopsis, [IV](#). 9.

Lutraria, [IV](#). 1-3.

Lyonsia, [II](#). 4.

Mactra, [III](#). 21-26.

Mangelia, [XIX](#). 4-26.

Margarita, [XI](#). 21-24.

Marginella, Erato.

Megathyris, Argiope.

Modiola, [VII](#). 6-11.

Montacuta, [VI](#). 1-3.

Murex, [XVIII](#). 3, 4.

Mya, [I](#). 19-20.

Mytilus, [VII](#). 18-21.

Nassa, [XIX](#). 1-3.

Natica, [XVI](#). 13-19.

Neæra, [I](#). 26-28.

Neritina, [XII](#). 5.

Nucula, [VIII](#). 1-5.

Odostomia, [xvii](#). 1-31.

Ostrea, [viii](#). 17.

Otina, [xvi](#). 25.

Ovula, [xx](#). 2, 3.

Paludina, [xii](#). 8, 9.

Pandora, [ii](#). 2, 3.

Panopæa, [i](#). 21, 21*.

Patella, [x](#). 18-21.

Pecten, [ix](#). 1-16.

Pectunculus, [viii](#). 13.

Petricola, [i](#). 17.

Phasianella, [xi](#). 27.

Philine, [xx](#). 20-25.

Pholas, [i](#). 8-11, 13.

Pholadidea, [i](#). 12.

Physa, [xxi](#). 12, 13.

Pileopsis, [x](#). 28.

Pilidium, [x](#). 24.

Pinna, [viii](#). 16.

Pisidium, [vi](#). 21-28.

Planorbis, [xxi](#). 18-28.

Pleurobranchus, [xx](#). 29.

Pleurotoma, Mangelia.

Poromya, [ii](#). 1, 1*, [ix](#).

Propilidium, [x](#). 25.

Psammobia, [iii](#). 1-4.

Puncturella, [xi](#). 3.

Pupa, [xxiv](#). 5-16.

Purpura, [xviii](#). 5.

Recluzia, [xvi](#). 20.

Rissoa, [xiii](#). 3-28; [xiv](#).

Saxicava, [i](#). 15, 16.

Scalaria, [xv](#). 16-20.

Scaphander, [xx](#). 26, 27.

Scrobicularia, [III](#). 18.
Scissurella, [XI](#). 26.
Segmentina, Planorbis, [XXI](#). 27, 28.
Skenea, [XIV](#). 19-22.
Solen, [II](#). 12-15.
Solecurtus, [II](#). 17, 18.
Spirialis, [X](#). 2-4.
Spirula, [XX](#). 31.
Stylifer, [XV](#). 21.
Succinea, [XXII](#). 1-3.
Syndosmya, [II](#). 19-22.
Sphænia, [I](#). 25.

Tapes, [IV](#). 4-8.
Tellina, [III](#). 5-16.
Terebratula, [IX](#). 18-20.
Teredo, [I](#). 1-6.
Testacellus, [XXII](#). 13, 14.
Thracia, [II](#). 5-9.
Tornatella, [XX](#). 1.
Trichotropis, [XVI](#). 26.
Triton, [XVIII](#). 1, 2.
Trochus, [XI](#). 8-20.
Trophon, [XVIII](#). 20-24.
Truncatella, [XVI](#). 12.
Turritella, [XV](#). 1-3.
Turtonia, [VI](#). 4.

Unio, [VII](#). 1-3.

Valvata, [XII](#). 10, 11.
Velutina, [XVI](#). 21, 22.
Venerupis, [I](#). 18.
Venus, [IV](#). 12-16.
Vertigo, Pupa.
Vitrina, [XXII](#). 15-16.

Xylophaga, [I](#). 7.

Zonites, [XXII](#). 17-29.

Zua, [XXIV](#). 22.

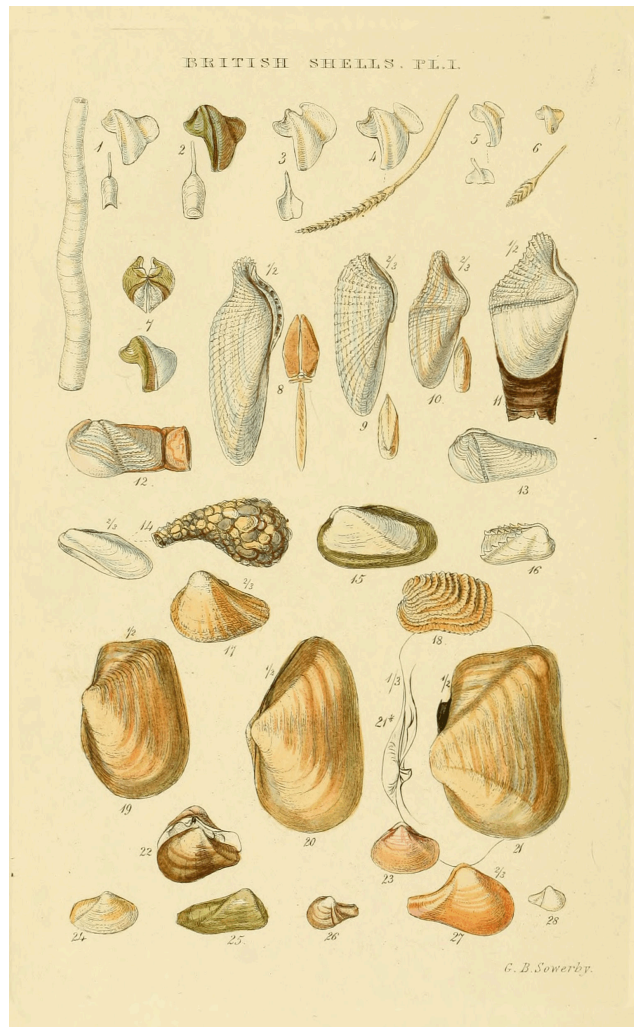


PLATE I.

TEREDO. Pierces and inhabits wood.

Fig.

1. *T. NAVALIS*, *Linn.*—*T. Batavæ*.—Herne Bay, R. Pallets shelly, crescented; valves with auricles extended sidewise; tube long.
2. *T. NORVEGICA*, *Spengl.*—*T. Bruguieri*.—Port Patrick, Teignmouth, etc., Mc. Pallets shelly, not crescented; auricles not extended.

3. *T. MEGOTARA*, *Hanl.*—*T. nana* (young).—Herne Bay, Devon, Swansea, etc., R. Pallets shelly, spade-shaped; auricles raised and expanded.
4. *T. BIPENNATA*, *Turt.*—Ireland, Scarborough, etc., R. Pallets horny, long; auricles lobed.
5. *T. MALLEOLUS*, *Turt.*—Torquay, Rr. Pallets shelly, mallet-shaped; valves narrow.
6. *T. PALMULATA*, *Lamk.*—*T. bipalmulata*.—Floating wood, Ire., S. Eng. Pallets horny, short; valves small, shaped like *T. navalis*.

XYLOPHAGA. In fixed and floating timber.

7. *X. DORSALIS*, *Turt.*—Ayr., Dubl., Exm., Scarb., etc., C. Two dorsal plates, no tube, no pallets.

PHOLAS. Pierces and inhabits stone.

8. *P. DACTYLUS*, *Linn.*—*P. hians*, *P. callosa*.—Eng., Scot., Ire., C. Dorsal plates four; hinge pitted.
9. *P. CANDIDA*, *Linn.*—Eng., Scot., Ire., C. One dorsal plate; valves not beaked.
10. *P. PARVA*, *Penn.*—*P. callosa*, *P. tuberculata*.—Salcombe, Belfast, Cornwall, etc., M. C. One dorsal plate; valves beaked.
11. *P. CRISPATA*, *Linn.*—*P. bifrons*.—Hastings, Liverp., Scarb., Dubl., Forth, etc., C. No dorsal plate; valves broad, beaked, with oblique division.
13. *P. STRIATA*. Rare and doubtful as British. Valves closed.

PHOLADIDEA. Pierces and inhabits stone.

12. *P. PAPYRACEA*, *P. Loscombiana*, *P. Goodalli*.—S. Devon, N. Ire., R. Valves closed, with a cup. *P. lamellata*, *Turt.*, when young, not closed; no cup.

GASTROCHÆNA. Bores and inhabits stone, thick shells, etc.

14. *G. MODIOLINA*, Lamk.—*G. Pholadia*, *G. hians*, *Mya dubia*, *Mytilus ambiguus*, *Pholas faba*.—Torbay, Birterbuy, Galway, Weymouth, etc. Valves thin, gaping, enclosed in bottle-shaped tube.

SAXICAVA. Bores and inhabits stone.

15. *S. RUGOSA*, Linn.—*Mytilus r.*, *Hiatella r.*, *S. gallicana*, *S. pholadis*, *Byssomya pholadis*, *Mya byssifera*.—Eng., Scot., Ire., C. Valves oval.
16. *S. ARCTICA*, Linn.—(*Mya*) *Hiatella a.*, *Anatina a.*, *Hiatella minuta*, *Solen minutus*, *S. rhomboides*, *Agina purpurea*, *Solen p.*—Eng., Scot., Ire., C. Valves rhomboidal; angle serrated.

Omitted.—*S. fragilis*, Nyst. Not the young of *S. rugosa*. Vigo Bay.

PETRICOLA. Inhabits stone.

17. *P. LITHOPHAGA*, Retz.—*P. striata*, *costellata*, *rocellaria*, *ruperella*, Lamk., *Mya* and *Sphænia decussata*.—Cork, Bristol, R. as British.

VENERUPIS. Bores and inhabits stone.

18. *V. IRUS*, Linn. (*Donax*).—*Tellina cornubiensis*, *Cuneus foliaceus*, *Petric.*, and *Pullastra*, I.—In limestone at Plymouth and other southern coasts. Wedge-shaped.

MYA. Burrows in mud and sand.

19. *M. TRUNCATA*, Linn.—*M. ovalis* and *Sphænia Swainsoni* (young).—Various localities and depths, C. Truncated at end.
20. *M. ARENARIA*, Linn.—Herne Bay, Portsmouth, etc., C. Tapering at end.

PANOPÆA. Burrows in mud.

21. *P. NORVEGICA*, Spengl.—*P. glycimeris*, *P. arctica*.—Scarborough, Zetland (90 fath.), Rr. Quadrate.

21*. P. ALDROVANDI, *Lamk.*—Cornwall, one specimen, *Jeffreys*. Oblique.

CORBULA. Dredged in mud and sand.

22. C. NUCLEUS, *Lamk.*—*C. striata*, *Mya*, and *C. inæquivalvis*.—Dublin, Torquay, Forth, 4 to 80 fath., C. Valves unequal, beaked.

23. C. ROSEA, *Brown.*—Weymouth, Rr. as Brit. Valves not beaked.

24. C. OVATA, *Forbes.*—Isle of Man, Rr. Oval, truncated at end.

SPHÆNIA. Burrows in foliaceous shells.

25. S. BINGHAMI, *Turt.*—*Corbula B.*—Torquay, Guernsey, Swansea, Forth, etc., R.

NEÆRA. Lives incrustated in sand.

26. N. COSTELLATA, *Desh.*—*Corbula c.*, *N. sulcata*.—Loch Fyne, Rr. Radiately 3-ribbed.

27. N. CUSPIDATA, *Olivi.*—Forth, Northumberland, Cape Clear, etc., Mr. Pear-shaped.

28. N. ABBREVIATA, *Forbes.*—Loch Fyne, Mc. Radiately 1-ribbed.

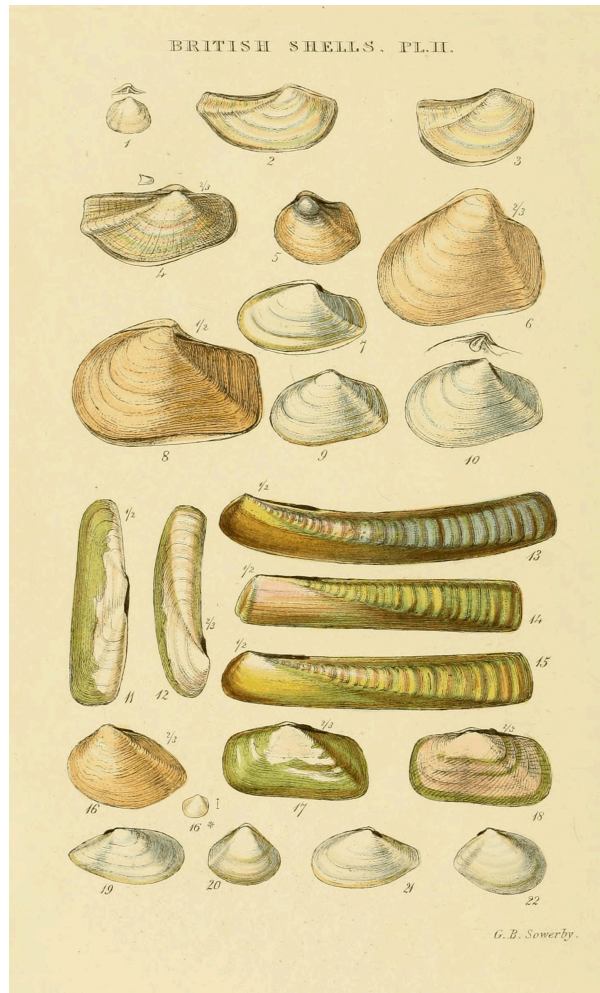


PLATE II.

POROMYA. On mud, in deep water.

Fig.

1. *P. GRANULATA*, Nyst.—*P. anatinoides*, Forbes.—Skye, R.
- 1*. *P. SUBTRIGONA*, Jeffreys.—Shetland, Mr. Minute, posteriorly expanded.
See Pl. IX.

PANDORA.

2. *P. ROSTRATA*, Lamk.—*Tellina inæquivalvis*, Linn.; *P. margaritacea* and *inæquivalvis*.—S. Devon?, Guernsey, R. as Brit. Valves long.

3. *P. OBTUSA*, *Leach*.—*Solen Pinna*, Mont.—S. Devon, Dorset, Mc. Ire., R. Valves short and broad.

LYONSIA. With byssal attachment.

4. *L. NORVEGICA*, *Chemn.*—*Mya* and *Anatina n.*, *Mya* and *Anatina striata*, *M. nitida*, *Lyonsia elongata*, *Myatella*, and *Osteodesma*.—Tenby, Bantry, Oban, etc., 5 to 80 fath. With movable ossicle on hinge.

THRACIA. Laminarian and Coralline.

5. *T. DISTORTA*, *Mont.*—(*Mya*) *Anatina* and *Amphidesma d.*, *Th. truncata*.—Distribution general, but not common. Short, truncated, irregular.
6. *T. CONVEXA*, *Wood.*—(*Mya*) *Anatina* and *Amphidesma c.*, *T. declivis*.—S. Devon, Bantry Bay, Arran, etc., Mc. Inflated.
7. *T. PHASEOLINA*, *Lamk.*—*Amphidesma p.*, *T. declivis*.—Various localities and depths, C. Oval, obliquely truncated.
8. *T. PUBESCENS*, *Pult.*—(*Mya*) *Anatina* and *Amphidesma p.*, *T. declivis*.—S. Devon, Cornwall, Belfast, etc., C. Oblong, straightly truncated.
9. *T. VILLIOSULCA*, *Macgillivray.*—*T. ovata*, Brown.—Exmouth, Swansea, Bantry Bay, etc. Various depths, Mc.

COCHLODESMA. Laminarian and Coralline.

10. *C. PRÆTENUA*, *Pult.*—(*Mya*) *Anatina*, *Ligula*, and *Amphidesma pr.*—Dorset, Devon, Man, Bantry B., Oban, etc., (25 to 60 fath.), R. Spoon-shaped process on hinge.

CERATISOLEN. Burrows in sand.

11. *C. LEGUMEN*, *Linn.*—(*Solen*) *Psammoiia* and *Solenicurtus l.*—Exmouth, Swansea, Bantry, etc., C. Hinge nearly central.

SOLEN. Burrows perpendicularly in sand.

12. *S. PELLUCIDUS*, *Penn.*—Various British localities, 6 to 100 fath. Curved; hinge not terminal.
13. *S. ENSIS*, *Linn.*—Eng., Scot., Ire., generally 5 to 15 fath., C. Arched and slender.
14. *S. MARGINATUS*, *Pult.*—*S. vagina*, *Penn.*, not *Linn.*—Exmouth, Clyde, Cork, etc., littoral to 10 fath., Mc. Straight, with terminal constriction.
15. *S. SILIQUA*, *Linn.*—*S. novacula*, *S. ligula*.—Eng., Scot., Ire., many places; littoral and laminarian, C. Straight, end not constricted.

DIODONTA.

16. *D. FRAGILIS*, *Linn.*—(*Tellina*) *Psammobia f.*, *Tellina jugosa* and *ochroleuca*.—Dublin, Tenby, Weymouth, R. Oval or wedge-shaped.
- 16*. *D. BARLEEI*, *Jeffr.*—Arran, Rr. Minute, subquadrate.

SOLECURTUS.

17. *S. COARCTATUS*, *Gmel.*—*S. cultellus*, *S. emarginatus*, *S. antiquatus*, etc.—Torbay, Anglesea, Man, Skye, etc., R. Oblong, smooth.
18. *S. CANDIDUS*, *Renieri.*—(*Solen*) *S. strigilatus*, *Turt.*, not *Linn.*, *Psammobia scopula*.—S. Devon, Dublin, etc., R. Striated.

SYNDOSMYA.

19. *S. PRISMATICA*, *Mont.*—(*Ligula*) *Mya* and *Amphidesma p.*—Weymouth, Forth, Dublin, Antrim, etc., littoral, Mc. More pointed and narrow than *S. intermedia*.
20. *S. TENUIS*, *Mont.*—(*Mactra*) *Amphidesma t.*—Dorset, Scarborough, Antrim, etc., 5 to 100 fath., Mc. Short, subtrigonal.
21. *S. INTERMEDIA*, *Thompson.*—*Mya* and *Syndosmya nitida*, *Abra profundissima*.—In mud, 6 to 100 fath., Zetland, Skye, Cape Clear, etc., R. Broader and more pointed than *S. prismatica*.

22. *S. ALBA*, Wood.—(*Mactra*) *M.* and *Amphidesma Boysii* and *A. album*.—
Brit. co. generally, C. Short, rounded, oval.

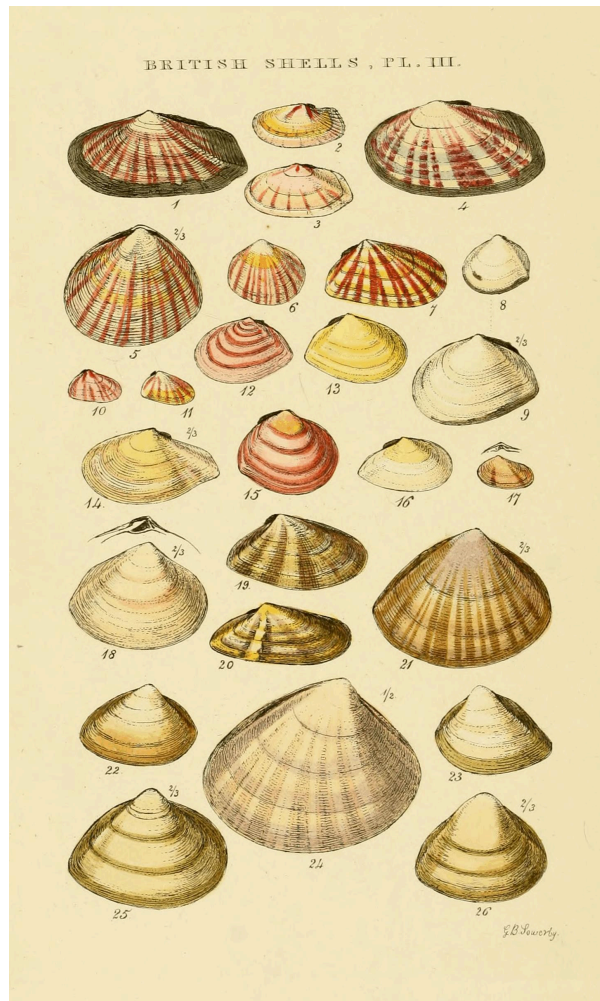


PLATE III.

PSAMMOBIA. Inhabits sand and mud.

Fig.

1. *P. FERROENSIS*, Chemn.—(*Tellina*) *T. incarnata*, *T. radiata*, *T. trifasciata*, *T. truncata*.—Brit. co. generally; coralline, C. Oblong, keeled and angulated behind.
2. *P. COSTULATA*, Turt.—*P. discors*.—Torbay, Cornwall, Skye, etc., R. Ribbed behind.

3. P. TELLINELLA, LAMK.—*P. florida*.—Exmouth, Guernsey, Tenby, Bantry B., Oban, Zetland, etc., 5 to 50 fath., C. Oval, smooth.
4. P. VESPERTINA, Chemn.—(*Lux V.*) *Solen*, *Sanguinolaria V.*, *Tellina depressa*, *variabilis*.—Weymouth, Scarborough, Tenby, Bantry, etc., Guernsey, C. Not angular nor keeled behind.

TELLINA.

5. T. CRASSA, Penn.—*T. maculata*, *T. rigida*.—S. Devon, Guernsey, Man, Tenby, Bantry, Forth, etc., low-water to 15 fath., C. Thick, orbicular.
6. T. BALAUSTINA, Linn.—Birterbuy, Skye, Exmouth, Rr. as British.
7. T. DONACINA, Linn.—*T. trifasciata*, *T. variegata*, *T. Llantivyi*.—Various British loc., and S. Europe, 5 to 50 fath., C. Wedge-shaped.
- 8, 9. T. PROXIMA, Brown.—Var. *T. calcarea* (9), *T. ovalis*, *T. sordida*, *Sanguinolaria sordida*, *Macroma tenera*.—Skye, etc., R.
- 10, 11. T. PYGMÆA, Phil.—Devon, Cork, Shetland, etc., 8 to 50 fath., C. Smaller, more convex, and more truncate than *T. donacina*.
- 12, 13. T. TENUIS, Da Costa.—*T. planata*, *T. polita*, *T. exigua*.—On sandy shores, in various localities, C. Flat, thin.
14. T. INCARNATA, Linn.—*T. squalida*, *T. depressa*.—Devon, Bantry, Tenby, R. Flat, beaked.
15. T. SOLIDULA, Pult.—*T. rubra*, Da Costa, *T. carnaria*, *T. zonata*, *Psammobia solidula*, *T. Balthica*.—All British coasts; littoral, C. Thick, inflated.
16. T. FABULA, Gronovius.—*T. discors*.—On sandy shores generally, C. Valve with oblique striæ (omitted by error).

ERVILLIA.

17. E. CASTANEA, Mont.—(*Donax*) *Mesodesma c.*, *Capsa c.*—Cornwall and Scilly Isles, R. Ligament within.

SCROBICULARIA.

18. *S. PIPERITA*, *Gmel.*—Generic syn.: *Mya*, *Venus*, *Trigonella*, *Macra*, *Lutraria*, *Listera*.—Specific syn.: *hispanica*, *borealis*, *plana*, *gaditana*, *Listeri*, *compressa*.—Scarborough, Southend, Swansea, Cork, Clyde, C. Broad, compressed.

DONAX.

19. *D. ANATINUS*, *Lamk.*—*D. trunculus*, *Linn.?*, *Cuneus vittatus*, *D. ruber* (young).—Brit. co. generally, sandy shores, C. Broad; margin crenulated.
20. *D. POLITUS*, *Poci.*—(*Tellina*) *D.* and *Capsa complanata*.—S. Devon, Bantry Bay, etc., Mr. Smooth; inner edge not crenated.

MACTRA. Lives on sandy beaches.

21. *M. STULTORUM*, *Linn.*—(*Cardium*) *Tellina radiata*, *Trigonella r.*, *M. cinerea*, *M. magna*.—British co. generally, C.
22. *M. ELLIPTICA*, *Brown.*—British co. generally, C. Oval.
23. *M. SUBTRUNCATA*, *Costa.*—*M. lactea*, *M. triangula*, *M. crassatella*.—Brit. co. generally, C. Obliquely cuneiform.
24. *M. HELVACEA*, *Chemn.*—*M. glauca*, *M. neapolitana*.—Odd valves found dead in Cornwall, R. Large, obliquely oval.
25. *M. SOLIDA*, *Linn.*—*Trigonella zonaria* and *gallina*.—Brit. co. generally, C. Trigonal, thick.
26. *M. TRUNCATA*, *Mont.*—Forth, Cork, Cornwall, etc., C. Trigonal, deep, thick.

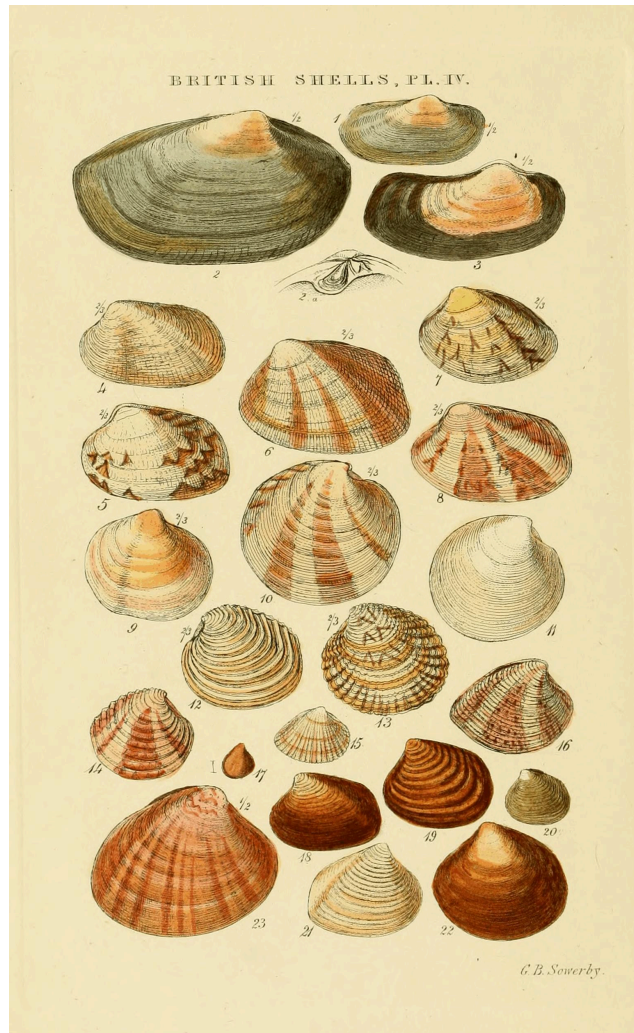


PLATE IV.

LUTRARIA. Inhabits mud.

Fig.

1. *L. INTERMEDIA?*, Sowb.—Perhaps only the young of *L. elliptica*.
2. *L. ELLIPTICA*, Lamk.—*Mactra lutraria*, Linn.—Scarborough, Torbay, Dublin, Forth, Lerwick, etc., C. Broad, oval.
3. *L. OBLONGA*, Chemn.—(*Mya*) *L. solenoides*, *L. hians*, *Mactra hians*, *Chama magna*.—Cornwall, Devon, etc., Mc. Arched, narrow.

TAPES. Burrows in mud and stone.

- 4, 5. T. PULLASTRA, *Wood*.—*Venus pullastra*, *V. perforans* (var. f. 4), *Venerupis nucleus*, *Pullastra vulgaris*, etc.—Plymouth, Scarborough, Swansea, Forth, etc., C.
6. T. DECUSSATA, *Linn.*—(*Venus*) *V. litterata*, *V. florida*, *Venerupis* and *Pullastra d.*—S. Devon, Weymouth, Tenby, etc., Mc. Oblong, coarsely decussated.
7. T. AUREA, *Gmel.*—(*Venus*) *V. nebulosa*, *V. ænea*, *V. nitens*, *V. sinuata*, *Pullastra a.*—Cornwall, Sussex, Bantry Bay, Loch Ryan, Mc. Yellow within.
8. T. VIRGINEA, *Linn.*—(*Venus*) *V. rhomboides*, *V. sarniensis*, *V. virago*, *Cuneus fasciatus*.—Various British loc., C. Pink within.

LUCINOPSIS, *Forbes and Hanley*. Sandy ground.

9. L. UNDATA, *Pennant*.—(*Venus*) *V. sinuosa*, *V. incorrupta*, *Lucina undata*.—Exm., Scarborough, Man, etc., 5 to 80 fathoms, M.R. Orbicular, flexuous at side.

ARTEMIS.

10. A. EXOLETA, *Linn.*—(*Venus*) *Cytherea e.*, *Pectunculus capillaceus*.—Sandy ground, on Brit. co. generally, low-water to 80 fath., C. Compressed; striæ coarse.
11. A. LINCTA, *Pult.*—(*Venus*) *V. sinuata*, *Cytherea sinuata*.—Various Brit. loc., C. Dorsal edge sloping; striæ fine.

VENUS.

12. V. CASINA, *Linn.*—*V. reflexa*, *V. lactea*, *V. discina*.—Northumberland, S. Devon, Man, Bantry, Skye, etc., Mc. Concentrically laminated.
13. V. VERRUCOSA, *Linn.*—*V. erycina*, *V. cancellata*, etc.—Devon, Cornwall, Channel Isls., etc., C. Ventricose, warted.

14. *V. FASCIATA*, *Donov.*—Various southern Brit, loc., C. Round, with thick ribs.
15. *V. OVATA*, *Pennant.*—Brit. co. generally, C. Oval, radiately ribbed.
16. *V. STRIATA*, *Donov.*—*V. gallina*, *V. Pennantii*, *V. rugosa*.—Brit. co. generally. Shores and deep water, C.

ASTARTE.

17. *A. TRIANGULARIS*, *Mont.*—*Mactra* and *Goodallia*, *minutissima* and *triangularis*.—Northern parts of Britain, 15 to 20 fathoms, C. Minute, triangular.
18. *A. ELLIPTICA*, *Brown.*—*Crassina e.*, *Crassina ovata*, and *C. sulcata*. Scotland, 5 to 10 fath., mud, M. C. Oblong, smooth near the margin.
19. *A. SULCATA*, *Costa.*—(*Pectunculus*) *V. scotica* and *Damnoniensis*, *Crassina sulcata*, etc.—Devon, Cornwall, Bantry, etc., C. Thickly ribbed.
20. *A. COMPRESSA*, *Mont.*—(*Venus*) *Venus* and *Crassina Montagu*, *Cyprina comp.*—Scarborough, Northumbria, Bantry Bay, Firth of Forth, 7 to 40 fath., M. C.
21. *A. CREBRICOSTATA*, *Forbes.*—Zetland and Skye (dead), R. Ribs more numerous than in *A. sulcata*.
22. *A. ARCTICA*, *Gray.*—(*Crassina*) *Venus borealis*, *Astarte compressa*.—Aberdeen, Zetland, St. Andrew's bay, etc., Rr. Lenticular, not ribbed.

CYTHEREA.

23. *C. CHIONE*, *Linn.*—(*Venus.*)—Plymouth, S. Devon, Falmouth, Carnarvon, 12 to 20 fath., Mr.

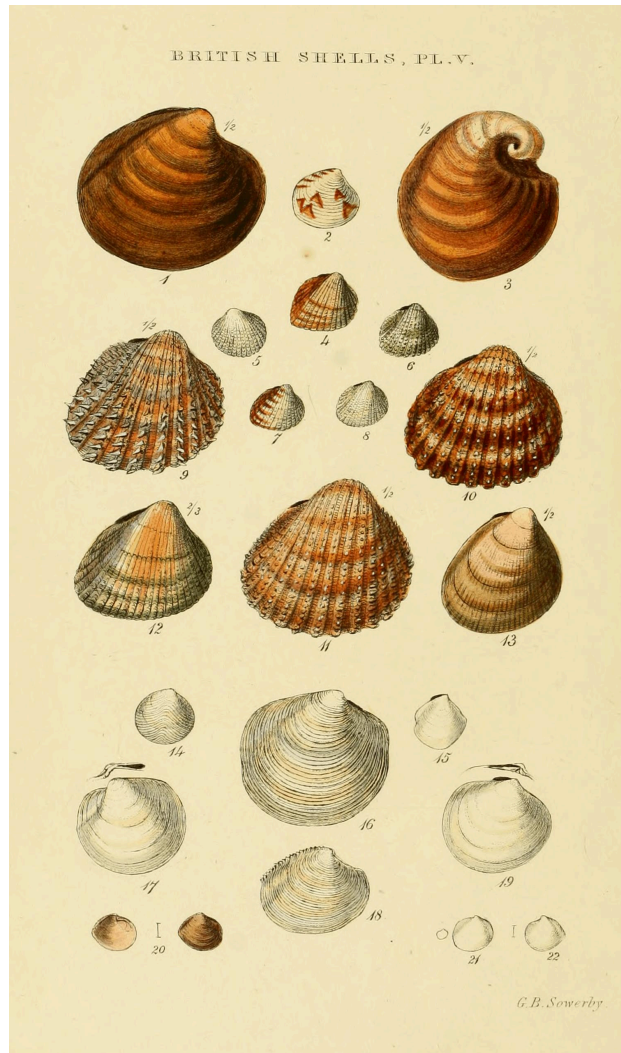


PLATE V.

CYPRINA.

Fig.

1. *C. ISLANDICA*, Linn. (*Venus*.)—*C. vulgaris*, *V. mercenaria*.—Brit. co. generally, most abundant north, 5 to 80 fath. Oval, with thick epidermis.

CIRCE.

2. *C. MINIMA*, Mont. (*Venus*.)—Gen. *Cyprina* and *Cytherea triangularis*, *apicalis*, *Cyrellii*, *minuta*.—Guernsey, Cornwall, Milford Haven, Clyde, Bantry Bay, etc., 10 to 50 fath., C.

ISOCARDIA.

3. *I. COR*, Linn. (*Chama*.)—*I. hibernica*, *Cardita cor*.—Dublin, Skye, etc., Mc. Heart-shaped, beaks coiled.

CARDIUM, or *Cockle*.

4. *C. PYGMÆUM*, Donovan.—*C. exiguum*.—Herne, Sark, Weymouth, Zetland, Bantry Bay, 5 to 50 fath., Mc. Hinder side large, angular.
5. *C. PAPILLOSUM*, Poli.—Guernsey, St. Martin's Port, *Jeffreys*, R. Rather rounded, equally ribbed and grained.
6. *C. PUNCTATUM*, Broc., *C. NODOSUM*, Turt.—*C. scabrum*, Ph.—Exmouth, Falmouth, Galloway, etc., Mc. Angular behind.
7. *C. FASCIATUM*, Mont.—*C. elongatum*, *C. ovale*, *C. parvum*, *C. rubrum* (young).—Various Brit. loc., Mr.
8. *C. SUECICUM*, Loven.—*C. Swediense*, Reeve, *C. Loveni*, Thompson.—Down, Oban, Zetland, 30 to 70 fath., Rr. Oval, equally ribbed.
9. *C. ACULEATUM*, Linn.—*C. ciliare* and *parvum* (young).—S. Devon, Mc. Thin, oblique, with thorn-like tubercles; interstices of ribs smooth.
10. *C. RUSTICUM*, Linn.—*C. tuberculatum*, Linn.?—Paignton, Torbay, Velvet Strand, Dublin co., Mc. Thick, not oblique, coarsely wrinkled.
11. *C. ECHINATUM*, Linn.—*C. mucronatum*, *C. spinosum*.—Brit. co. generally, mud and sand, 7 to 80 fath. Oblique, wrinkled; hinder tubercles sharp.
12. *C. EDULE*, Linn.—*C. vulgare*, *C. zonatum*, *C. obliquum*, *C. crenulatum*, *C. rusticum*.—Brit. co. generally. Common eatable, Cockle.

13. *C. NORVEGICUM*, *Spengler*.—*C. lævigatum*, *C. serratum*, *C. oblongum*, *C. Pennantii*.—Brit. co. generally. Sand and gravel, 15 to 80 fath. Smooth, variegated when young.

LUCINA.

14. *L. DIVARICATA*, *Linn. (Tellina.)*—*Cardium arcuatum*, *L. arcuata*, *commutata*.—Near Falmouth, 20 fath., Rr. Rounded, with diverging ribs.
15. *L. FLEXUOSA*, *Mont. (Tellina.)*—*Venus sinuosa*, *Amphidesma flexuosa*, *L. sinuata*; gen. *Cryptodon*, *Axinus*, and *Ptychina*.—Clyde, Bantry, Anglesea, etc., 7 to 80 fath., Mc. Thin, with a fold behind.
16. *L. BOREALIS*, *Linn. (Venus.)*—*Tellina radula*, *Lucina r.*, *L. alba*.—Many Brit. loc., 1 to 90 fath., C. Lenticular, concentrically laminated.
17. *L. LEUCOMA*, *Turt.*—*L. lactea*, not *Linn.*, *Loripes lactea*.—Torbay, Scarborough, Bantry Bay, etc., low-water to 80 fath., C. Round, smooth.
18. *L. SPINIFERA*, *Mont. (Venus.)*—*Myrtea s.*, *L. hiatelloides*.—Dartmouth, Plymouth, Hebrides, etc., 8 to 100 fath., M. R. Oval; dorsal margin spinose.

DIPLODONTA, *Mont.*

19. *T. ROTUNDATA*, *Mont. (Tellina.)*—*T. undata*, *Lucina r.*, *D. dilatata*.—Dorset, Devon, Tenby, Bantry Bay, etc., 7 to 25 fath., R.

CLAUSINA, *Jeffreys.*

20. *C. FERRUGINOSA*, *Forbes, (Kellia.)*—Skye, Loch Fyne, etc., 30 to 100 fath., C. Concentrically ribbed, minute, oval.
21. *C. CROULINENSIS*, *Jeffreys.*—Croulin Island, Shellands, R. Suborbicular.
22. *C. ABYSSLICOLA*, *Jeffreys.*—Not ribbed, R.

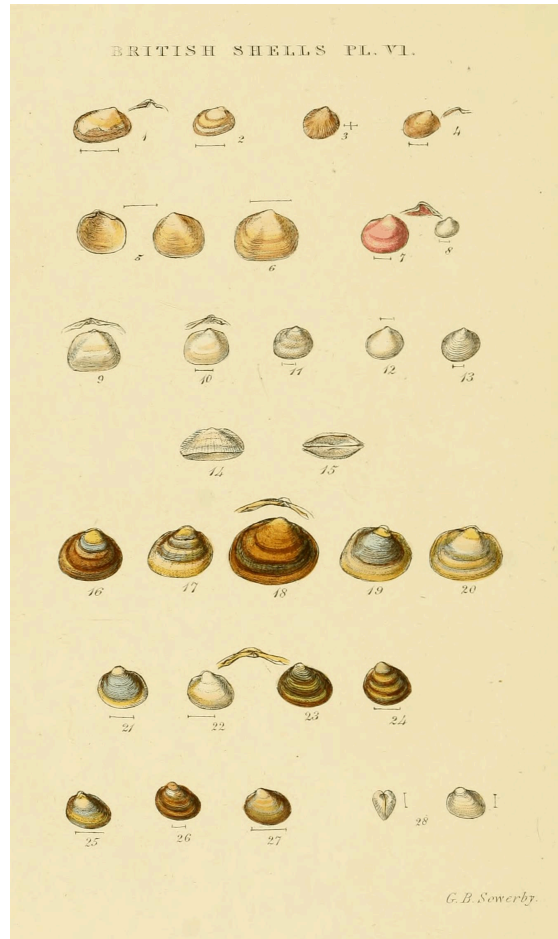


PLATE VI.

MONTACUTA.

Fig.

1. *M. FERRUGINOSA*, Mont. (*Mya*.)—*M. oblonga*, *M. glabra*, *Tellimya elliptica*, *Erycina* f.—S. Devon, Scarborough, Tenby, Cape Clear, etc., Mr. Wedge-shaped, long.
2. *M. BIDENTATA*, Mont. (*Mya*.)—*Erycina*, *Petricola*, and *Tellimya b.*—Newcastle, Weymouth, Dublin, etc. Burrows in dead shells, Mr. Short, subquadrate.
3. *M. SUBSTRIATA*, Mont. (*Mya*.)—*Erycina* and *Tellimya s.*—Devon, Tenby, Durham, Zetland, Bantry Bay, etc., on spines of *Spatangus*, deep

water, C. Minute, with diverging striæ.

TURTONIA.

4. T. MINUTA, *O. Fabr. (Venus.)*—*Mya purpurea*, etc.—Scarborough, Tenby, Cork, Oban, etc., C.

KELLIA.

5. K. SUBORBICULARIS, *Mont. (Mya.)*—*Bornia inflata*, *Tellina* and *Tellimya* s.—Anglesea, Portland, Zetland, etc., low-water to 60 fath., Mr.
6. K. LACTEA, *Brown, (Tellimya.)*—Brit. co. generally. More oblong than *K. sub.*
- 7, 8. K. RUBRA, *Mont. (Cardium.)*—*Poronia r.*, etc.—Numerous localities; littoral, C.

LEPTON.

9. L. SQUAMOSUM, *Mont.*—*Solen* and *Lutraria* s.—Salcombe Bay, Tenby, Cork, etc., R. Flat, subquadrate.
10. L. NITIDUM, *Turt. (Kellia.)*—Torbay, Skye, R. Less square and flat than *L. squamosum*.
11. L. CONVEXUM, *Alder.*—Bantry Bay, Rr. Convex, granulated.
12. L. CLARKIÆ, *Clark.*—Exmouth, Plymouth, Zetland, Rr. Minute, oval, smooth.
13. L. SULCATULUM, *Jeffer.*—Annals, Feb. 1859. Guernsey, Mc. Concentrically wrinkled.

GALEOMMA. Adheres to rocks by byssus.

- 14, 15. G. TURTONI, *Sowb.*—Channel Islands, and Exmouth. Oblong, gaping.

CYCLAS. Inhabits fresh-water.

16. C. CORNEA, *Linn. (Tellina.)*—*Sphærium c.*, Scopoli, *C. rivalis*, etc.—Ponds and ditches everywhere. Equilateral, subquadrate.
17. C. PISIDIOIDES, *Gray.*—*C. cornea*, var., Brit. Moll.—Ditches and ponds, C. Oblique, and light coloured.
18. C. RIVICOLA, *Leach.*—*Cardium nux*, etc.—Thames, etc. Large, oval, wrinkled.
19. C. CALICULATA, *Drap.*—*C. lacustris*, etc.—Newcastle, Bristol, etc., Mr. With a ridge at the apex.
20. C. LACUSTRIS, *Drap.*—*pallida*, Gray.—Regent's Park, etc., Mr. Oval, light.

PISIDIUM. Inhabits fresh-water.

21. P. OBTUSALE, *Pfeiffer.*—*P. pusillum*, var.?
22. P. PUSILLUM, *Turton, (Tellina.)*—*Cyclas fontinalis* and *gibba*.—In marshes and ponds at various places, C. Includes *P. obtustale*?
23. P. AMNICUM, *Müller, (Tellina.)*—*T. rivalis*, *Cyclas palustris*, *C. obliqua*.—British rivers, etc., generally. *C. obliqua* ribbed.
24. P. NITIDUM, *Jennyns.*—Battersea, Swansea, etc., C. Thick, dark, blunt.
25. P. HENSLOWIANUM, *Sheppard, (Tellina.)*—*Cyclas appendiculata*, *P. acutum*.—Cambridge, Swansea, etc., R. With a projection on the umbones.
26. P. PULCHELLUM, *Jennyns.*—*Cyclas fontinalis*, etc.—Various localities, C. Ridged.
27. P. CINEREUM, *Alder.*—*Cyclas c.*—Various localities, Mc. Smooth.
28. P. ROSEUM, *Scholtz.*—Various loc., with *P. pulchellum* and *P. nitidum* M. C. Subquadrate; animal pink.

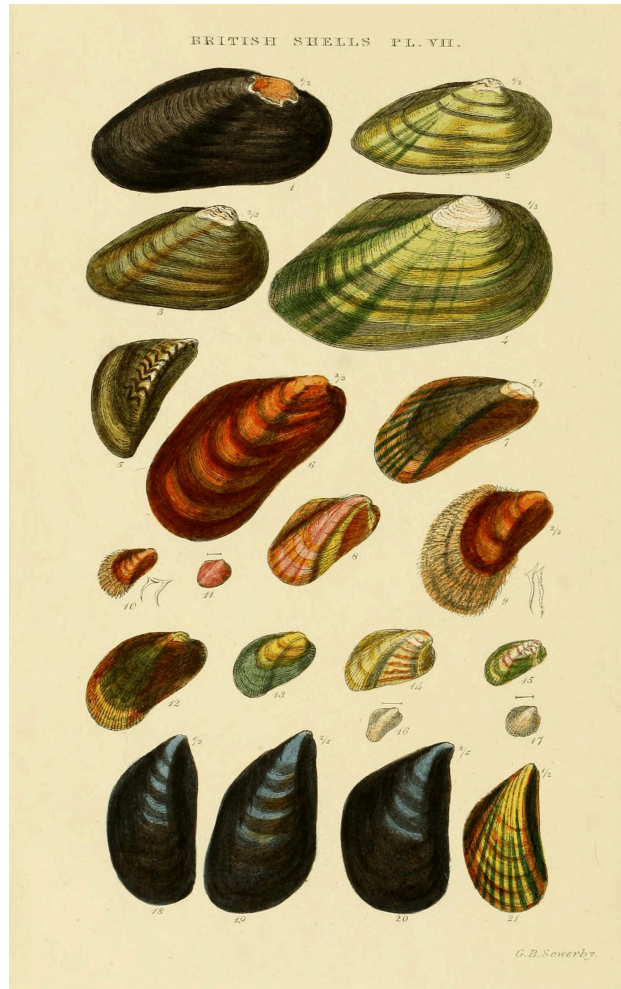


PLATE VII.

UNIO. Fresh-water Mussel with hinge-teeth.

Fig.

1. *U. MARGARITIFERUS*, Linn. (*Mya*.)—*Alasmodon m.*, *A. arcuata*, *U. elongata*, *U. sinuata*.—R. Conway; rivers in Cumberland, Wales, Ireland, etc., Mc. Long, thick, arched.
2. *U. PICTORUM*, Linn. (*Mya*.)—*U. ovalis*, *U. rostrata*, *U. limosus*, *U. Deshayesii*, etc.—Various British rivers, C. Long, oval, not wedge-shaped behind.

3. U. TUMIDUS, Retz.—*Mya* and *Unio ovalis*, *M. depressa*, *M. ovata*, *Myscei o.*, *M. solida*.—Thames, and various British rivers, C. More solid, deep, and wedge-shaped than *U. pictorum*.

ANODONTA. Fresh-water Mussel, without teeth.

4. A. CYGNEA, Linn. (*Mytilus*.)—*M. anatinus*, *M. stagnalis*, *M. paludosa*, *Symphonota cy.*—Britain and Europe generally, in rivers and canals, C.

DREISSINA, Beneden.

5. D. POLYMORPHA, Pallas, (*Mytilus*.)—London Docks, Edinburgh Canal, C.

MODIOLA, Lamarck.

6. M. MODIOLUS, Linn. (*Mytilus*.)—*M. umbilicatus*, *M. curtus*, *M. vulgaris*.—British coasts generally, C. Large, brown, not rayed.
7. M. OVALIS?, Sowb.—*M. tulipa (radiata)*, var., Brit. Moll., now named provisionally. Less tumid, and with longer and more gradually sloped dorsal line than *M. radiata*, Rr.
8. M. RADIATA, Hanley, Brit. Marine Conch.—*M. tulipa*, Brit. Moll., Mc. Tumid, radiated.
9. M. BARBATA, Linn. (*Mytilus*.)—*M. Gibbsii*.—Jersey, Torbay, Milford Haven, Youghal, R. as Brit. Barbs of epidermis serrated.
10. M. PHASEOLINA, Philippi.—Guernsey, Falmouth, Scotland, Mc. Small barbs of epidermis not serrated.
11. M. CUPREA, Jeffreys, Ann. Nat. Hist. Jan. 1859.—In the stomach of a bird shot at Scarborough, Rr. Minute, rhomboidal, smooth.

CRENELLA, Brown.

12. C. NIGRA, Gray, (*Modiola*.)—*Mytilus* and *Modiola discrepans*, *depressa*, *compressa*, *discors*.—Scotland, R. Broad and compressed behind.

13. *C. DISCORS*, *Linn.* (*Mytilus*.)—*Modiola d.* and *discrepans*.—Brit. co. generally, among seaweeds, C. Wedge-shaped and lobed.
14. *C. MARMORATA*, *Forbes*.—*Mytilus* and *Modiola discors*, *discrepans*, and *tumida*.—Brit. co. generally, Cc. Rhomboidal, marbled.
15. *C. COSTULATA*, *Risso*.—Exmouth, Cornwall, Swansea, Herm. Wedge-shaped, marbled, strongly-ribbed.
16. *C. RHOMBEA*, *Berkley*, (*Modiola*.)—*M. Prideauxiana*.—Guernsey, Torbay, Penzance, etc.; coralline, R.
17. *C. DECUSSATA*, *Montagu*, (*Mytilus*.)—*C.* and *Modiola elliptica*, *cicercula*, and *faba*.—Scarborough, and a few northern localities, 5 to 50 fath., Mc.; Norway, etc. Subquadrate, decussated, minute.

MYTILUS, *Linn.*

18. *M. EDULIS*, *Linn.*—*M. vulgaris*.—Brit. co. generally, Cc. Byssal margin swelled.
19. *M. UNGULATUS*, *Linn. Syst. Nat.*—*M. edulis*, var. *M. incurvatus* (Hoof-Mussel).—Cornwall, Guernsey, etc., Mr. Byssal margin bent inwards, lower margin broad and straight.
- 20, 21. *M. GALLOPROVINCIALIS*, *Lamk.*—*M. edulis*, var. *M. dilatatus?*, *M. pellucidus?* (young, f. 21), *M. subsaxatilis?*—British Channel, etc., C. Dorsal margin dilated, beak advancing.

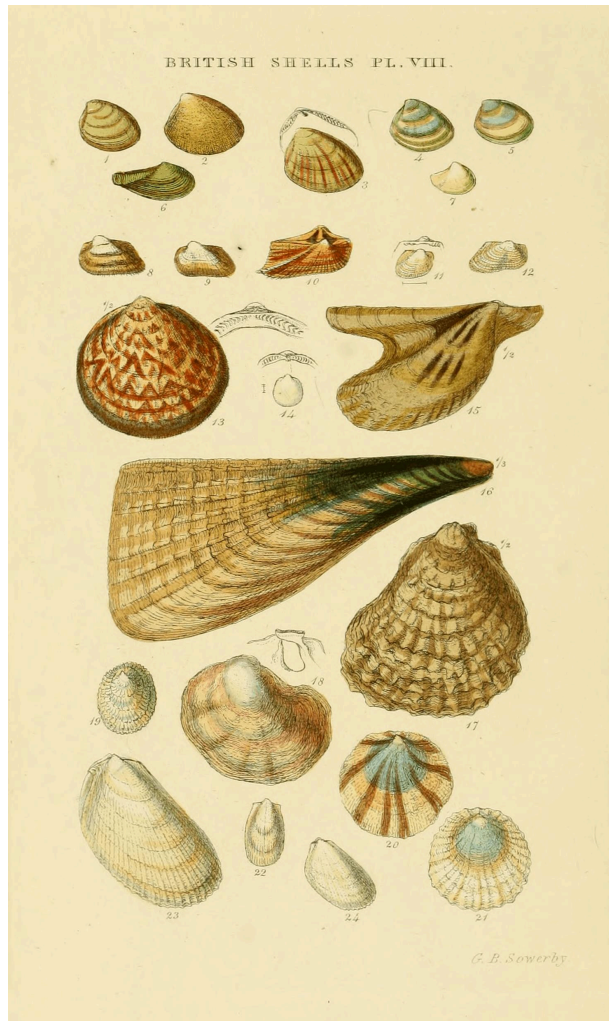


PLATE VIII.

NUCULA, *Linn.*

Fig.

1. *N. NUCLEUS*, *Linn.* (*Arca.*)—*N. margaritacea*, *Glycimeris argentea*.—Brit. co. generally, 7 to 90 fath., C. Thick, obtuse; beaks terminal.
2. *N. DECUSSATA*, *Sowerby*.—*N. sulcata*, *N. Polii*.—Hebrides, Loch Fyne, Dublin Bay, 30 to 50 fath., Mr. Large, striated; beak not terminal.
3. *N. RADIATA*, *Hanley*.—*N.* and *Arca nucleus* and *radiata*.—Torbay, Milford Haven, Arran, etc., 10 to 70 fath., Mc. Prolonged anteriorly, rayed.

4. *N. NITIDA*, Sowerby.—*N. nucleus*, var.—Numerous British loc., 1 to 35 fath., C. Shining, ends rather pointed.
5. *N. TENUIS*, Montagu, (Arca.)—North British co., 40 to 100 fath., sand, C. Thin, inner edge not crenated; beak terminal.

LEDA, Schum. Beaked behind.

6. *L. CAUDATA*, Donovan, (Arca.)—*Arca* and *Nucula minuta* and *rostrata*.—North co., C.; south co., R. 10 fath. to deep water. Ridged.
7. *L. PYGMÆA*, Munster, (*Nucula*.)—*N.* and *L. tenuis*, *gibbosa*, and *lenticula*.—Skye, Hebrides, Croulin Isl., 25 to 50 fath. Minute, smooth. Size (omitted) one-fifth of an inch.

ARCA, Linnæus.

- 8, 9. *A. LACTEA*, Linn.—*A. barbata*, *A. Modiolus*, *A. crinita*, *A. perforans*, *A. Gaimardii*, var. f. 8; *A. Quoyi*, var. f. 9.—Penzance, Ilfracombe, Bantry Bay, Guernsey, etc. Adhering to shells and stones, 15 to 20 fath., C. Beaks approaching, radiately striated.
10. *A. TETRAGONA*, Poli.—*A. Noæ* and *imbricata*.—Many Brit, loc., C. (12 to 50 fath.). Broad area between beaks.
11. *A. RARIDENTATA*, S. Wood.—*A. pectunculoides*.—Skye, Hebrides, etc., Scot. Cape Clear, 40 to 60 fath., Mr. Minute, teeth few.
12. *A. NODULOSA*, Ph.—E. Zetland, Shetland Isls., Rr.

PECTUNCULUS.

13. *P. GLYCIMERIS*, Linn. (Arca.)—*A.* and *P.* and *Glycimeris pilosa*, *orbicularis*, *undata*, and *marmorata*.—Brit. co. generally, 15 to 60 fath., C.

LIMOPSIS.

14. *L. PELLUCIDA*, *Jeffreys, Ann. Nat. Hist. Jan. 59.*—Dredged in sand at Guernsey, Rr.

AVICULA.

15. *A. TARENTINA*, *Lamk.*—*Mytilus* and *Avicula hirundo*, etc.—Bantry Bay, Devon, etc., R.

PINNA.

16. *P. PECTINATA*, *Linn.*—*P. fragilis, muricata, ingens, lævis, papyracea, rotundata, elegans.*—Dorset, Milford Haven, Hebrides, Cape Clear; deep water, Mc.

OSTREA.

17. *O. EDULIS*, *Linn.*—*O. vulgare*, etc.—Various natural and artificial beds, in England, Scotland, and Ireland. The common Oyster.

ANOMIA.

18. *A. EPHIPPIMUM*, *Linn.*—*A. electrica, A. squamula, cepa*, etc.—Brit. co. generally, 1 to 30 fath., C. Pearly and irregular; beak marginal, curved.
19. *A. ACULEATA*, *Müller.*—*A. striolata.*—Brit. co. generally, Mc. Small, with spinous sculpture.
20. *A. STRIATA*, *Lovén.*—Hebrides, R. Rayed, striated, with apex not marginal.
21. *A. PATELLIFORMIS*, *Linn.*—*A. undulata.*—Cape Clear, etc., 1 to 50 fath., Mr. Flat, ribbed.

LIMA.

22. *L. SUBAURICULATA*, *Montagu, (Pecten.)*—*L. nivea, sulcata*, etc.—Plymouth, Man, Zetland, Belfast, etc., Mr. Nearly straight.

23. L. HANS, *Gmel. (Vitræa.)*—*L. tenera, fragilis, aperta*, etc.—Channel Isls., Penzance, Oban, Belfast, etc., Mc. Oblique, gaping at sides.
24. L. LOSCOMBII, *Sowerby.*—*Pecten* and *Ostrea* and *Lima fragilis* and *bullata*.—Many Brit. loc., 10 to 50 fath., Mc. Oblique, closed at sides.



PLATE IX.

PECTEN. Free, except 1.

Fig.

1. P. PUSIO, *Pennant, (Ostrea.)*—*P., O.*, and *Hinnites distortus* and *sinuosus*.—Brit. co. generally, 15 to 90 fath., C. Irregular, attached to stones, etc.

- 2, 3. *P. VARIUS*, Linn. (*Ostrea*).—*P. monotis*.—Brit. co. generally, 3 to 35 fath.; C. Vertically elongate; auricles very unequal.
4. *P. NIVEUS*, Macgillivray.—Scotland, R. (15 fath.) Ribs more numerous than *P. varius*; white.
- 5, 6, 7. *P. OPERCULARIS*, Linn. (*Ostrea*).—*O.*, and *P. subrufus*, *pictus*, *lineatus* (var. f. 6), and *sanguinea*.—Brit. co. generally, 5 to 100 fath., C. As broad as long; ribs rounded.
8. *P. AUDOUINII*, Payraudeau.—*P. opercularis*, var., B. Moll.—Ribs angular and squamose.
- 9, 10. *P. DANICUS*, Chemn.—*P.* and *O. pes-felis*, *glaber*, *triradiata*, *adpersus*, *septemradiata*, etc.—Several Scotch localities, 25 to 90 fath., Mc. Very variable, extreme vars. f. 9, 10. Auricles more equal than *P. tigrinus*.
- 11, 12. *P. TIGRINUS*, Müller.—*P. obsoletus*, *domesticus*, *lævis*, *parvus*, etc.—Brit. co. generally, 12 to 60 fath., C. Variable; one auricle minute, the other large and ribbed.
13. *P. MAXIMUS*, Linn. (*Ostrea*).—*P. vulgaris*.—Brit. co. generally, 3 to 40 fath. The common eatable Scallop. Valves unequal.
14. *P. SIMILIS*, Laskey.—*O.* and *P. tumidus*.—Fr. Forth, Isl. Man, Lerwick, Cape Clear, etc., Mc. Minute, smooth, with undulating marks.
15. *P. STRIATUS*, Müller.—*P. aculeatus*, *fuci*, etc.—Scotland, C.; England, R.
16. *P. FURTIVUS*, Lovén.—Zetland, Guernsey, etc., 40 to 50 fath., Mr. More rounded, and with auricles less broadly based than *P. striatus*.

HIPPOTHYRIS. Fixed by byssus.

17. *H. PSITTACEA*, Gmel. (*Anomia*).—*A. rostrum-psittaci*, Chem.; *Terebratula psittacea*.—Northumberland, Fr. Forth; deep water, Rr. as Brit. "The Parrot's-beak *Anomia*."

TEREBRATULA. Fixed by byssus.

18. T. CRANIUM, *Müller*.—*Anomia c.*, and *T. vitrea*.—Zetland; deep water, unique as Brit. Smooth, oval.
19. T. CAPSULA, *Jeffreys*, *Ann. Nat. Hist.*, *Jan.* 1859.—Belfast Bay, R. Minute, punctured.

ARGIOPE, or MEGATHYRIS.

20. T. CAPUT-SERPENTIS, *Linn.* (*Anomia*.)—*A.* and *T. retusa*, *pubescens*, *costata*, and *aurita*.—Oban, Skye, etc., 20 to 50 fath., Mr.
21. A. CISTELLULA, *S. Wood.* (*Terebratula*.)—*A. seminulum?*, T.
22. A. DECOLLATA, *Chemn.*—*A. detruncata*.—Guernsey, R.
23. A. SEMINULUM, *Philippi*.—Croulin, near Skye, *Jeffreys*.

CRANIA.

24. C. ANOMALA, *Müller.* (*Patella*.)—*Pat. Anomia*, *Discina*, and *Criopus distorta*, *ostreoides*, *personata*, and *rostrata*.—Zetland, Arran, Lerwick, Cork, Mr.

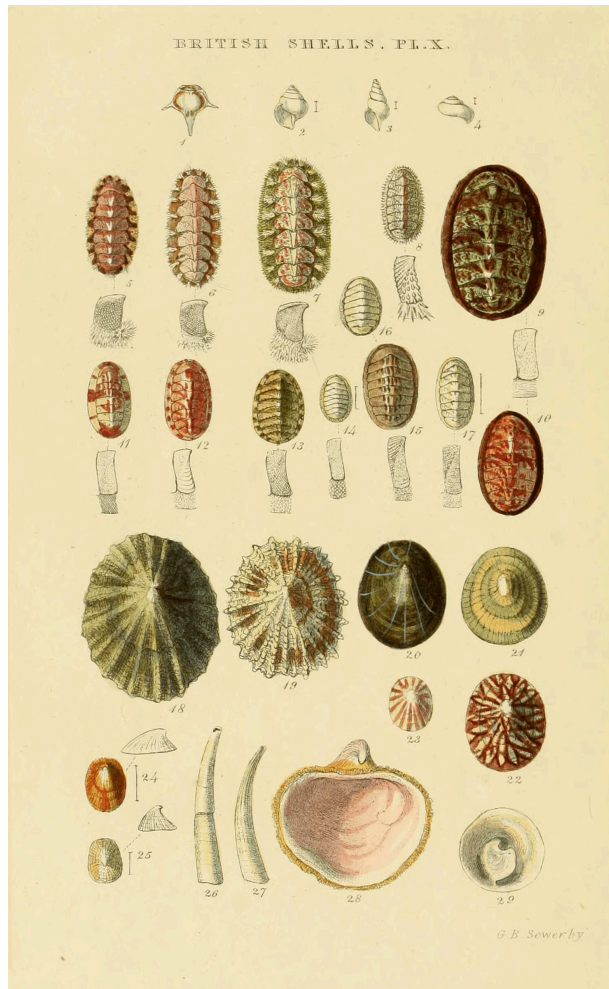


PLATE X.

HYALÆA. *Fam.* Pteropoda.

Fig.

1. *H. TRISPINOSA*, *Lesueur*.—Once taken on floating timber, Dublin. Doubtful as British.

SPIRIALIS, or PERACLE. *Fam.* Pteropoda.

2. *S. FLEMINGII*, *Forbes*.—*Fusus retroversus*, Fl.—Zetland, Skye, Arran, Donegal, M. C. Snail-shaped.
3. *S. MACANDREI*, *F. and H.*—Near Mizen Head, S. Ire., Rr. Turreted.

4. *S. JEFFREYSII*, *F. and H.*—Brit. Channel, Rr. Discoid.

CHITON. Sedentary on rocks.

5. *C. FASCICULARIS*, *Linn.*—Brit co. generally, Mc. Tufted margin, small, granules large.
6. *C. GRACILIS*, *Jeffer. Ann. Nat. Hist., Jan. 1859.*—Weymouth. Narrow; granules very small; two postero-terminal tufts.
7. *C. DISCREPANS*, *Brown.*—*crinitus*, Pennant?—Tenby, Guernsey, C. Tufted margin, large: granules very small.
8. *C. HANLEYI*, *Bean.*—Scarborough, Skye, W. Ireland, R. Spinose margin.
- 9, 10. *C. MARMOREUS*, *O. Fabr.*—*C. latus*, Lowe, *C. lævigatus*, *ruber*, and *pictus*.—Scarborough, Forth, Skye, Zetland, W. Ire., R. Margin smooth.
11. *C. LÆVIS*, *Pennant.*—*C. marginatus*, Pult., *C. Achatina*, Brown.—Dorset, Devon; Brit. co. generally, but rare. Dorsally elevated; margin reticulated.
12. *C. RUBER*, *Linn.*—Brit, seas generally, Mr. Valves smooth; margin coriaceous.
13. *C. CINEREUS*, *Linn.*—*C. marginatus*, Penn.?—Brit. co. generally, C. Valves carinated; margin and valves granulated.
14. *C. ALBUS*, *Linn.*—*C. asselloides*, Lowe.—Northern coasts, R. Marginal granules scaly and large; dorsal gr. smaller than *C. cinereus*.
- 15, 16. *C. ASSELUS*, *Chemn.*—*C. minimus*, *Cinereus albus*.—Brit. co. generally, C. Broad, granules on valves beaded.
17. *C. CANCELLATUS*, *Sowerby.*—*C. albus*, Pult.—Devon, Guernsey, Isl. Man, Oban, Zetland, Loch Carron, R. Narrow, back elevated, else like *C. assellus*.

PATELLA. (Limpet.) Sedentary.

18. *P. VULGATA*, *Linn.*—Brit, generally, Cc. Ribs not toothed.
19. *P. ATHLETICA*, *Bean.*—Devon, Northumberland, etc., Cc. Ribs toothed.
20. *P. PELLUCIDA*, *Linn.*—*P. lævis, cærulea, Acmaea pel.*—British co. generally, C. smooth.
21. ——— var. *lævis*, Penn.

ACMÆA, or LOTTIA.

22. *A. TESTUDINALIS*, *Müller.*—*P. Clealandi.*—North Ire., C. Oval, with tortoise-shell markings.
23. *A. VIRGINEA*, *Müller.*—*P. parva, pulchella.*—Brit. co. generally, Mc. Pale, pink-rayed, shallow.

PILIDIUM, or TECTURA.

24. *P. FULVUM*, *Müller.*—*P. Forbesii.*—Clyde, Cork, Zetland, etc., R.

PROPILIDIUM.

25. *P. ANCYLOIDE*, *Forbes.*—Lamlash, Ayrshire, etc., R.

DENTALIUM, *Linn.*

26. *D. ENTALIS*, *Linn.*—Brit. co. generally, C. Smooth, small end notched.
27. *D. TARENTINUM*, *Lamk.*—*D. vulgare, striatum, labiatum, politum, læve.*—South coasts, C. Striated and entire at small end.

PILEOPSIS, or CAPULUS.

28. *P. HUNGARICUS*, *Linn.*—Fool's-cap Limpet.—S. Devon, and Br. co. generally, C.

CALYPTRÆA, *Linn.*

29. *C. SINENSIS*, Linn. (*Patella*.)—*C. chinensis*, *albida*, *lævigata*.—S. Brit., and Chan. Isls., R. With oblique septum.

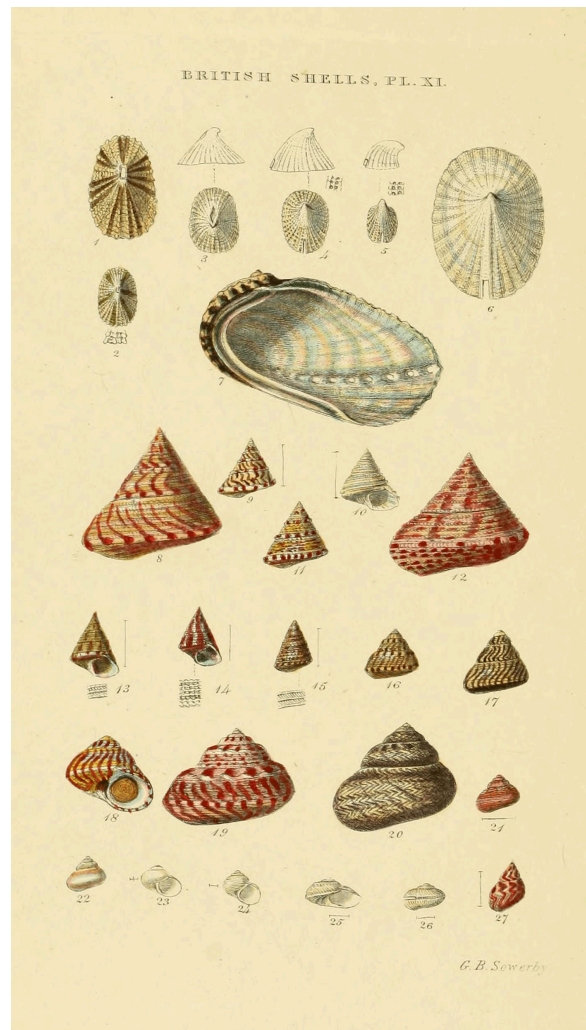


PLATE XI.

FISSURELLA.

Fig.

1. *F. RETICULATA*, Donovan. (*Patella*.)—*P.* and *F. Græca*, *apertura* (young), *cancellata*, *europæa*.—Hebrides, Thanet, Devon, Ireland. On shells, etc., 1 to 15 fath., C.

2. *F. COSTARIA*, *Deshayes*.—Guernsey, R. More fine and numerous ribs than *F. reticulata*.

PUNCTURELLA.

3. *P. NOACHINA*, *Linn.* (*Patella*.)—*Cemoria Flemingii*, *Fissurella N.*, *Rimula Flemingii*, etc.—Scotland and Northumberland, 20 to 100 fath., R. With slit under apex.

EMARGINULA.

4. *E. RETICULATA*, *Sowerby*.—*Patella fissura*, etc.—Brit. co. generally, C. Conical, with slit in margin.
5. *E. ROSEA*, *Bell.*—*E. pileolus*, *conica*, *capuliformis*, etc.—Channel Isls., Exmouth, Cornwall, etc., 7 to 25 fath., Mc. Curved, coarsely cancellated. Red within.
6. *E. CRASSA*, *Sowerby*.—Loch Fyne, Oban and Belfast, 20 to 25 fath., Rr. Large, not cancellated.

HALIOTIS, or "Ear-shell."

7. *H. TUBERCULATA*, *Leim.*—*H. vulgaris*, Channel Isls., Cc.

TROCHUS.

8. *T. ZIZYPHINUS*, *Linn.*—*C. conulus*, *C. discrepans*, *C. Lyonsii* (white, ribbed var.), etc. Brit. co. generally, low-water to 50 fath., C.
9. *T. CONULUS*, *Linn.*—Lincolnshire, Rr. More simply conical than the most extreme forms of *T. zizyphinus*.
10. *T. ALABASTRUM*, *Beck.*—*T. formosus*.—Zetland, Lerwick, etc., 45 to 80 fath., R. White, with deep, plain, and beaded ridges.
11. *T. MILLEGRANUS*, *Philippi.*—*T. Clealandi*, *T. Martini*.—Brit. co. generally, 15 to 90 fath., Mr. Granulated, whorls with prominent margin.

12. *T. GRANULATUS*, *Born.*—*T. papillosus*, *T. fragilis*.—Channel Isls., Weymouth, Dublin, etc., R. Like 8, but grained.
13. *T. STRIATUS*, *Linn.*—*T. parvus*, etc.—S. Brit., C. More finely ribbed than *T. exiguus*.
14. *T. EXIGUUS*, *Pult.*—*T. exasperatus*, *minutus*, etc. S. co., C.
15. *T. MONTAGUI*, *Gray.*—*T. striatus*, *Forbes.*—Scarborough, Exmouth, Cape Clear, etc., C. Elevated, whorls tumid.
16. *T. TUMIDUS*, *Mont.*—*T. patholatus*, *Rackettii*, etc. Brit. co. generally, C. Short, whorls tumid, umbilicated.
17. *T. CINEREUS*, *Linn.*—*T. lineatus*, *T. perforatus*, etc.—Brit. co. generally, C. Littoral, small umbilicus.
18. *T. UMBILICATUS*, *Mont.*—*T. cinerarius*, etc.—Guernsey, S. and W. Brit. co., C. Littoral, umbilicus large.
19. *T. MAJUS*, *Linn.*—*T. tuberculatus*.—Brit. co. generally; littoral, C. Broad, tuberculated. Large umbilicus.
20. *T. LINEATUS*, *Costa.*—*T. crassus*, etc.—Channel Isls., Devon. W. Ire., Mc. Smooth, imperforate.

MARGARITA, OR TROCHUS.

21. *M. UNDULATUS*, *Sowerby.*—*Turbo carneus*, *M. striata*, etc.—Scotland, 5 to 50 fath., Mc. Striated and undulated.
22. *M. HELICINUS*, *Fabricius.*—*Turbo* and *Trochus neritoideus*, *margarita*, *M. vulgaris*, etc.—Many Brit. loc.; littoral, C.
23. *M. PUSILLUS*, *Jeffreys.*—Lerwick, R., 10 to 40 fath. Minute, white, depressed.
24. *M. EXILIS*, *Phil.*—*Skenea Cutleriana*.—Guernsey, Skye, etc., R. Globose, striated.

ADEORBIS.

25. *A. SUBCARINATA*, *Mont.*—(*Helix*) *Trochus rugosus*, *Cingula subc.*, etc.—
S. Brit, co., Mr. Depressed, keeled.

SCISSURELLA.

26. *S. CRISPATA*, *Flem.*—Zetland, etc., in sand, Mc.

PHASIANELLA.

27. *P. PULLUS*, *Linn.* (*Turbo.*)—*T. pictus*, *Cingula p.*, etc.—Brit. co.
generally, C.

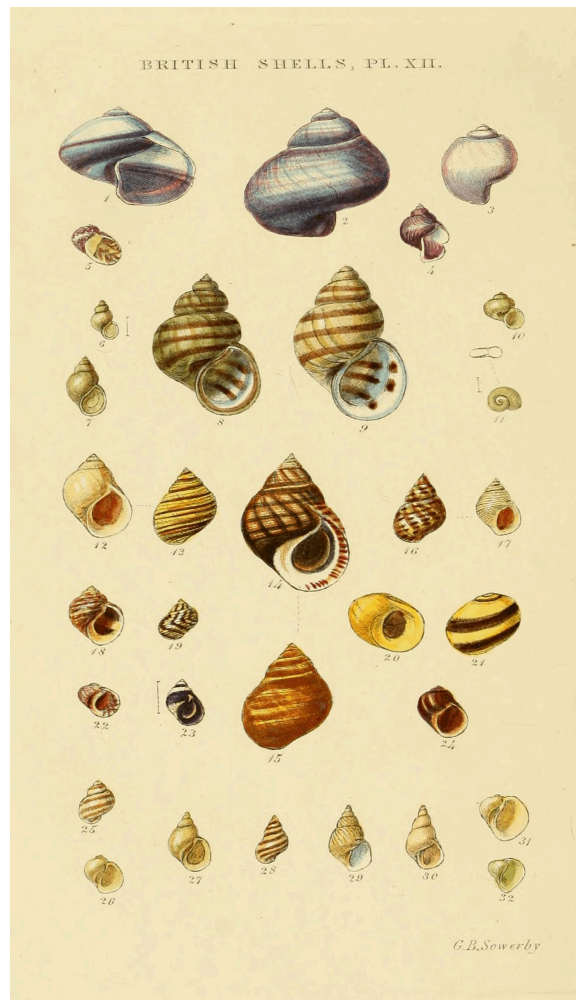


PLATE XII.

IANTHINA. Floats on ocean.

Fig.

1. I. COMMUNIS, *Lamk.*—*Helix Ianthina*, *I. fragilis*, etc.—Cornwall, Bristol, etc.; oceanic, C. Sides depressed.
2. I. BRITANNICA, *Leech. Reeve.*—Ireland, Cornwall, etc., Mr. Darkly coloured, not compressed.
3. I. PALLIDA, *Harvey.*—*I. patula.*—Clare, Rr. as Brit.
4. I. EXIGUA, *Lamk.*—Cornwall, Clare, etc., Mr.

NERITINA. In rivers.

5. N. FLUVIATILIS, *Linn.*—*N. fontinalis.*—Thames, Trent, Humber, etc., Scotch lakes, Cc.

BITHINIA. In ponds.

6. B. LEACHII, *Shepp. (Turbo.)*—*B. ventricosus*, *Paludina acuta*, *similis*, etc.—Rivers S. Brit., Mr. Narrow, with rounded whorls.
7. B. TENTACULATA, *Linn. (Helix.)*—*T. nucleus*, *Paludina impura*, etc.—Fresh-water, throughout Brit. Oval.

PALUDINA. In ponds.

8. P. LISTERI, *Forbes and Hanley.*—*Helix vivipara*, etc.—Lakes and ponds generally, Mc. Whorls more separate, umbilicus larger than *P. vivipara*.
9. P. VIVIPARA, *Linn.*—*Pal. achatina*, etc.—Thames, etc., C.

VALVATA. In ponds.

10. V. PISCINALIS, *Müller.*—*Nerita*, *Turbo*, *Cyclostoma fontinalis*, *obtusa*, *thermalis*, etc.—Fresh-water, C. Subglobose.

11. *V. CRISTATA*, Müller.—*V. planorbis*, etc.—Many loc., Mc. Orbicular.

LITTORINA. On the sea-shore.

12, 13. *L. RUDIS*, Donovan. (*Turbo.*)—*L. littoreus*, etc., *L. nigro-lineatus* (var. f. 13).—British shores generally, Cc.

14, 15. *L. LITTOREA*, Linn. (*Turbo.*)—*L. vulgaris*, etc.—British shores generally. The common eatable Winkle.

16, 17. *L. TENEBROSA*, Mont. (*Turbo.*)—British shores generally, C. Not thickened at bottom of mouth like *L. rudis*.

18. *L. PATULA*, Jeffreys.—*L. labiata*, *L. jugosa*, etc.—Common on our shores.

19. *L. SAXATILIS*, Johnston.—*L. neglecta*.—Probably *L. patula* var.?

20, 21. *L. LITTORALIS*, Linn. (*Turbo.*)—*T. neritoides*, *T. ustulatus*, etc.—Brit. co. generally. Depressed above.

22. *L. FABALIS*, Turt.—*L. Beanii*.—Young of *L. littoralis*?

23. *L. NERITOIDES*, Linn. (*Turbo.*)—*T. petræa*, *L. cærulea*, etc.—Kent, Sussex, etc., C. With white band.

24. *L. PALLIATA*, Say, (*Turbo.*)—Yarmouth, Mr.

LACUNA.

25, 26. *L. PUTEOLUS*, Turt. (*Turbo.*)—*Cochlea*, *Helix*, and *Nerita parva*, *fasciata*, *lacuna*, *rufa*, etc.—Brit. shores, C. Globose.

27, 28. *L. VINCTA*, Mont. (*Turbo.*)—*T. canalis*, *quadrifasciatus*, etc.—Brit. co. generally, C.

29. *L. CRASSIOR*, Mont. (*Turbo.*)—*L. pallidus*.—Numerous Brit. co. Thick, conical.

30. — var. ? of this rather than of *C. vincta*.

31. *L. PALLIDULA*, Costa, (*Turbo*.)—Brit. co. generally, C.

32. *L. PATULA*.—Fry of *L. pallidula*?

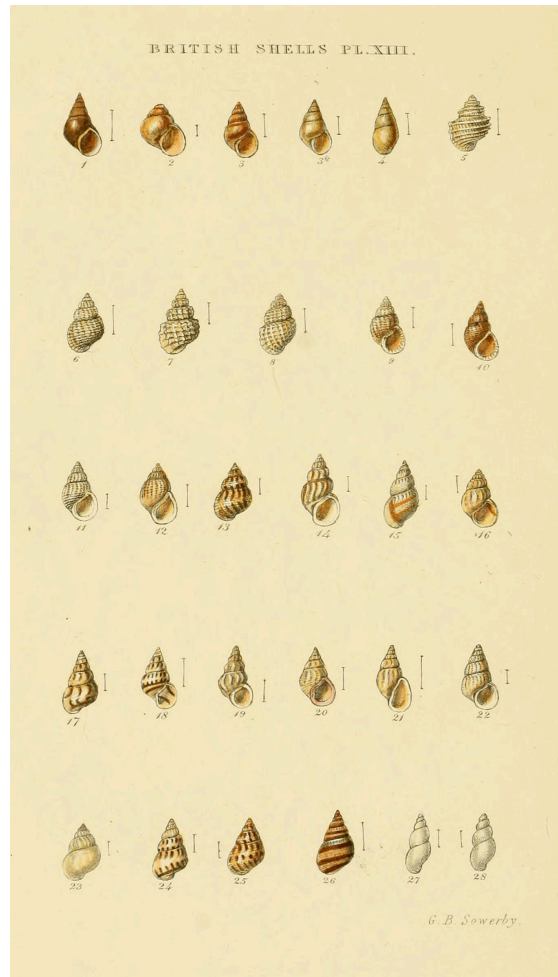


PLATE XIII.

ASSIMINEA.

Fig.

1. *A. GRAYANA*, Jeffreys.—*Limneus* and *Paludina* G.—Greenwich and other marshes.
2. *A. LITTOREA*, Delle Chiaje, (*Helix littorina*.)—*Rissoa globularis* and *littorea*.—On shore at Weymouth, Serk, Portland, Mr.

RISOA, or CINGULA.

3. *R. ULVÆ*, Pennant, (*Turbo.*)—*Helix*, *Cingula*, *Paludina*, and *Littorina*.—In brackish water, on British shores, C.
- 3*. ——— var. *subumbilicata*, R.
4. *R. BARLEEI*, Jeffreys.—*R. ulvæ* var.—Loch Carron, Skye, R.
5. *R. STRIATULA*, Mont.—*R.*, *Turbo*, *Cingula*, and *Littorina carinata* and *monilis*.—Devon, Margate, Galway, Herm, etc., 0 to 7 fath., Mr.; S. Eu. Carinated.
6. *R. CIMICOIDES*, Forbes.—*R. sculpta*.—W. Scotland. Like *R. zetlandica*, but with outer lip crenulated; finely decussated.
7. *R. ZETLANDICA*, Mont. (*Turbo.*)—*R.*, *Cyclostoma*, *Cingula scalariformis*, *Cyclostrema z.*—N. Scotland, 0 to 60 fath., Mc. Whorls angulated.
8. *R. CRENULATA*, Michaud.—*Turbo* and *R. cimex* and *cancellatus*.—Falmouth, Cornwall, Skye, Galway, C. Very coarsely cancellated.
9. *R. CALATHUS*.—Forbes and Hanley.—Sheerness, Whitesand Bay, 15 to 50 fath., R. Finely cancellated.
10. *R. BEANII*, Hanley.—Cornwall, Plymouth, Orkney, Galway, C.; N. Eu. Striated.
11. *R. ABYSSICOLA*, Forbes.—N. Scotland; deep water, R.
12. *R. LACTEA*, Michaud.—*Turbo cancellatus*.—Jersey and Medit., Rr. Oval, plicated and striated.
13. *R. PUNCTURA*, Montagu.—*Turbo* and *Cingula* and *R. reticulata*.—Brit. co. generally, 10 to 20 fath.
14. *R. COSTATA*, Adams.—*R.*, *Turbo*, *Cingula plicata*, *costata*, and *exigua*.—South co. Brit., S. Wales, Orkneys, Ireland, Channel Isls., 0 to 25 fath., C.
15. *R. STRIATA*, Mont.—*R.*, *Turbo*, and *Cingula semicostata*, *minutissima*, *communis*, *gracilis*, etc.—Brit. co. generally, N. and S. Eu.; littoral C.

- 16, 17, 18. *R. PARVA*, *Costa*.—*T.*, *C.*, and *R. sublutea*, *lactea*, *alba*, etc., *R. interrupta* (var. f. 17), *R. Sarsii*, var. f. 18 (unique).—Brit. shores generally, *C.*
19. *R. COSTULATA*, *Alder*.—*Turbo variabilis*, *R. similis*.—Devon, Channel Isls., W. Ireland, etc., *C.*; S. Eu.
20. *R. RUFILABRUM*, *Leach*.—*R.* and *C. hyalina* and *punctata*.—Devon, Belfast, etc., *Mc.*
21. *R. LABIOSA*, *Mont. (Helix)*.—*T.*, *C.*, and *R. membranacea* and *costata*.—Brit. co. generally, *C.*
- 22, 23, 24, *R. INCONSPICUA*, *Alder*.—*R. albula* and *supra-costata*, var. f. 23, *R. similis*, var. 24, *R. maculata*.—Brit. co. generally, *C.*
25. *R. SEMISTRIATA*, *Mont.*—*T.* and *R. pulchra* and *tristriata*.—Brit. shores generally, *C.*
26. *R. CINGILLUS*, *Mont. (Turbo)*.—*R.*, *T.*, and *C. vittata*, var., *R. graphica* and *R. rupestris*.—Brit co. generally.
27. *R. VITREA*, *Mont. (Turbo)*.—*Helix glabrata*, *R. crystallina*.—Exmouth, Oban, Birterbuy, etc., *Mr.* Elongated, smooth.
28. *R. PROXIMA*, *Alder*.—A rare Irish species. Elongated, striated.



PLATE XIV.

RISSEA,—continued.

Fig.

1. *R. PULCHERRIMA*, *Jeffreys*.—Channel Isls., R.
2. *R. SOLUTA*, *Philippi*.—Exmouth, Bute, Cork, etc., R. Like *R. pulch.*, but with spiral striæ.
3. *R. ALDERI*, *Jeffreys*, *Annals*, Aug. 1858.—Skye, Rr. Larger and more conical than *R. soluta*.
4. *R. FULGIDA*, *Adams*, (*Helix*).—*Turbo* and *Cingula* f.—Weymouth, Arran, Cork, etc., Mr. Whorls rounded and banded.

5. R. ANATINA, *Drap.* (*Cyclostoma*.)—*Paludina* and *Littorina a.*—Very rare, Greenwich marshes.
6. R. EXIMIA, *Jeffreys.*—*Chemnitzia Barleei*, Clark.
7. R. VENTROSA, *Mont.* (*Turbo*.)—*Cyclostoma*, *Paludina*, and *Cingula acutus*, *ventricosus*, *octona*, *stagnorum*, etc.—Marine shores and brackish-water marshes, Mc.
8. R. DENTICULATA, *Mont.* (*Turbo*.)—Herm, Rr.
9. R. UNICA, (*Turbo*.)—*Aclis u.*, Brit. Moll., *Turritella*, *Pyramis*, and *Chemnitzia*, R.—Devon, Cork, etc., Mr.
10. R. GLABRATA, *Jeffreys.*—N. sp.
11. R. CASTANEA, *Jeffreys.*—N. sp.

BARLEEIA.

12. B. RUBRA, *Flem.* (*Cingula*.)—*Turbo*, C., and *Rissoa ruber*, *unifasciatus*, *fulva*.—Herm, Tenby, Dunbar, etc., S. Eu., Mr.

JEFFREYSIA.

13. J. DIAPHANA, *Alder*, (*Rissoa*.)—*R. glabra*.—Dublin, Swansea, Northumberland, etc.; littoral, R.
14. J. OPALINA, *Jeffreys*, (*Rissoa*.)—Channel Isls., R.
15. J. GLOBULARIS, *Jeffreys.*—Skye.
16. J. GULSONÆ, *Clark*, (*Odostomia*.)—*Chemnitzia G.*—Sandwith, Lerwick, Weymouth, etc., Guernsey.

EUOMPHALUS.

17. E. NITIDISSIMUS, *Adams*, (*Helix*.)—*Skenea*, B. Moll., *Truncatella atomus*.—Shetland to Channel Isls., Mc.

18. E. ROTA, *Forbes and Hanley*, (*Skenea*.)—Donegal, rare. Like a minute Ammonite.

SKENEA.

19. S. PLANORBIS, *O. Fabr. (Helix)*.—*Turbo*, *Sk. depressa*.—Brit. shores generally, on coastal Corallines, C. Flat, like a *Planorbis*.
20. S. DIVISA, *Flem. (Turbo)*.—*Helix* and *Sk. serpuloides*, *Adeorbis striatus*.—Scarborough, Weymouth, Orkney, Galway, etc., 7 to 25 fath., Mr. Striated.
21. S. COSTULATA, *Möller, (Margarita)*.—Loch Fyne, Rr.
22. S. LÆVIS, *Philippi, (Delphinula)*.—Newcastle-on-Tyne, Rr.

ACLIS.

23. A. ASCARIS, *Turt. (Turbo)*.—*Turritella*, *Alvania*, and *Pyramis supranitida* and *acutissima*.—Ireland, Tenby, Zetland, Mr.
24. A. SUPRA-NITIDA, *S. Wood (Alvania)*.—Tenby, Dublin, etc., Rr.
25. A. NITIDISSIMA, *Mont. (Turbo)*.—*Turritella*, *Pyramis*, and *Chemnitzia*.—Falmouth, Burrow Isl., R.

EULIMELLA.

26. E. SCILLÆ, *Lovén, (Turbonilla)*.—*Eulima*, *Odostomia*, *Chemnitzia*, and *Eulimella crassula*, and *Macandræi*.—N. Scotland, Croulin, etc., 20 to 90 fath., R.
27. E. ACICULA, *Philippi, (Melania)*.—*E.*, *Chemn.*, *Odost.*, and *Eulima clavula*.—Skye, Rr.
28. E. AFFINIS, *Philippi, (Eulima)*.—*Pyramis*, *Odost.*, and *E. gracilis, lævis*.—Skye, Torbay, R.
29. E. CLAVULA, *Lovén (Turbonilla)*.—*Odostomia c.*—Torbay, Rr. Guernsey.

30. E. OBELISCUS, *Jeffreys, Ann. Nat. Hist. Jan. 1858.*—Shetland.

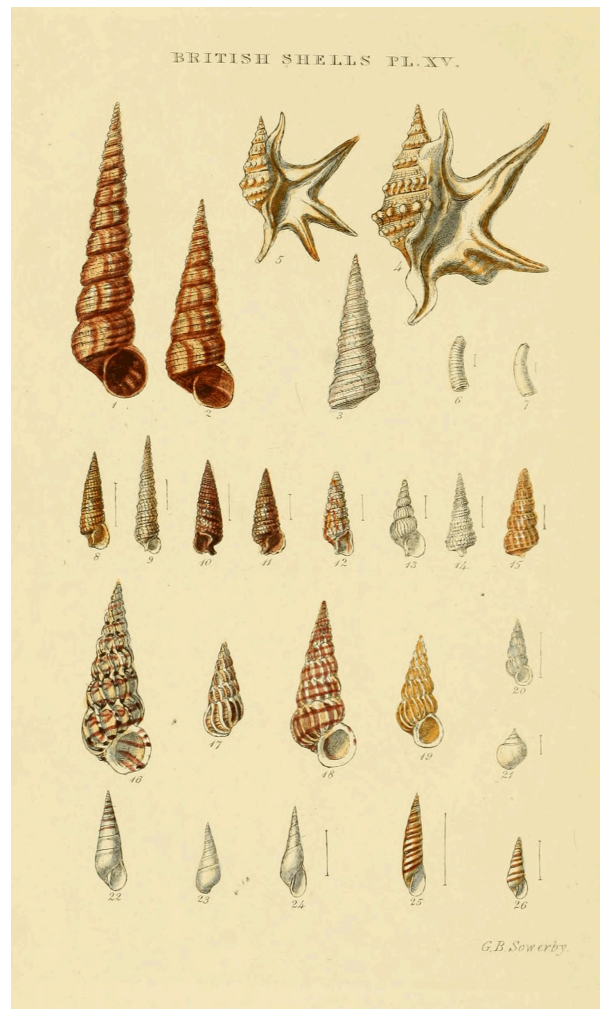


PLATE XV.

TURRITELLA.

Fig.

1. *T. CORNEA*, *Lamk.*—*T. communis*, var., B. M.—Cork and Zetland, Mr. Whorls less flat and less rapidly enlarged than *T. communis*. A white var. exists (*T. nivea*).
- 2, 3. *T. COMMUNIS*, *Risso.*—*T. cornea*?, *T. Terebra.*—Brit. co. generally; 4 to 100 fath., Cc. Aperture rather square.

APORRHAI.

4. A. PES-PELICANI, *Linn. (Strombus.)*—*A. quadrifidus*, *Tritonium*, and *Rostellaria p.*—Brit. co. gen.; N. and S. Eu.; 4 to 100 fath., C. Outer lip with one anterior lobe.
5. A. PES-CARBONIS, *Brongniart.*—*Rostellaria* and *A. p.*, and *serressiana.*—Zetland, 70 to 100 fath., R. Outer lip with two anterior lobes.

CÆCUM.

6. C. TRACHEA, *Mont.*—*Dentalium* and *Orthocera imperforatum*, *C. annulatus.*—Devon, Weymouth, Scotland, Cork, etc., R.; S. Eu. Ringed and striated.
7. C. GLABRUM, *Mont.*—*C.*, *Dentalium*, and *Orthocera minutum*, and *lævissimum.*—Exmouth, Weymouth, Swansea, Lerwick, Bantry Bay. Rare, smooth.

CERITHIUM.

8. C. RETICULATUM, *Costa, (Strombiformis.)*—*Murex* and *Terebra r.*—Brit. co. gen., 0 to 20 f., C. Narrow, and closely reticulated.
9. C. METAXA, *Delle Chiaje.*—*C. angustissimum.*—Guernsey, R.; S. Eu. Lengthened, white.
10. C. ADVERSUM, *Mont. (Murex.)*—*C.*, *Triphoris*, *Turbo*, and *Terebra a.*, *perversus*, and *reticulatus.*—W. and S. Brit., Channel Isls., etc., 5 to 25 fath., Mr.

CERITHIOPSIS.

11. C. TUBERCULARE, *Mont. (Murex.)*—*Cerithium* and *Terebra t.*—Brit. co. gen.; S. Eu.; 4 to 40 fath., Mc. Three rows of tubercles in the penultimate whorl.
12. C. CLARKII, *Hanley.*—Exmouth, Rr. Two rows of tubercles.

13. C. NIVEA, *Jeffreys, Ann. Nat. Hist. 1859.*—In shell sand, Belfast, R. Rather Buccinum-shaped.
14. C. METULA, *Lovén, (Cerithium.)*—*C. nitidum*, Forbes.—Zetland, 50 to 80 fath., R. White, with angular whorls.
15. C. PULCHELLA, *Jeffreys.*—Falmouth, Plymouth, Guernsey, R. Closely whorled and cancellated.

SCALARIA.

16. S. COMMUNIS, *Lamk.*—*Turbo* and *Strombiformis clathrus* and *clathratus*.—S. co. principally, C. Laminarian zone, S. Eu. Ridges raised; whorls separated.
17. S. GRÆNLANDICA, *Chemn. (Turbo.)*—*S. planicostata*, *subulata*.—Fragments only, N. Scotland, N. Eu. Grooved between ridges.
18. S. TURTONIS, *Turton.*—Scarborough, Exmouth, Cork, etc., N. and S. Eu., R. as Brit. Ridges flat.
19. S. TREVELYANA, *Leach.*—N. Brit., 15 to 85 fath., Mr. Thin, with ribs narrower than *Sc. Turtonis*.
20. S. CLATHRATULA, *Mont.*—S. Brit. and Channel Isls., Mr. Small, white, with thin ribs.

STYLIFER.

21. S. TURTONI, *Brod.*—*Phasianella stylifera*, *Velutina s.*, *S. globosus* and *astericola*.—On spines of *Echinus sphæra*, Torbay and Berwick, R. Glassy, thin.

EULIMA.

22. E. POLITA, *Linn. (Helix.)*—*Strombiformis*, *Turbo*, *Rissoa*, and *Phasianella p.* and *albus*, *Boscii* and *anglica*.—Ireland, S. Brit., E. Scot., N. and S. Eu., 7 to 50 fath., Mr. Rather solid.

23. *E. DISTORTA*, *Deshayes*.—Brit. co. gen., Mc.; 12 to 90 fath.; N. and S. Eu. Small, tortuous.
24. *E. STENOSTOMA*, *Jeffreys*, *Ann. Nat. Hist.* 1858.—Zetland and Norway, Rr. Thin, with rapidly increasing whorls.
25. *E. SUBULATA*, *Donov. (Turbo.)*—*Strombiformis*, *Helix*, and *Melania s.* and *glaber*, *trifasciatus*, *Donovani*, and *lineata*.—S. and W. Brit., Ire., Zetland, Medit., 10 to 80 fath., Mr. Whorls deep, banded.
26. *E. BILINEATA*, *Alder*.—Or *E. subulata*, var.—Northumberland, etc. Whorls more swelled than *E. su.*, two-banded.

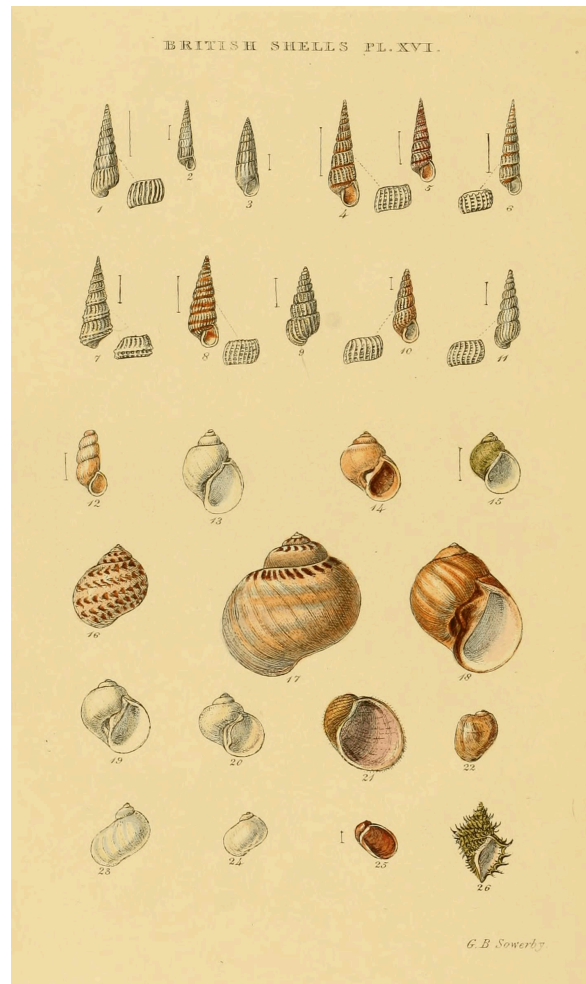


PLATE XVI.

CHEMNITZIA.

Fig.

1. *C. ELEGANTISSIMA*, *Mont. (Turbo.)*—*T. lacteus*, Linn.?, *T. acutus*.—Brit. co. generally; not deep, C. Interstices of longitudinal ribs smooth.
2. *C. SIMILLIMUS*, *Mont. (Turbo.)*—*C. gracilis*, Phil. More slender and minute than f. 1.
3. *C. PUSILLA*, *Phil. Moll. Sic. 2. p. 224. pl. 28. f. 21.*—More stout than f. 1, with the ribs straight and broad. *Medit., Brit.?*
- 4, 5. *C. RUFA*, *Phil.*—And var. *C. fulvocincta* (f. 5).—Brit. co. generally, 15 to 50 fath., Mc. Spiral grooves between ribs.
6. *C. FORMOSA*, *Jeffreys, (Odostomia.)*—Swansea, Shellness, Bantry Bay, etc., Mr. Grooved between ribs, excavated at suture.
7. *C. FENESTRATA*, *Forbes and Jeffr. (Odostomia.)*—Torrey, etc., R. With pits between spiral ridges.
8. *C. RUFESCENS*, *Forbes.*—*Odostomia* and *Turritella indistincta* and *scalaris*.—Clyde, Hebrides, Oban, 15 to 50 fath., R. More stout than *C. rufa*. Three-banded.
9. *C. SCALARIS*, *Philippi, (Melania.)*—Exmouth, Guernsey, *Medit., R.* Ribbed; whorls angular above.
10. *C. CLATHRATA*, *Jeffreys.*—Birterbuy Bay, R. Lower part of whorls spirally grooved between ribs.
11. *C. INDISTINCTA*, *Mont. (Turbo.)*—*Odost.*, *Turritella*, *Terebra*, and *Pyramis truncata* and *speciosa*, etc.—Brit. co. generally, 5 to 40 fath., Mr.; S. Eu. Ribs curved, striated between.

TRUNCATELLA.

12. *T. SUBCYLINDRICA*, *Linn. (Helix.)*—*T. Montagu*, *Turbo truncatus*, *subtruncatus*, *Cyclostoma truncatulum*, *Turritella t.* and *s.*, etc.—Weymouth, S. Devon, Scarborough, etc., C.

NATICA.

13. N. HELICOIDES, *Johnston*.—Scotland, Scarborough, etc., 15 to 40 fath., R.; N. Eu. Thin-grooved suture.
14. N. MONTAGUI, *Forbes*.—*Nerita rufa*, *Natica r.* and *rutilla*.—Principally northern loc. In sand and gravel, 12 to 90 fath., Mr. Small, stout.
15. N. KINGII, *Forbes and Hanley*.—Northumberland, unique, *Jeffreys* coll.
16. N. NITIDA, *Donovan*.—*Nerita* and *Natica catena*, *nitida*, *Alderi*, *glaucina*, etc.—Brit. co. generally, 4 to 90 fath., C. Oblique, marked with undulating lines.
17. N. MONILIFERA, *Lamk*.—*Nerita glaucina*, *catena*, etc.—Near water-mark, in sandy places generally, C. Whorls rounded.
18. N. SORDIDA, *Phil*.—*N. fulva*, *plumbea*, etc.—Scilly, Cape Clear, Zetland, etc., 25 to 90 fath., R.
19. N. PUSILLA, *Say*.—*N. Grænlandica*, *livida*, *alba*.—Northumberland, Yorkshire, N. Eu., 45 fath., R. White; umbilicus small.

RECLUZIA.

20. R. APERTA, *Jeffreys*.—*Natica aperta*, Lovén?—Zetland, Rr.

VELUTINA.

21. N. LÆVIGATA, *Linn. (Helix)*.—*Bulla Velutina*, etc.—Brit. co. generally, 1 to 30 fath., N. Eu., C.
22. N. FLEXILIS, *Mont. (Bulla)*.—*Coriocella* and *Sigaretus plicatilis*.—N. Scot., 25 fath., Rr.

LAMELLARIA.

23. L. PERSPICUA, *Linn*.—*Helix*, *Bulla haliotoidea*, *Sigareta* and *Coriocella neritoideus*.—Brit. co. generally, but R.; 1 to 15 fath., Mr. Thin, glassy, more ventricose than *L. tentaculata*.

24. *L. TENTACULATA*, Mont.—*Sig.* and *Cor.*—Falmouth, Swansea, Arran, etc.
Laminarian, Mr. Upper whorls less full than *L. perspicua*.

OTINA.

25. *O. OTIS*, Turt. (*Helix*.)—*Velutina*, *Galericulum ovatum*.—Wales, Devon,
Channel Isls., Clare, etc., Mc. Littoral.

TRICHOTROPIS.

26. *T. BOREALIS*, Broderip and Sowb.—*T.* and *Murex carinatus*, *umbilicatus*,
acuminatus, etc.—N. Scotland, W. Ireland, 15 to 80 fath., R.

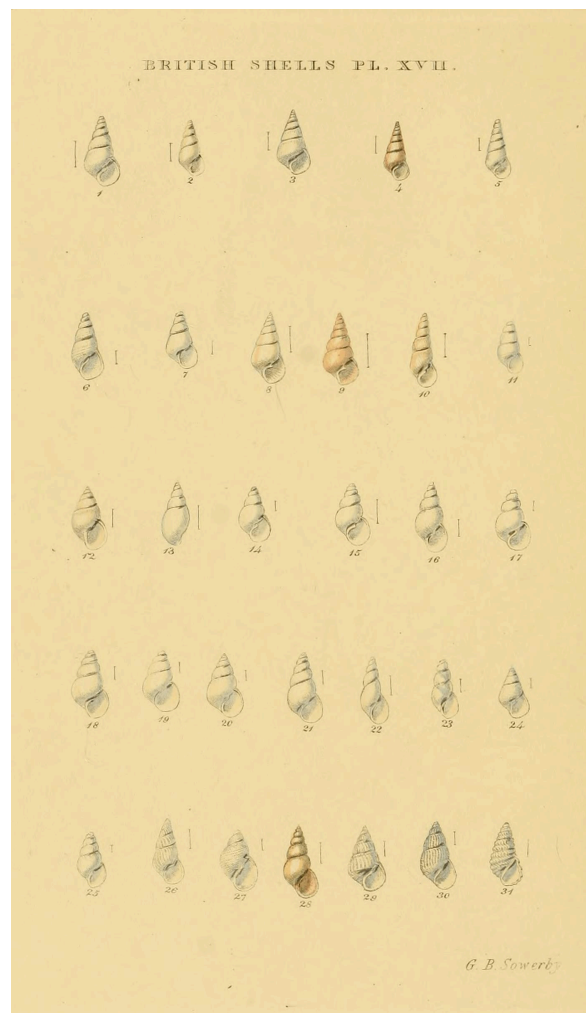


PLATE XVII.

ODOSTOMIA.

Fig.

1. *O. UNIDENTATA*, *Mont. (Turbo.)*—*Voluta u.*, *O. plicata*.—Brit. co. generally, Mc. Rather conical. Tooth strong; no epidermis.
2. *O. TURRITA*, *Hanley, Brit. Mar. Conch.*—*O. unidentata*, var. B. M.—R. More lengthened than No. 1.
- 3, 4. *O. ACUTA*, *Jeffreys.*—Torbay, Loch Fyne, Galway, etc., Mc. Strong, conical. Suture deep.
5. *O. PLICATA*, *Mont. (Turbo.)*—*Voluta p.*, *V. plicatula*, *O. Annæ*, *Jaminia p.*, etc., Mc.—Salcombe, Swansea, Dublin, Aberdeen, Mc. Turreted. Narrow, thin.
6. *O. INSCULPTA*, *Mont. (Turbo.)*—*Voluta*, *Jaminia*, and *O. ins.*—S. Devon, Oban, Hebrides, etc., R. Conical, spirally striated.
7. *O. PALLIDA*, *Mont. (Turbo.)*—*Vol.*, *Cingula*, *Phasianella p.*, *O. ambigua*.—S. Devon, Rr. Broad. Thin, pillar straight.
8. *O. CONOIDEA*, *Brock.*—*Turbo*, *Odost.*, etc.—N. Scotland, Mc. S. Devon, Mr. N. and S. Eu. White. Whorls straight; throat striated.
9. *O. CONSPICUA*, *Alder.*—Isl. Man, Whitburn, Herm. Coloured. Whorls rounded; throat striated.
10. *O. TRUNCATULA*, *Jeffreys.*—Plymouth. Elongated.
11. *O. CYLINDRICA*, *Alder.*—*Turbo nivosus?*—S. Devon, Ilfracombe, Guernsey, Cork, R. Cylindrical, few whorls.
- 12, 13. *O. EULIMOIDES*, *Hanley.*—*O. crassa*, *O. pallida* (var. f. 13), *O. notata*.—Brit. co. generally. Coralline zone, Mr. Ample body-whorl.
14. *O. DUBIA*, *Jeffr.*—S. Devon, Lerwick, Guernsey, Mr. Rather large body-whorl.
- 15, 16. *O. ALBA*, *Jeffreys*, (and var.)—Swansea, Skye, Zetland, Guernsey, Mr. Thin, whorls rounded.

17. *O. NITIDA*, *Alder*.—S. Devon. Littoral, Mc. Aperture differently formed from No. 17.
18. *O. LUKISII*, *Jeffreys, Ann. Nat. Hist. 1859*.—Guernsey. Sublittoral and coralline, Mc. Whorls tumid; a slight umbilicus.
19. *O. GLABRATA*, *Muhlfeldt, (Helix.)—Rissoa punctulum*, etc.—Zetland, R. Pillar-lip thin, arched.
20. *O. RISSOIDES*, *Hanley*.—*O. scalaris*.—Channel Isls., Devon, Cornwall, Tenby, etc., Mc. Thin, conical, no umbilicus.
21. *O. ALBELLA*, *Lovén*.—*O. rissoides*, var., B. M.—Guernsey, S. Devon, etc. Turreted, upper whorls more cylindrical than No. 20.
22. *O. OBLIQUA*, *Alder*.—Tynemouth, Guernsey, 7 fath. to deep water, Rr. Thin, oblique.
23. *O. DIAPHANA*, *Jeffer. Ann. Nat. Hist. New Ser. vol. ii. p. 341*.—R. Manifestly distinct from No. 22.
24. *O. STRIALATA*, *Alder*.—Northumberland, Rr. Conical, striated.
25. *O. MINIMA*, *Jeffreys, Ann. Nat. Hist. Jan. 1, 1858*.
26. *O. INTERSTINCTA*, *Mont. (Turbo.)—T., O., Voluta, and Jaminia canaliculata, oblonga, and obtusa*.—Brit. co. generally. Shallow water, C. With longitudinal ribs.
27. *O. DOLIOLIFORMIS*, *Jeffer*.—Scarborough, Exmouth, Swansea, Tenby, Rr., Scotland. Oval, body-whorl very tumid.
28. *O. WARRENII*, *Thompson, (Rissoa.)—Turbonilla and O. obliqua, O. decorata*.—Dublin, S. Devon, W. Scotland, R. Turreted. Mouth oblique; an umbilicus; no fold or columella.
29. *O. SPIRIALIS*, *Mont. (Turbo.)—Voluta and Rissoa s., pellucida, and plicatula*.—Brit. co. generally, not deep, Mc. Whorls ribbed longitudinally above, spirally beneath.

30. *O. DECUSSATA*, *Mont. (Turbo.)*—*T. Helix*, *Rissoa* and *O. arenaria*, *pellucida?*, and *d.*—Exmouth, Lerwick, etc., R. Ribbed and spirally striated.
31. *O. EXCAVATA*, *Phil. (Rissoa.)*—*R.*, *O.*, and *Cingula Harveyi* and *Deshayesiana*.—Dublin, Arran, Guernsey, R. Largely sculptured.



PLATE XVIII.

TRITON.

Fig.

1. T. CUTACEUS, *Linn.*—Channel Isls., Rr. as Brit. Mr. Lukis's specimen taken living off Guernsey.
2. T. NODIFERUS, *Lamk.*—Guernsey; three living specimens, one incrustated with *Lepralia coccinea* (truly British). Mr. Jeffrey's specimen reduced.

MUREX.

3. M. ERINACEUS, *Linn.*—*Purpura scalata*, var. *M. Tarentinus*.—Brit. co. generally; 5 to 30 fath., C. N. and S. Eu.
4. M. CORALLINUS, *Scacchi.*—*Fusus* and *M. lavatus*, *inconspicuus*, *gyrinus*, and *badius*?—Channel Isls. and S. Eu., C.

M. RUDIS. See *Trophon*.

PURPURA.

5. P. LAPILLUS, *Linn. (Buccinum.)*—*B.* and *P. filiosum*, *imbricatum*, and *bizonalis*.—In crevices on shores, Brit. co. generally, N. and S. Eu., C.

LACHESIS.

6. L. MINIMA, *Mont. (Buccinum.)*—*B. brunneum*, *B. rubrum*, *Nesæa granulata*, *mamillata*, *Fusus minimus*, etc.—Channel Isls. and S. Brit., on shores. S. Eu.

BUCCINUM.

7. B. HOLBOLLII.—*Mangelia H.*—Cornwall, Rr.
8. B. UNDATUM, *Linn.*—*B. striatum*, *anglicanum*, *pyramidale*, *imperiale*, etc.—Brit. co. generally, shores, Cc. N. Eu. and Am.
9. B. ACUMINATUM, *Brod.*—*B. undatum* var., B. M.—Rr.
10. B. CILIATUM, *Fabr. (Tritonium.)*—Shetland, deep water, R. More thin than No. 9; with striæ finer, and hairs of epidermis tufted minutely.

11. *B. FUSIFORME*, *Brod.*—Cork, Wexford, Rr. Turreted, fusiform, white, ribbed, striated.
12. *B. DALEI*, *Sowerby.*—*B. ovum*, *Halia Flemingiana.*—Torbay, Northumberland, Aberdeen, etc., R. Ventricose.
13. *B. HUMPHREYSIANUM*, *Bennett.*—*Tritonium H.*—Cork, Skye, Bantry Bay, R. Light, smooth.

FUSUS.

14. *F. BERNICIENSIS*, *King.*—Northumberland, Rr. Carinated, with open pink mouth.
15. *F. NORVEGICUS*, *Chemn. (Strombus.)*—*Tritonium n.*—Scarborough, Durham, Rr. Oval, smooth, with large aperture.
16. *F. ANTIQUUS*, *Linn. (Murex.)*—*M.*, *B.*, and *Tritonium despectus*, *contrarius*, *carinatus*, *babylonicus*, etc.—Various Brit. co., 5 to 30 fath., C.
17. *F. ISLANDICUS*, *Chemn.*—*Murex corneus*, *Tritonium gracile*, etc.—Brit. co. generally; 5 to 80 fath.; N. Eu., C. (See *F. propinquus*.)
18. *F. TURTONI*, *Bean.*—Yorkshire, Northumberland, 60 fath., Rr.
19. *F. PROPINQUUS*, *Alder.*—*F. corneus* and *islandicus*, var.—N. Scotland, Bantry Bay, Northumberland, deep water, Mr. Shorter than *F. islandicus*, with deep suture and symmetrical spire.

TROPION.

20. *T. RUDIS*, *Phil. (Fusus.)*—*Murex r.*, *Jeffreys.*—Tenby; unique as British. Whorls and ribs rounded.
21. *T. MURICATUS*, *Mont.*—*Fusus* and *Murex m.*—Brit. co. generally, Channel Is., 15 to 50 fath., Mr. Whorls angular, ribs intersected.
22. *T. SCALARIFORMIS*, *Gould.*—Belfast, in sand, Rr. Ribs smooth.

23. *T. CLATHRATUS*, Linn. (*Murex*.)—*Tritonium*, *Fusus*, and *Pleurotoma Bamfius*.—Irish sea, Zetland, Mc. Ribs laminated.
24. *T. BARVICENSIS*, Johnston.—*Fusus* and *Tritonium b.*—Hebrides, Cork, Northumberland, 60 fath., Mr. Ribs aculeated.



PLATE XIX.

NASSA.

Fig.

1. *N. RETICULATA*, Linn. (*Buccinum*.)—*N. pullus*, *Tritonium r.*—Brit. and Eu. co. generally. Littoral, Cc.

2. *N. INCRASSATA*, Müller, (*Buccinum*.)—*N.* and *B. minutum*, *macula*, *asperulum*, etc.—Brit. shores generally, 0 to 50 fath., gravel, C.
3. *N. PYGMÆA*, Lamk. (*Ranella*.)—*B. tuberculatum*, *Tritonia varicosa*, etc.—S. Brit. co., S. Eu., 4 fath. to deep water, C. Small, with white varices on the body.

MANGELIA.

4. *M. TURRICULATA*, Mont. (*Murex*.)—*Bela*, *Fusus*, and *Pleurotoma angulatus*, etc.—Brit. co. generally, 3 to 100 fath., N. Eu., C. Whorls flat above.
5. *M. TREVELLIANA*, Turt.—*Pleurotoma*, *Bela*, and *M. reticulata*, *decussata*, and *leucostoma*.—Northumberland, Aberdeen, Zetland, etc., 15 to 40 fath., R. Whorls with angular slope above.
6. *M. RUFA*, Mont. (*Murex*.)—*F.*, *M.*, and *B. chordula*, *Ulideana* (var.), etc.—Brit. co. generally, Mc.
7. *M. TERES*, Forbes, (*Pleurotoma*.)—*P. boreale*.—Various loc., 15 to 50 fath. S. Eu., Mr. Carinated, aperture deeply notched.
8. *M. PURPUREA*, Mont. (*Murex*.)—*Fusus* and *Pleu. p.*, *Philberti versicolor* and *variegatum*.—Brit. co. gen., 3 to 60 fath., C. S. Eu.
9. *M. CANCELLATA*, Sowerby (*Fusus*.)—*M. purpurea* var., *F. asperrimus*,—Cornwall, Cork, R. Sharply sculptured.
10. *M. RETICULATUM*, Brown, (*Pleurotoma*.)—*M. cancellata* var.—Shetland, Guernsey, Rr. Sculpture between 8 and 9; inner lip not thickened.
11. *M. LEUFROYI*, Michaud, (*Pleurotoma*.)—*P.*, *F.*, and *M. inflata*, *Boothii*.—Arran, Orkneys, Guernsey, etc.; S. Eu., 5 to 50 fath., Mc.
12. *M. LINEARIS*, Mont. (*Murex*.)—*M.*, *Pl.*, *Fus. elegans*, *concinna*, etc.—Brit. co. generally, sand and shells, 1 to 80 fath., C.
13. *M. SCABRA*, Jeffreys, (*Pleur.*)—*M. linearis* var.—Guernsey, Plymouth, R.

14. *M. NEBULA*, *Mont. (Murex.)*—*M.*, *P.*, and *F. Bertrandi*, *varians*, and *pyramidatus*.—Various places and depths, N. and S. Eu., C.
15. *M. LÆVIGATA*, *Phil. (Pleu.)*—*M. nebula* var.—South Brit., S. Eu. Small, fusiform, shiny, banded.
16. *M. GUINNIANA*, *Phil. (Pleu.)*—*M. nebula* var.—Tenby, etc., Mr. S. Eu. Whorls longer than No. 14.
17. *M. BRACHYSTOMA*, *Phil. (Pleu.)*—*M. tiarula* and *Clavatula b.*—Brit. co. gen., Mr. Turreted; ribs crossed by raised lines.
18. *M. NANA*, *Lovén, (Tritonium.)*—*Fusus albus*.—Lerwick, Zetland, Orkneys, etc., 45 fath., and deep, R. Ventricose, cancellated.
- 19, 20. *M. STRIOLATA*, *Scacchi, (Pleu.)*—*P.* and *Murex Smithii, Farranti*.—S. Devon, Clyde, Guernsey, Bantry, etc., Medit., R. as Brit. Ribs angularly curved above.
- 21, 22. *M. COSTATA*, *Pennant, (Murex.)*—*Buccinum, Fusus, Pl. c.*, etc.; var. f. 22, *M. Metcalfei*.—Brit co. generally, 5 to 50 fath., N. and S. Eu., C. Body-whorl larger than No. 24.
23. *M. COARCTATA*, *Forbes.*—*Pl.*, *M. costata* var., B. M.—Distribution as 21 and 22; not so common.
24. *M. SEPTANGULARIS*, *Mont. (Murex.)*—*M.*, *F.*, and *Pl. costatus, heptagona*, etc.—Torbay, Bristol Channel, Forth, Galway, 5 to 30 fath., Mc.
25. *M. ATTENUATA*, *Mont. (Murex.)*—*M. aciculatus, Pl. gracilis, Villiersii*, etc.—S. Devon, Oban, Hebrides, Cork, etc., Mr. S. Eu.
26. *M. GRACILIS*, *Mont. (Murex.)*—*M. emarginatus, suturalis, elegans*, etc., S. Brit., Galway, S. Eu., Mr.

M. HOLBOLLII. See *Buccinum*.

ERATO.

27. *E. LÆVIS*, *Donovan, (Voluta.)*—*E.*, *Marginella, Bulla, Volvaria*, and *Columbella fusiformis, Cypræola, Donovanii, Muscaria*, etc.—Brit. co.

generally, 1 to 50 fath., Mc.

CYPRÆA.

28. C. EUROPÆA, Mont.—Cypr. and Trivia, Pediculus arctica, bullata, and diaphana.—Brit. co. generally, 0 to 50 fath., C.

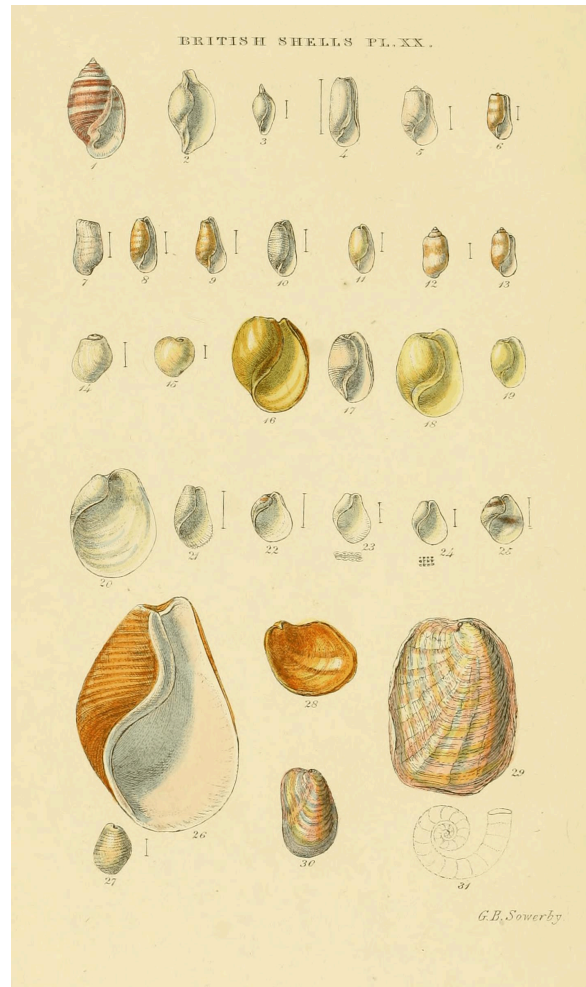


PLATE XX.

TORNATELLA.

Fig.

1. T. FASCIATA, Lamk.—Voluta tornatilis, Linn., Bulimus T., T. pellucida and pusilla (young).—Brit. co. generally. N. and S. Eu.

OVULA.

2. O. PATULA, *Penn. (Bulla)*.—Exmouth, Arran, etc., 20 fath. S. Eu. Wide aperture, Mc.
3. O. ACUMINATA, *Brug.*—*Bulla* and *Volvula a.*—Loch Fyne, Lamlash, Galway, etc., R. S. Eu. Narrow, pointed.

CYLICHNA.

4. C. CYLINDRACEA, *Penn. (Bulla)*.—*Bullina* and *Volvaria c., oliva*, and *umbilicata*.—Brit. co. generally, C. N. and S. Eu.
5. C. OBTUSA, *Montagu (Bulla)*.—*Utriculus*, *B.* and *C. Jerviscensis* and *discors*.—Brit. co. generally, C.
6. C. MAMILLATA, *Philippi*.—*Bulla* and *Tornatina m.* and *truncatula*.—Exm., Skye, Galway, etc., Mc. S. Eu. Spire sunk, apex mamillated.
7. C. TRUNCATA, *Adams, (Bulla)*.—*B. truncatula, retusa, semisulcata*.—Brit. co. generally, C. N. and S. Eu.
8. C. NITIDULA, *Lovén*.—*C.* and *Bulla n.* and *umbilicata*.—Skye, Loch Fyne, etc., R. N. Eu. Elongated, oval.
9. C. CONULUS, *Desh.*—Zetland, Rr. Conical.
10. C. STRIGILLA, *Lovén*.—*C.*, *Bulla*, and *Atys s.* and *ovulata*.—Scotland, Mr. Spirally striated.
11. C. UMBILICATA, *Mont. (Bulla)*.—*Bullina* and *Volvaria u.* and *subcylindrica*.—Brit. co. gen., Mc. With umbilicated apex.
- 12, 13. C. LAJONKAIREANA, *Basterot, (Bulla)*.—Guernsey, R. Narrower, and with more raised spire than *C. obtusa*.

AMPHISPIRA.

14. A. HYALINA, *Turt. (Bulla)*.—*A. diaphana* and *Utriculus pellucidus, minutus*, and *candidus*.—Scarborough, Weymouth, Cork, etc. Littoral, Mc. N. Eu. Rather square.

15. A. GLOBOSA, *Jeffreys, Ann. Nat. Hist. Jan. 1859.*—Skye.

AKERA.

16. A. BULLATA, *Müller.*—*Bulla Akeræ, B. norvegica, B. resiliens, fragilis, flexilis, Hanleyi.*—Southampton, Zetland, Galway, etc. Littoral, Mc. N. and S. Eu.

BULLA.

17. B. CRANCHII, *Leach.*—*B. punctura* and *striata.*—Brit. co. gen. Devon, Aberdeen, Cork, etc., 50 to 100 fath., Mr. Spirally puncto-striated.

18. B. CORNEA, *Lamk.*—*B. hydatis* (as of *Linn.*), *B. navicula, B. ampulla, Haminea hydatis.*—South Brit. co., 0 to 50 fath., C. S. Eu. Broad, thin.

19. B. HYDATIS, *Linn.*—*Haminea elegans.*—Guernsey and Mediterranean. Narrower and more solid than *B. cornea.*

PHILINE.

20. P. APERTA, *Linn.*—*P. Bulla, Bullæa, and Lobaria a., quadripartita, quadrilobata, and Planciana.*—Brit. co. gen., C. S. Eu. Large, smooth.

21. P. SCABRA, *Müller.*—*Bulla and Scaphander scabra, pectinata, dilatata, and catenulifera.*—Northumberland, Zetland, Ireland, etc., Mc., 5 to 50 fath. N. Eu. Oblong, serrated.

22. P. QUADRATA, *S. Wood.*—*Bullæa* and *P. scutulum.*—Zetland, etc., 10 to 100 fath., R. Striated.

23. P. CATENA, *Mont.*—*Bulla and Bullæa c., angustata, and punctata,*—Brit. co. generally, Mr., 0 to 40 fath. With chain-like sculpture.

24. P. PUNCTATA, *Clark.*—*Bulla and Bullæa.*—Devon, Swansea, Aberdeen, Clare. In sand, 60 fath., Mc. With punctured sculpture.

25. P. PRUINOSA, *Clark.*—*Bulla and Bullæa.*—Devon, Northumb., Hebrides, etc. Mud, 20 to 70 fath. N. Eu.

SCAPHANDER.

26. *S. LIGNARIUS*, *Linn. (Bulla.)*—*B. oblonga*.—Brit. co. generally. 1 to 50 fath. N. and S. Eu., C.
27. *S. ZONATUS*, *Turt. (Bulla)*—Young specimen taken at Zetland by Mr. Jeffreys, resembling older Norwegian specimens. More oval than *S. lignarius*. Spire umbilicated.

APLYSIA.

28. *A. HYBRIDA*, *Sowerby*.—*Laplysia depilans*, *A. depilans* and *punctata*.—Brit. co. generally. Among seaweeds, 0 to 6 fath., C.

PLEUROBRANCHUS.

29. *P. MEMBRANACEUS*, *Mont. (Lamellaria.)*—Devon, Arran, Cork, etc. Low-water, Mr. Broad, flexible.
30. *P. PLUMULA*, *Mont. (Bulla.)*—Guernsey, Devon, Skye, etc. Littoral, R. Narrow, less flexible.

SPIRULA.

31. *S. PERONII*—Once taken near Swansea, imperfect.

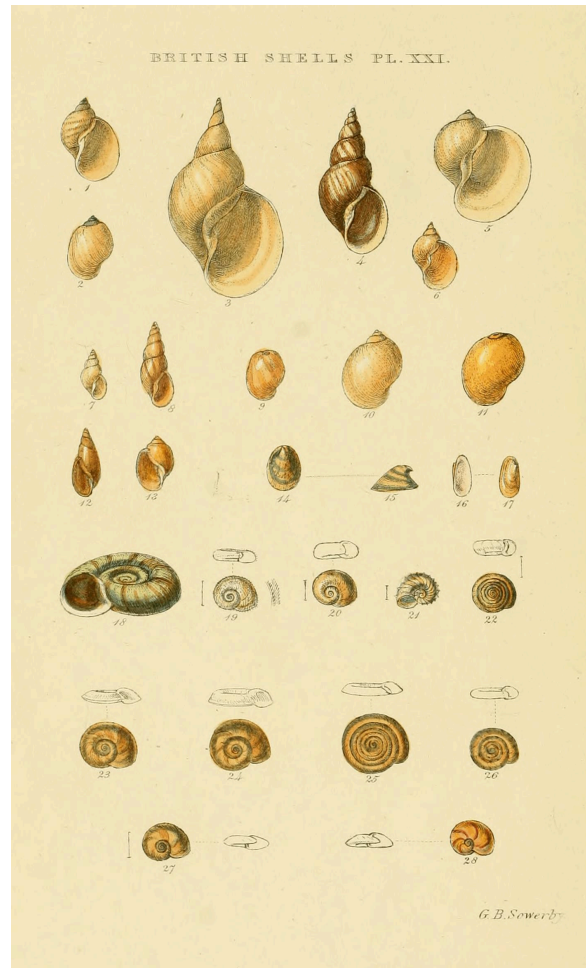


PLATE XXI.

LIMNÆUS. In and near fresh-water.

Fig.

- 1, 2. L. PEREGER, Müller (*Buccinum*.)—*Helix*, *Turbo*, *Bulimus*, and *L. putris*, *inflata*, *teres*, *lutea*, *limosa*, *ovata*, *vulgaris*, *marginata*, *intermedia*, *lacustris*, etc.—Springs and ponds, Cc.
3. L. STAGNALIS, Linn. (*Helix*.)—*Buccinum*, *Turbo*, etc., *stagnalis*, *fragilis*, and *major*, etc.—Ponds, etc., not so generally as No. 1, 2, C.
4. L. PALUSTRIS, Müller, (*Bucc.*)—*Helix* and *L. fragilis*, *fontinalis*, *stagnalis*, etc.—Ditches, ponds, etc., C. Whorls more numerous and less rapidly enlarged than No. 3.

5. *L. AURICULARIUS*, Linn. (*Helix*).—*L.*, *Helix*, *Bucc.*, *Turbo*, *Bulimus a.*, *patulus*, etc.—Near London, Swansea, etc., Mc. Aperture very wide, and outer lip partially reflected.
6. *L. ACUTUS*, Jeffreys.—*L. auricularius*, var. B. M.—Oban, Clumber Lake, Bloomfield, Kent, R. Apex sharp and tapering; aperture smaller than No. 5.
7. *L. TRUNCATULUS*, Müller, (*Bucc.*)—*H.*, *B.*, and *L. truncatus*, *fossarius*, *minutus*, *limosus*, etc.—Partly amphibious, generally diffused, C. Small, spire equal to body.
8. *L. GLABER*, Müller, (*Buccinum*).—*H.*, *B.*, and *L. octanfracta*, *peregrina*, *elongata*, *octona*, *leucostoma*, etc.—Fresh and brackish marshes, Stafford, Somerset, York, etc., R. Spire longer than aperture.
9. *L. INVOLUTUS*, Harvey.—*Amphipeplea i.*—Lake near Killarney, Belfast, Rr. Spire almost covered.
10. *L. GLUTINOSUS*, Müller, (*Buccinum*).—*Helix*, *Amphipeplea*, and *L.*—Windermere, etc., Mr. Oval.
11. *L. BURNETTI*, Alder.—Lakes in Dumfries and Breconshire. Inflated; spire flat; colour opaque-amber.

PHYSA. In and near fresh-water.

12. *P. HYPNORUM*, Linn. (*Bulla*).—*Helix*, *Physa*, *Limnæa*, and *Aplexa turrita* and *marmorata*. Pools and lakes, Brit. generally, C.
13. *P. FONTINALIS*, Linn. (*Bulla*).—*Turbo*, *Planorbis*, *Bulimus*, and *Ph. adversus*, *bulla*, *bulloides*, and *fluviatilis*.—Brit. pools, lakes, and ditches, C.

ANCYLUS. In fresh-water.

- 14, 15. *A. FLUVIATILIS*, Müller.—*Patella lacustris*.—Running streams, on stones and plants generally, C. Oval, apex turned to left.

- 16, 17. A. OBLONGUS, *Lightfoot*, (*Patella*).—A. and *P. lacustris*. More lacustrine than No. 14, Mc. Oblong, apex turned to right.

PLANORBIS. In fresh-water.

18. P. CORNEUS, *Linn.* (*Helix*).—*P.* and *H. cornu-arietis*, *purpureus*, and *similis*, *P. nana* (young).—Ponds and ditches generally, C.
19. P. ALBUS, *Müller*.—*Helix* and *P. hispidus*, *hirsutus*.—Distribution general, C. Finely ciliated in lines.
20. P. GLABER, *Jeffreys*.—*P. lævis*, *Alder*.—Penzance, Falmouth, Swansea, Belfast, etc., Mr.
21. P. NAUTILEUS, *Linn.* (*Helix*).—*P. imbricatus*, *Müller*; *P. cristatus*, *Turbo nautilus*, etc., Mc. With horny ridges.
22. P. CONTORTUS, *Linn.* (*Helix*).—*P.* and *H. crassa*, *umbilicata*.—Distribution general, C. Numerous close whorls; aperture semi-lunar.
23. P. CARINATUS, *Müller*.—*Helix Planorbis*, *Linn.*; *P. planata* and *lutescens*.—Generally distr., C. Whorls more rapidly increased than No. 24; keel more distinct.
24. P. MARGINATUS, *Drap.*—*H. Planorbis*, *H.*, *P. limbata*, *complanata*, *Draparnaldi turgidus*, *rhombeus*, etc.—Distr. gen., C. Whorls more rhomboidal than 23.
25. P. VORTEX, *Linn.* (*Helix*).—*Pl. compressa*.—Distr. gen., C. Many whorls, very compressed.
26. P. SPIORBIS, *Linn.* (*Helix*).—*P. vortex*, var.—Distr. general, C. Fewer whorls than No. 25; mouth not angular.
27. P. NITIDUS, *Müller*.—*Helix* and *P. fontanus*, *complanatus*, and *lenticularis*.—Most parts of Brit., Mc. Very compressed.
28. P. LACUSTRIS, *Lightfoot*.—*P. nitidus*, var., *Segmentina lineata*, *P. nautilus* and *clausulatus*.—Ponds and ditches near London, etc., Mc. With internal divisions; lower disc flat.



PLATE XXII.

SUCCINEA. Amphibious.

Fig.

1. *S. PUTRIS*, Linn. (*Helix*.)—*H.*, *S.*, and *Limnæa Succinea* and *amphibia*.—On plants by the side of rivers, ditches, etc., C. Rather oval in form.
2. *S. GRACILIS*, Alder.—*S. putris*, var. *gracilis*, *S. amphibia*, var. *Pfeifferi*.—By sides of streams and in damp meadows, C. Elongated, with spire produced.
3. *S. OBLONGA*, Drap.—*Helix elongata*, *Amphibulina o.* and *e.*—Swansea, Glasgow, etc. Sometimes at a distance from water, R. Spire with more

distinct whorls.

CONOVULUS. Partly marine, amphibious.

4. *C. BIDENTATUS*, *Mont. (Voluta.)*—*C.*, *N.*, *Actæon*, and *Auricula alba* (var.) and *erosa*.—Salt-water marshes, and on rocks near high-water mark all round our coasts, *C.* No tooth on outer lip.
5. *C. MYSOTIS*, *Drap.*—*C. denticulatus*, var., etc.—Brackish marshes, Thames, etc., *C.* Outer lip with one tooth.
6. *C. DENTICULATUS*, *Mont. (Voluta.)*—*C.*, *V.*, *Auricula*, and *Actæon ringens*, *reflexa* (var. f. 7), *Mysotis*, and *personata*.—Numerous places, near the sea, *C.* Outer lip with many teeth.
7. *C. REFLEXUS*.—*Auricula reflexa*, *C. denticulatus*, var. *Jeffreys*.

CARYCHIUM.

8. *C. MINIMUM*, *Müller.*—*Turbo Carychium*, *Auricula minima*.—Moist places, wet Mosses, etc. Generally distr., *C.*

LIMAX. Land Slug, with shell under shield.

9. *L. CINEREUS* *Müller.*—*L. maximus*, *maculatus*, *Limacella parma*. Brit. gen., *C.*; Eu. Shell oval, large, thin.
10. *L. AGRESTIS*, *Müller.*—*L. concava* and *obliquus*.—Brit, gen., *C. Foss cr.* Shell small, oblique.
11. *L. BRUNNEUS*, *Drap.*—*L. concava*.—Under stones, etc. Shell minute, square.
12. *L. FLAVUS*, *Linn.*—*Limacella unguiculata* and *variegata*.—The large Cellar Slug. Shell thick, rather square.

TESTACELLA. Slug with shell on tail.

13. T. MAUGEI, *Férussac*.—Clifton, Taunton, Plymouth, Swansea, etc., Mr. Large, and more square than 14.
14. T. HALIOTOIDEA, *Drap.*—*T. europæa*, *T. scutulum*.—Near London. Channel Isls., etc., Mc.

VITRINA. Shell not including the whole Slug.

15. V. PELLUCIDA, *Müller*, (*Helix*).—*H. diaphana*, *Helicolimax*, *Limacina major*, etc.—Various parts of the country, under stones, etc., Mc. Eu. Very transparent.
16. V. DRAPARNALDI, *Jeffreys*.—*V. oblonga*, *H. brevipes*?—Less globular than 15, R.

ZONITES. Transparent, thin-edged Snails.

- 17, 18. Z. CELLARIUS, *Müller*, (*Helix*).—*H.* and *Z. lucida* and *nitens*.—Brit. gen., C.
- 19, 20. Z. NITIDUS, *Müller*, (*Helix*).—*H.* and *Z. nitens*, *lucida*, *tennis*, *Succinea*, etc.—Less depressed than 17, 18, with larger umbilicus.
21. Z. CRYSTALLINUS, *Müller*, (*Helix*).—*H.*, *Z.*, *Discus*, and "*Polita*" *c.*, *eburnea*, *vitrea*.—In damp spots, among stones and Moss. Minute, white, transparent.
22. Z. NITIDULUS, *Drap.*—Generally distributed, Mc. Mouth oblique. Umbilicus small.
- 23, 24. Z. ALLARIUS, *Miller*, (*Helix*).—*H.* and *Z. glabra*, *alliacea*, *nitens*, *tenera*, *lurida*, etc.—Distr. gen., C. Smaller and more convex than 17, 18.
25. Z. EXCAVATUS, *Bean*, (*Helix*).—*H.* and *G. nitida*, *lucida*, etc.—Durham, Northumberland, Galway, etc., Mc. Umbilicus very large.
- 26, 27. Z. RADIATULUS, *Alder*, (*Helix*).—Distr. gen., C. Striated, minute.

28, 29. *Z. PURUS*, Alder, (*Helix*.)—*H.* and *Z. polita*, and *Helicella nitidosa*, *electrina*, etc.—Most common North Engl. Whorls less close and more oblique than 21.

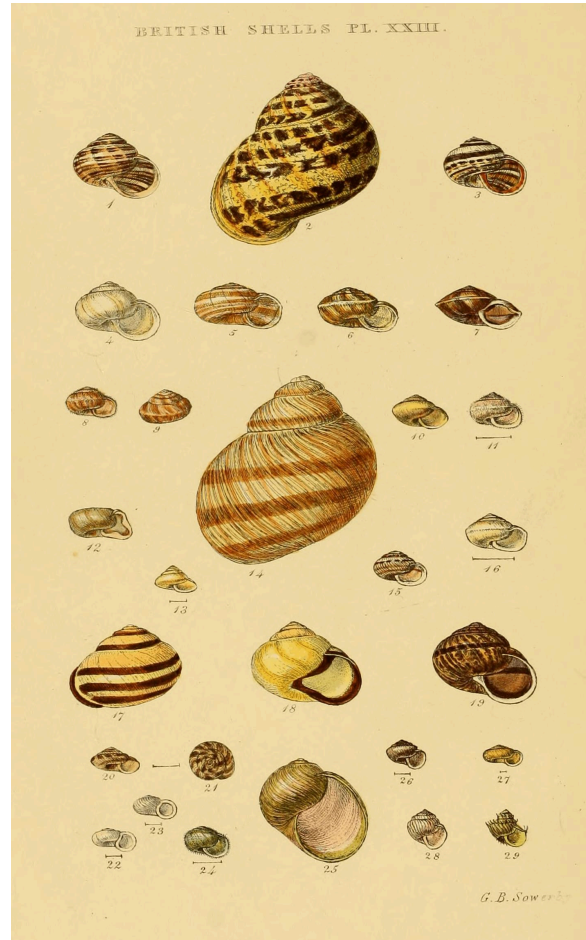


PLATE XXIII.

HELIX. Snails with thickened lips.

Fig.

1. *H. PISANA*, Müller.—*H. zonaria, cingenda, rhodostoma, strigata, albella*.
—Cornwall, S. Wales, Dublin, C. S. and N. Eu.
2. *H. ASPERSA*, Müller.—*H. grisea, hortensis, variegata, vulgaris*, etc.—
Common in all our gardens and hedges.

3. *H. VIRGATA*, Costa.—*H. variabilis, ericetorum, zonaria*, etc.—In sandy and chalky places, C. More compressed than No. 1.
4. *H. CANTIANA*, Mont.—*H. pallida*, etc.—South and west Engl., C.
5. *H. ERICETORUM*, Müller.—*H. itala, striata, albelli, erica, nivea, cæspita*, etc.—Chalky and sandy places, C. Depressed, delicately banded; large umbilicus.
6. *H. RUFESCENS*, Pennant.—*H. rufina, montana, cæolata, clandestina*, etc.—Most common in southern, chalky and limestone districts. Depressed, striated; whorls slightly angular. Generally lighter than our figure, sometimes nearly white.
7. *H. LAPICIDA*, Linn.—*H. acuta, affinis, Carocolla l.*—Chalk and limestone districts, C. S. and N. Eu. Sharply angular.
- 8, 9. *H. HISPIDA*, Linn.—*H. rufescens, sericea, concinna* (var. without hairs), etc., *plebeia*.—All parts of Brit., C. More depressed than *H. sericea*, with larger umbilicus.
10. *H. FUSCA*, Mont.—*H. subrufescens*.—Brit. gen., Mc.
11. *H. SERICEA*, Drap.—*H. hispida, globularis, granulata*, etc.—Granulated, hairy. Mouth wider than 8. Small umbilicus.
12. *H. OBVOLUTA*, Müller.—*H. trigonophora*, etc.—Ditcham Wood, Hampshire. In Moss at roots of trees, Rr.
13. *H. FULVA*, Müller.—*H. terrestris, trochiformis, nitidula*, etc.—Pretty general and common. Conical, minute, transparent.
14. *H. POMATIA*, Linn.—Southern chalky districts, S. Eu., C.
15. *H. CAPERATA*, Mont.—*H. striata, H. crenulata*, etc.—Chalk-pits, etc., C.
16. *H. CARTHUSIANA*, Müller.—*H. nitida, carthusianella, rufilabris*, etc.—Chalk-districts of Surrey and Kent, C.
- 17, 18. *H. NEMORALIS*, Linn.—*H. hortensis, fasciata, hybrida*, etc.—Cc. A peculiar flattening near the base of aperture.

19. H. ARBUSTORUM, *Linn.*—Common and general. Brown-marbled. Edge of aperture different from 17, 18.
- 20, 21. H. ROTUNDATA.—*H. radiata*, *Turtoni*, etc.—Common and general.
- 22, 23. H. PULCHELLA, *Müller.*—*H. costata*, *paludosa*, *crenella*, *minuta*, etc.—Common and general under stones, etc.
24. H. REVELATA, *Férussac.*—Cornwall, Devon, Torquay, Guernsey. Rare. A transparent, greenish, hairy shell.
25. H. APERTA.—*H. neritoides*, *naticoides*, etc.—A single specimen found in Guernsey. Not uncommon in S. Eu. Our specimen is foreign.
26. H. UMBILICATA, *Mont.*—*H. rupestris*, etc.—On perpendicular heights and under stones, C.
27. H. PYGMÆA, *Drap.*—*H. minuta*, etc.—Common in damp places.
28. H. LAMELLATA, *Jeffreys.*—*H. Scarburgensis*, *seminulum*.—Brit. and Germany, C.
29. H. ACULEATA, *Müller.*—*H. spinulosa*, etc.—Among Moss and stones, etc., C. General.

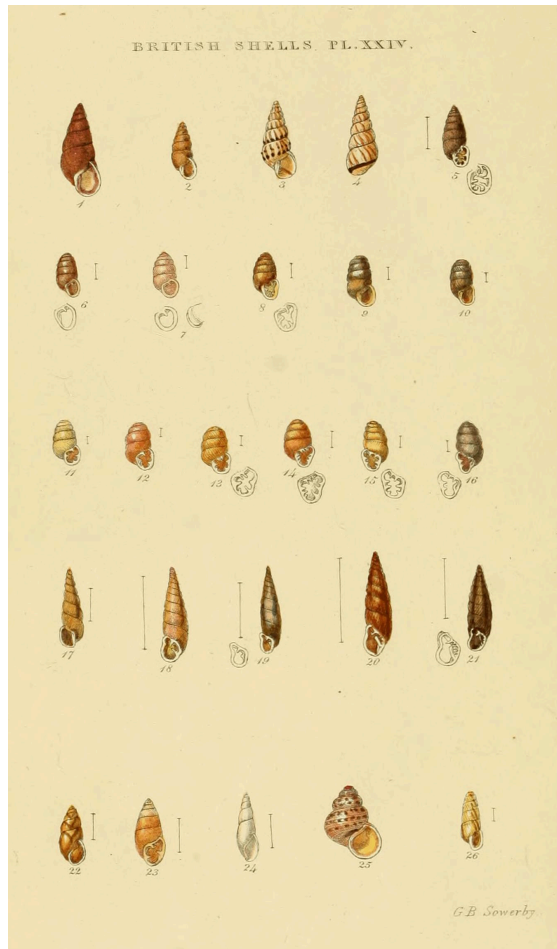


PLATE XXIV.

BULIMUS. Long Snail.

Fig.

1. *B. LACKHAMENSIS*, Mont.—*B. montanus*, Montacuti, *Eua montana*.—Kent, Surrey, Gloucester, Selborne, N. and S. Eu., R. as Brit. Shagreened.
2. *B. OBSCURUS*, Müller.—*Turbo rupium*, *B. hordaceus*, etc.—C. and general. Surface smooth.
- 3, 4. *B. ACUTUS*, Müller.—*Helix barbara*, *B. cretacea*, *articulatus*, *Turbo fasciatus*, etc.—On sand and limestone in marine neighbourhoods, C.

PUPA. Small Snails, among Moss, stones, etc.

5. *P. SECALE*, *Drap.*—*Turbo*, *Helix*, *Chondrus*, and *Vertigo juniperi* and *cylindricus*.—S. Engl., Channel Isls., C. S. Eu. Larger and more tapering than the other sp.
6. *P. UMBILICATA*, *Drap.*—*Turbo*, *Bulimus*, *Helix*, and *P. muscorum*, *cylindraceus*, *bidentata*, etc.—C. and general, Eu. Mouth oblique, long, one tooth.
7. *P. MUSCORUM*, *Linn.*—*T.*, *P.*, *H.*, *B. marginata*, *chrysalis*, *unidentatus*, etc.—C. and gen. A rib behind the outer lip.
8. *P. ANGLICA*, *Férussac.*—*Vertigo*, *Turbo*, and *P. a.* and *ringens*.—Scarborough, Newcastle, Guernsey, etc., Mr. More cylindrical than 5.
9. *P. EDENTULA*, *Drap.*—*H. exigua*, *Vertigo*, *Helix*, and *Jamimia e.*, *nitida*, etc.—C. and gen. No teeth in aperture; smooth.
10. *P. MINUTISSIMA*, *Hartmann.*—*P.*, *Vertigo*, *Alæa cylindrica*, *obtusa*, *muscorum*, etc.—Bristol, Salisbury, Skye, etc., R. as Brit.; C. in S. Eu. No teeth in aperture; striated.
- 11, 12. *P. PYGMÆA*, *Drap.*—*P.*, *T.*, *V.*, *H.*, *Alæa*, etc., *sexdentatus*, *quinquedentatus*, *vulgaris*, etc.—The lighter variety, *P. alpestris*, C. and general. Four or five teeth.
13. *P. SUBSTRIATA*, *Jeffreys.*—*Vertigo*, *Turbo*, *Alæa sexdentata*, *curta*, etc.—C. and gen. Very cylindrical; six teeth; striated.
14. *P. ANTIVERTIGO*, *Drap.*—*P.*, *V.*, *Alæa octodentata*, *septemdentata*, *palustris*, etc.—Near Swansea and London; Bristol, Devon, Northumberland, Ayr, Mr. Ireland. Aperture subtrigonal, eight or nine teeth.
15. *P. PUSILLA*, *Müller.*—*Helix vertigo*, and *V. heterostropha*.—S. Wales, Northumberland, Belfast, etc., Mc. Sinistral, with seven teeth.
16. *P. VENETZII*, *Charpentier.*—*Turbo*, *Vertigo*, and *P. vertigo* and *angustior*.—Swansea, Cork, Clare, R. Sinistral; four teeth and a flat fold on columella.

BALEA.

17. B. FRAGILIS, *Drap. (Pupa.)*—*Bulimus, Pupa, Clausilia f., perversus.*—C. and gen. Under bark on trees, and under stones.

CLAUSILIA. With a twisted *clausium* on columella.

18. C. LAMINATA, *Mont.*—*Helix, Turbo, Bulimus,* and *Pupa bidens* and *perversus* and *derugata.*—S. Brit. principally. Among dead leaves, and on Beech-trunks, Mc. Smooth, broad.
19. C. NIGRICANS, *Maton and Rackett.*—*Strombiformis, Turbo,* and *Odostomia perversus, bidens, parvula,* etc.—C. and gen.; under stones, on walls and trees, etc. Narrow, striated.
20. C. BIPPLICATA, *Mont.*—*Helix, Turbo, Odost., C. perversa, plicata, ventricosa,* etc.—Near London, etc., Mc. Two folds on columella.
21. C. PLICATULA, *Drap.*—*Turbo* and *C. conversus.*—Charlton Wood, Kent, Hastings, Gloucestershire, etc., R. Numerous plaits on columella above fold.

ZUA.

22. Z. LUBRICA, *Müller, (Helix.)*—*Turbo, Bulimus l., subcylindrica, muscorum,* etc.—C. and gen. Among leaves and under stones.

AZECA.

23. A. TRIDENS, *Pulteney, (Turbo.)*—*Helix, Pupa* and *Cionella Goodallii, Britannica,* etc.—C. and general. Habits as *Zua* and *Pupa.*

ACHATINA.

24. A. ACICULA, *Müller.*—*Buccinum, Bulimus, Helix octona, terrestre,* etc.—In marshy places, among Grass, etc., C. S. Eu.

CYCLOSTOMA. Snail with operculum.

25. *C. ELEGANS*, Müller, (*Nerita*.)—*Turbo reflexus*, *T. tumidus*, *striatus*, *elegans*.—Chalky districts of S. Engl., C.

ACME. Cylindrical Snail with operculum.

26. *A. LINEATA*, Drap. (*Auricula*.)—*Turbo*, *Carychium*, *Cyclostoma*, *Bulimus fuscus*, *Cochlea minuta*.—Bristol, Pembroke, Clare, Dublin, etc., R.
-

G. B. SOWERBY, JUNR.,

Supplies single Specimens or large parcels of

RECENT SHELLS AND FOSSILS

Parcels will be sent for selection, to those who forward suitable credentials, in any part of the world, so that those Collectors whose supplies have been interrupted by a recent lamented death can now have them renewed.

G. B. S., Junr., has a very large stock of Shells, purchased from the VAUTERIAN and DENNISONIAN Collections, and from the stores of the late M. CUMING, as well as from the sale of Mr. REEVE'S stock, and other sources.

COLLECTIONS OF BRITISH SHELLS.

Seventy-five named species, arranged in a box, having the appearance of a book in two volumes, One Guinea. Or a more extensive series prepared in the same manner.

THE RECENTLY DESCRIBED FOSSILS FROM THE LINGULA
FLAGS, SOUTH WALES, SUPPLIED SINGLY OR IN SETS.

BOOKS BY G. B. SOWERBY, F.L.S.

(Discount to direct purchasers.)

THESAURUS CONCHYLIIORUM;

OR MONOGRAPHS OF RECENT SHELLS.

Parts 24 and 25 will be published before Christmas, 1865, containing additions to former Monographs, and some new Monographs;

Completing Vol III. and the First Series.

The New Series, to be commenced early next year, will appear under improved conditions, and will be carried to a completion in as small a compass and as short a time as possible.

ILLUSTRATED INDEX OF BRITISH SHELLS.

Cloth gilt, Imperial.

700 coloured figures of 600 species, with names, localities, and other information.

NEW PROPOSED WORK, BY G. B. SOWERBY, F.L.S.

GENERIC FORMS OF SHELLS ILLUSTRATED.

The Author has long been engaged collecting materials for this Work, which is to contain a harmonization of existing systems, in which a balance will be struck between a too negligent and too minute subdivision. Comparative susceptibility of *definition* will be the guiding principle in deciding the claims of certain groups to generic separation. The illustrations will be copious and of full size, embracing every marked form, whether recent or fossil, and whether adopted as a genus or not. The price is not yet fixed, but is expected to be about £3 3s. To appear between March and May, 1866.

G. B. S., jun., will be glad to receive the names and addresses of those who desire to see the Work when published.

45, GREAT RUSSELL STREET, BLOOMSBURY.

NATURAL HISTORY AGENCY OFFICES,

CONDUCTED BY

G. B. SOWERBY, F.L.S.,

45, GREAT RUSSELL STREET, BLOOMSBURY.

Collections Bought, Sold on Commission, or Prepared for Auction.

STOCK ON HAND.

FOSSILS.

Crag.—Numerous fine Teeth, Bones, and Shells.

LONDON CLAY.—Fish-heads, Teeth, Fruits, Teredinæ, etc.

CHALK and GAULT.—A magnificent series, including fine specimens of Goniaster, Cidaris, Ventriculites, Crustacea, Shells, etc.

GREENSAND.—A comprehensive series from Blackdown, Cambridge, etc.

OOLITES.—A fine specimen of the New Cidaris (block with five individuals and many long spines), extensive Minchinhampton series, etc.

LIAS.—Ichthyosaurus,—a fine museum specimen, mounted 6 ft. 6 in.,—Plesiosaurus, Teleosaurus, a most beautiful specimen of Lepidotus, and numerous other Fishes in fine condition.

COAL.—Fine specimens of Lepidodendron, Sigillaria, Ferns, etc.

DEVONIAN Fishes, Plants, etc.

RECENT SHELLS.

Among a most extensive range of genera and species are included *Voluta Rossiana*, *V. fulgetrum*, *Conus Aurisiacus*, *C. princeps*, *C. marchionatus*, a fine series of *C. venulatus*, etc., *Halia Priamus*, *Cypræa picta*, *Cardium hians*, *C. Æolicum* (true), *Galathæa concamerata*, *Velorita*, *Anastoma*, beautiful *Spondyli*, etc. Series of British Shells made up.

MINERALS.

A very extensive series of specimens; also illustrative collections.

BIRDS AND INSECTS.

A fine collection of British and other Birds in 31 cases.

Beautiful rosewood cabinet (48 drawers), with fine Butterflies, etc.

Beautiful mahogany cabinet (40 drawers), with fine Butterflies, etc.

BOOKS, NEW AND SECOND-HAND.

Works on Crustacea, including Milne-Edwards' 'Histoire;' Desmarests' 'Considérations;' Roux, 'Crust. de la Méditerranée;' 'Crustacea of the Fauna Japonica,' etc.; on Zoology, the 'Historia de Chili;' 'Journal de Conchyliologie,' by Crosse; Bernardi's 'Monog. des Galathées,' etc.; Supplement to Sowerby's 'Cones;' Deshayes' 'Coquilles Fossiles de Paris;' Sowerby's *Thesaurus Conchyliorum*, Pt. 1 to 22; Sowerby's 'Wild Flowers;' and new edition of Sowerby's 'English Botany.'

THESAURUS CONCHYLIIORUM, 22 parts at 25s. each.

ILLUSTRATED INDEX OF BRITISH SHELLS, 700 Coloured Figures, with Names, Localities, and other information; by G. B. Sowerby, F.L.S., etc. Impl. cloth gilt, 30s. Labels, 6d.

* * * *Further particulars will be furnished. Sets made up to illustrate Formations, Classes, or Genera, and specimens forwarded on approval.*

*** END OF THE PROJECT GUTENBERG EBOOK ILLUSTRATED
INDEX OF BRITISH SHELLS ***

Updated editions will replace the previous one—the old editions will be renamed.

Creating the works from print editions not protected by U.S. copyright law means that no one owns a United States copyright in these works, so the Foundation (and you!) can copy and distribute it in the United States without permission and without paying copyright royalties. Special rules, set forth in the General Terms of Use part of this license, apply to copying and distributing Project Gutenberg™ electronic works to protect the PROJECT GUTENBERG™ concept and trademark. Project Gutenberg is a registered trademark, and may not be used if you charge for an eBook, except by following the terms of the trademark license, including paying royalties for use of the Project Gutenberg trademark. If you do not charge anything for copies of this eBook, complying with the trademark license is very easy. You may use this eBook for nearly any purpose such as creation of derivative works, reports, performances and research. Project Gutenberg eBooks may be modified and printed and given away—you may do practically ANYTHING in the United States with eBooks not protected by U.S. copyright law. Redistribution is subject to the trademark license, especially commercial redistribution.

START: FULL LICENSE

THE FULL PROJECT GUTENBERG™ LICENSE

PLEASE READ THIS BEFORE YOU DISTRIBUTE OR USE THIS WORK

To protect the Project Gutenberg™ mission of promoting the free distribution of electronic works, by using or distributing this work (or any other work associated in any way with the phrase “Project Gutenberg”), you agree to comply with all the terms of the Full Project Gutenberg License available with this file or online at www.gutenberg.org/license.

Section 1. General Terms of Use and Redistributing Project Gutenberg electronic works

1.A. By reading or using any part of this Project Gutenberg electronic work, you indicate that you have read, understand, agree to and accept all the terms of this license and intellectual property (trademark/copyright) agreement. If you do not agree to abide by all the terms of this agreement, you must cease using and return or destroy all copies of Project Gutenberg electronic works in your possession. If you paid a fee for obtaining a copy of or access to a Project Gutenberg electronic work and you do not agree to be bound by the terms of this agreement, you may obtain a refund from the person or entity to whom you paid the fee as set forth in paragraph 1.E.8.

1.B. “Project Gutenberg” is a registered trademark. It may only be used on or associated in any way with an electronic work by people who agree to be bound by the terms of this agreement. There are a few things that you can do with most Project Gutenberg electronic works even without complying with the full terms of this agreement. See paragraph 1.C below. There are a lot of things you can do with Project Gutenberg electronic works if you follow the terms of this agreement and help preserve free future access to Project Gutenberg electronic works. See paragraph 1.E below.

1.C. The Project Gutenberg Literary Archive Foundation (“the Foundation” or PGLAF), owns a compilation copyright in the

collection of Project Gutenberg electronic works. Nearly all the individual works in the collection are in the public domain in the United States. If an individual work is unprotected by copyright law in the United States and you are located in the United States, we do not claim a right to prevent you from copying, distributing, performing, displaying or creating derivative works based on the work as long as all references to Project Gutenberg are removed. Of course, we hope that you will support the Project Gutenberg mission of promoting free access to electronic works by freely sharing Project Gutenberg works in compliance with the terms of this agreement for keeping the Project Gutenberg name associated with the work. You can easily comply with the terms of this agreement by keeping this work in the same format with its attached full Project Gutenberg License when you share it without charge with others.

1.D. The copyright laws of the place where you are located also govern what you can do with this work. Copyright laws in most countries are in a constant state of change. If you are outside the United States, check the laws of your country in addition to the terms of this agreement before downloading, copying, displaying, performing, distributing or creating derivative works based on this work or any other Project Gutenberg work. The Foundation makes no representations concerning the copyright status of any work in any country other than the United States.

1.E. Unless you have removed all references to Project Gutenberg:

1.E.1. The following sentence, with active links to, or other immediate access to, the full Project Gutenberg License must appear prominently whenever any copy of a Project Gutenberg work (any work on which the phrase “Project Gutenberg” appears, or with which the phrase “Project Gutenberg” is associated) is accessed, displayed, performed, viewed, copied or distributed:

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg™ License included with

this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

1.E.2. If an individual Project Gutenberg electronic work is derived from texts not protected by U.S. copyright law (does not contain a notice indicating that it is posted with permission of the copyright holder), the work can be copied and distributed to anyone in the United States without paying any fees or charges. If you are redistributing or providing access to a work with the phrase “Project Gutenberg” associated with or appearing on the work, you must comply either with the requirements of paragraphs 1.E.1 through 1.E.7 or obtain permission for the use of the work and the Project Gutenberg trademark as set forth in paragraphs 1.E.8 or 1.E.9.

1.E.3. If an individual Project Gutenberg electronic work is posted with the permission of the copyright holder, your use and distribution must comply with both paragraphs 1.E.1 through 1.E.7 and any additional terms imposed by the copyright holder. Additional terms will be linked to the Project Gutenberg License for all works posted with the permission of the copyright holder found at the beginning of this work.

1.E.4. Do not unlink or detach or remove the full Project Gutenberg License terms from this work, or any files containing a part of this work or any other work associated with Project Gutenberg.

1.E.5. Do not copy, display, perform, distribute or redistribute this electronic work, or any part of this electronic work, without prominently displaying the sentence set forth in paragraph 1.E.1 with active links or immediate access to the full terms of the Project Gutenberg License.

1.E.6. You may convert to and distribute this work in any binary, compressed, marked up, nonproprietary or proprietary form, including any word processing or hypertext form. However, if you provide access to or distribute copies of a Project Gutenberg work in a format other than “Plain Vanilla ASCII” or other format used in the official

version posted on the official Project Gutenberg website (www.gutenberg.org), you must, at no additional cost, fee or expense to the user, provide a copy, a means of exporting a copy, or a means of obtaining a copy upon request, of the work in its original “Plain Vanilla ASCII” or other form. Any alternate format must include the full Project Gutenberg License as specified in paragraph 1.E.1.

1.E.7. Do not charge a fee for access to, viewing, displaying, performing, copying or distributing any Project Gutenberg works unless you comply with paragraph 1.E.8 or 1.E.9.

1.E.8. You may charge a reasonable fee for copies of or providing access to or distributing Project Gutenberg electronic works provided that:

- You pay a royalty fee of 20% of the gross profits you derive from the use of Project Gutenberg works calculated using the method you already use to calculate your applicable taxes. The fee is owed to the owner of the Project Gutenberg trademark, but he has agreed to donate royalties under this paragraph to the Project Gutenberg Literary Archive Foundation. Royalty payments must be paid within 60 days following each date on which you prepare (or are legally required to prepare) your periodic tax returns. Royalty payments should be clearly marked as such and sent to the Project Gutenberg Literary Archive Foundation at the address specified in Section 4, “Information about donations to the Project Gutenberg Literary Archive Foundation.”
- You provide a full refund of any money paid by a user who notifies you in writing (or by e-mail) within 30 days of receipt that s/he does not agree to the terms of the full Project Gutenberg™ License. You must require such a user to return or destroy all copies of the works possessed in a physical medium and discontinue all use of and all access to other copies of Project Gutenberg™ works.
- You provide, in accordance with paragraph 1.F.3, a full refund of any money paid for a work or a replacement copy, if a defect in the electronic work is discovered and reported to you within 90 days of receipt of the work.

- You comply with all other terms of this agreement for free distribution of Project Gutenberg™ works.

1.E.9. If you wish to charge a fee or distribute a Project Gutenberg™ electronic work or group of works on different terms than are set forth in this agreement, you must obtain permission in writing from the Project Gutenberg Literary Archive Foundation, the manager of the Project Gutenberg™ trademark. Contact the Foundation as set forth in Section 3 below.

1.F.

1.F.1. Project Gutenberg volunteers and employees expend considerable effort to identify, do copyright research on, transcribe and proofread works not protected by U.S. copyright law in creating the Project Gutenberg™ collection. Despite these efforts, Project Gutenberg™ electronic works, and the medium on which they may be stored, may contain “Defects,” such as, but not limited to, incomplete, inaccurate or corrupt data, transcription errors, a copyright or other intellectual property infringement, a defective or damaged disk or other medium, a computer virus, or computer codes that damage or cannot be read by your equipment.

1.F.2. LIMITED WARRANTY, DISCLAIMER OF DAMAGES - Except for the “Right of Replacement or Refund” described in paragraph 1.F.3, the Project Gutenberg Literary Archive Foundation, the owner of the Project Gutenberg™ trademark, and any other party distributing a Project Gutenberg™ electronic work under this agreement, disclaim all liability to you for damages, costs and expenses, including legal fees. YOU AGREE THAT YOU HAVE NO REMEDIES FOR NEGLIGENCE, STRICT LIABILITY, BREACH OF WARRANTY OR BREACH OF CONTRACT EXCEPT THOSE PROVIDED IN PARAGRAPH 1.F.3. YOU AGREE THAT THE FOUNDATION, THE TRADEMARK OWNER, AND ANY DISTRIBUTOR UNDER THIS AGREEMENT WILL NOT BE LIABLE TO YOU FOR ACTUAL, DIRECT, INDIRECT, CONSEQUENTIAL, PUNITIVE OR INCIDENTAL DAMAGES

EVEN IF YOU GIVE NOTICE OF THE POSSIBILITY OF SUCH DAMAGE.

1.F.3. LIMITED RIGHT OF REPLACEMENT OR REFUND - If you discover a defect in this electronic work within 90 days of receiving it, you can receive a refund of the money (if any) you paid for it by sending a written explanation to the person you received the work from. If you received the work on a physical medium, you must return the medium with your written explanation. The person or entity that provided you with the defective work may elect to provide a replacement copy in lieu of a refund. If you received the work electronically, the person or entity providing it to you may choose to give you a second opportunity to receive the work electronically in lieu of a refund. If the second copy is also defective, you may demand a refund in writing without further opportunities to fix the problem.

1.F.4. Except for the limited right of replacement or refund set forth in paragraph 1.F.3, this work is provided to you 'AS-IS', WITH NO OTHER WARRANTIES OF ANY KIND, EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO WARRANTIES OF MERCHANTABILITY OR FITNESS FOR ANY PURPOSE.

1.F.5. Some states do not allow disclaimers of certain implied warranties or the exclusion or limitation of certain types of damages. If any disclaimer or limitation set forth in this agreement violates the law of the state applicable to this agreement, the agreement shall be interpreted to make the maximum disclaimer or limitation permitted by the applicable state law. The invalidity or unenforceability of any provision of this agreement shall not void the remaining provisions.

1.F.6. INDEMNITY - You agree to indemnify and hold the Foundation, the trademark owner, any agent or employee of the Foundation, anyone providing copies of Project Gutenberg™ electronic works in accordance with this agreement, and any volunteers associated with the production, promotion and distribution of Project Gutenberg™ electronic works, harmless from all liability, costs and expenses, including legal fees, that arise directly or indirectly from any of the following which you do or cause to occur: (a)

distribution of this or any Project Gutenberg work, (b) alteration, modification, or additions or deletions to any Project Gutenberg work, and (c) any Defect you cause.

Section 2. Information about the Mission of Project Gutenberg

Project Gutenberg is synonymous with the free distribution of electronic works in formats readable by the widest variety of computers including obsolete, old, middle-aged and new computers. It exists because of the efforts of hundreds of volunteers and donations from people in all walks of life.

Volunteers and financial support to provide volunteers with the assistance they need are critical to reaching Project Gutenberg's goals and ensuring that the Project Gutenberg collection will remain freely available for generations to come. In 2001, the Project Gutenberg Literary Archive Foundation was created to provide a secure and permanent future for Project Gutenberg and future generations. To learn more about the Project Gutenberg Literary Archive Foundation and how your efforts and donations can help, see Sections 3 and 4 and the Foundation information page at www.gutenberg.org.

Section 3. Information about the Project Gutenberg Literary Archive Foundation

The Project Gutenberg Literary Archive Foundation is a non-profit 501(c)(3) educational corporation organized under the laws of the state of Mississippi and granted tax exempt status by the Internal Revenue Service. The Foundation's EIN or federal tax identification number is 64-6221541. Contributions to the Project Gutenberg Literary Archive Foundation are tax deductible to the full extent permitted by U.S. federal laws and your state's laws.

The Foundation's business office is located at 41 Watchung Plaza #516, Montclair NJ 07042, USA, +1 (862) 621-9288. Email contact

links and up to date contact information can be found at the Foundation's website and official page at www.gutenberg.org/contact

Section 4. Information about Donations to the Project Gutenberg Literary Archive Foundation

Project Gutenberg™ depends upon and cannot survive without widespread public support and donations to carry out its mission of increasing the number of public domain and licensed works that can be freely distributed in machine-readable form accessible by the widest array of equipment including outdated equipment. Many small donations (\$1 to \$5,000) are particularly important to maintaining tax exempt status with the IRS.

The Foundation is committed to complying with the laws regulating charities and charitable donations in all 50 states of the United States. Compliance requirements are not uniform and it takes a considerable effort, much paperwork and many fees to meet and keep up with these requirements. We do not solicit donations in locations where we have not received written confirmation of compliance. To SEND DONATIONS or determine the status of compliance for any particular state visit www.gutenberg.org/donate.

While we cannot and do not solicit contributions from states where we have not met the solicitation requirements, we know of no prohibition against accepting unsolicited donations from donors in such states who approach us with offers to donate.

International donations are gratefully accepted, but we cannot make any statements concerning tax treatment of donations received from outside the United States. U.S. laws alone swamp our small staff.

Please check the Project Gutenberg web pages for current donation methods and addresses. Donations are accepted in a number of other ways including checks, online payments and credit card donations. To donate, please visit: www.gutenberg.org/donate.

Section 5. General Information About Project Gutenberg electronic works

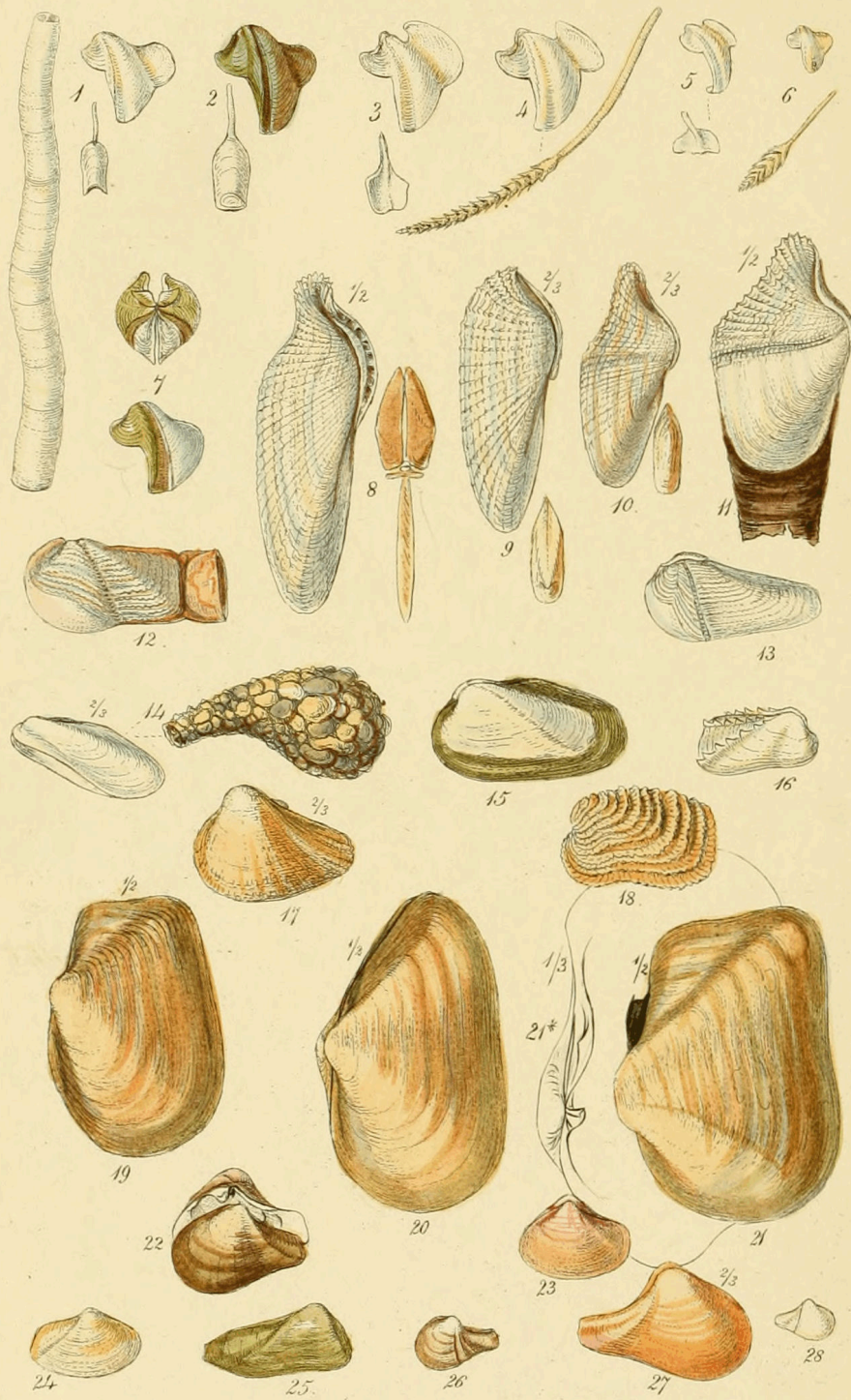
Professor Michael S. Hart was the originator of the Project Gutenberg concept of a library of electronic works that could be freely shared with anyone. For forty years, he produced and distributed Project Gutenberg eBooks with only a loose network of volunteer support.

Project Gutenberg eBooks are often created from several printed editions, all of which are confirmed as not protected by copyright in the U.S. unless a copyright notice is included. Thus, we do not necessarily keep eBooks in compliance with any particular paper edition.

Most people start at our website which has the main PG search facility:
www.gutenberg.org.

This website includes information about Project Gutenberg, including how to make donations to the Project Gutenberg Literary Archive Foundation, how to help produce our new eBooks, and how to subscribe to our email newsletter to hear about new eBooks.

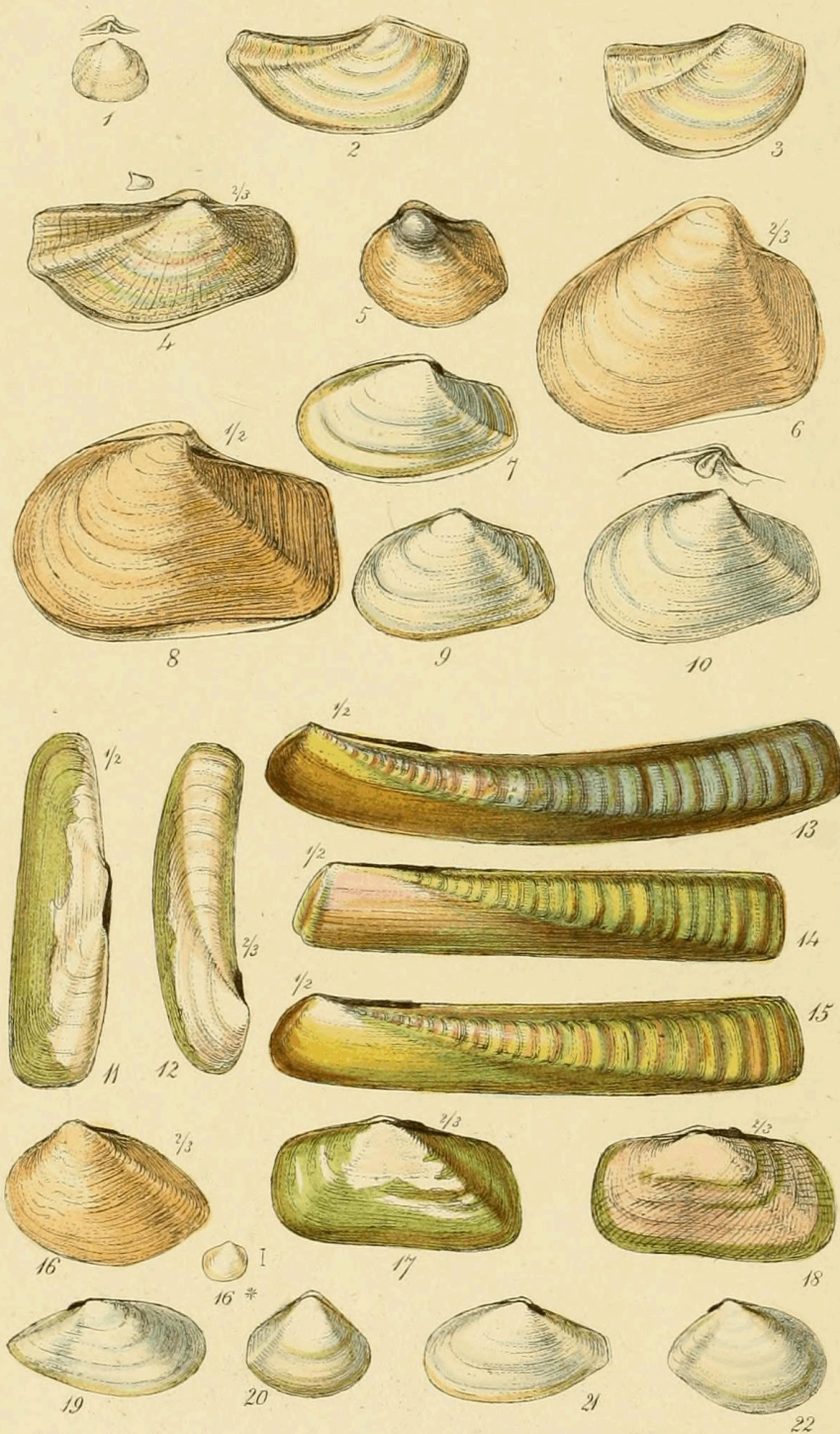
BRITISH SHELLS. PL. I.



G. B. Sowerby.

[back](#)

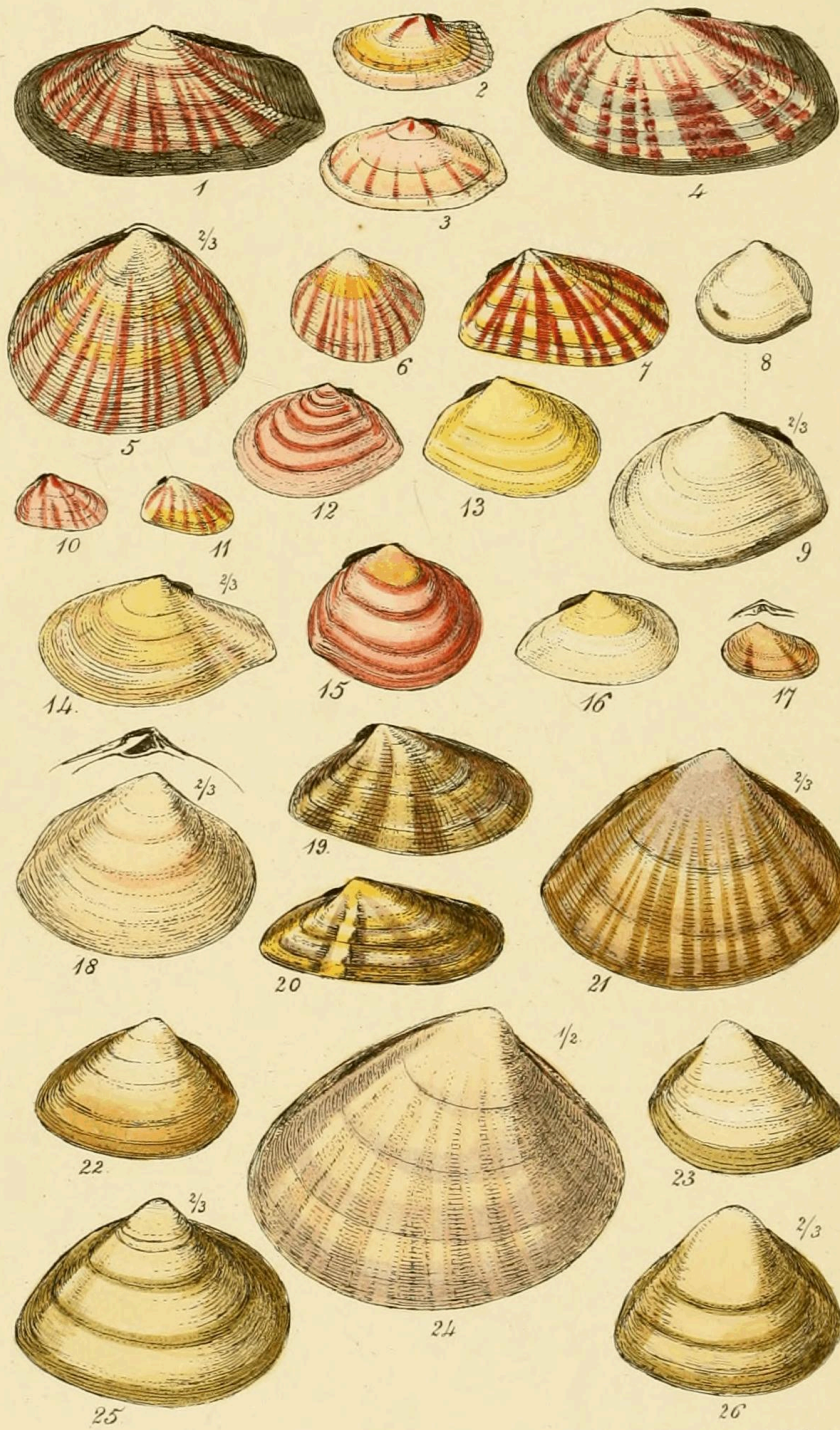
BRITISH SHELLS. PL. II.



G. B. Sowerby.

[back](#)

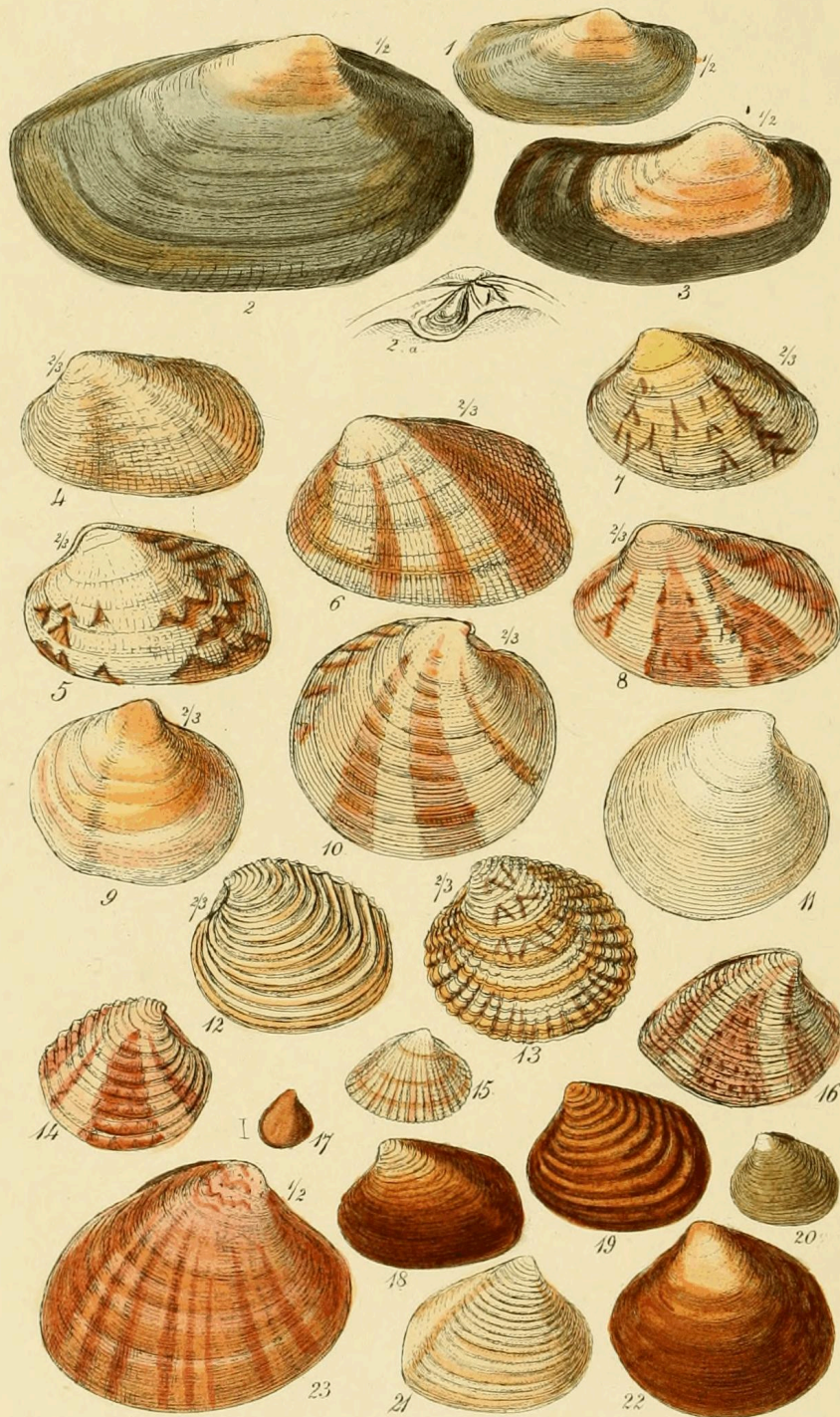
BRITISH SHELLS, PL. III.



G.B. Sowerby.

[back](#)

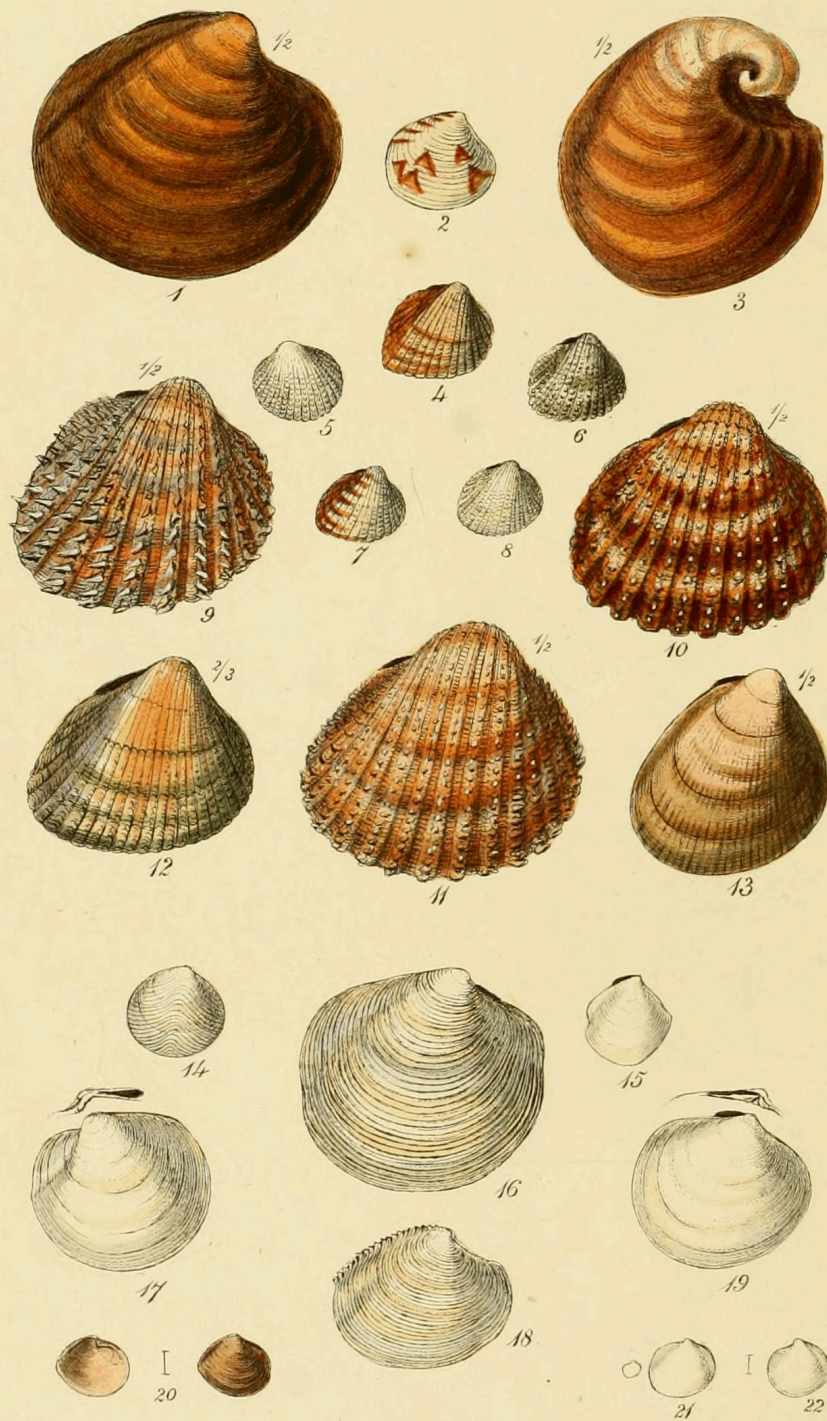
BRITISH SHELLS, PL. IV.



G. B. Sowerby.

[back](#)

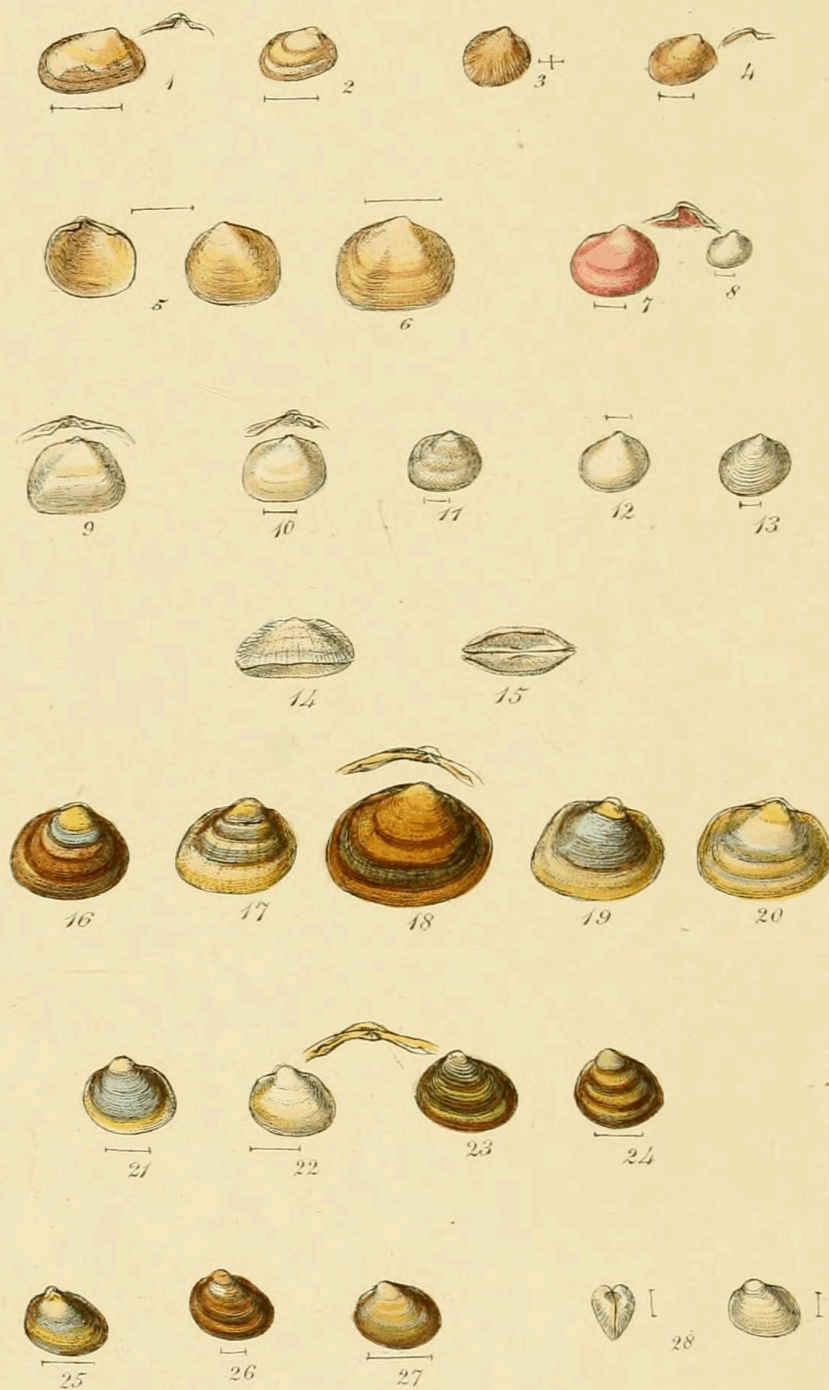
BRITISH SHELLS, PL. V.



G.B. Sowerby.

[back](#)

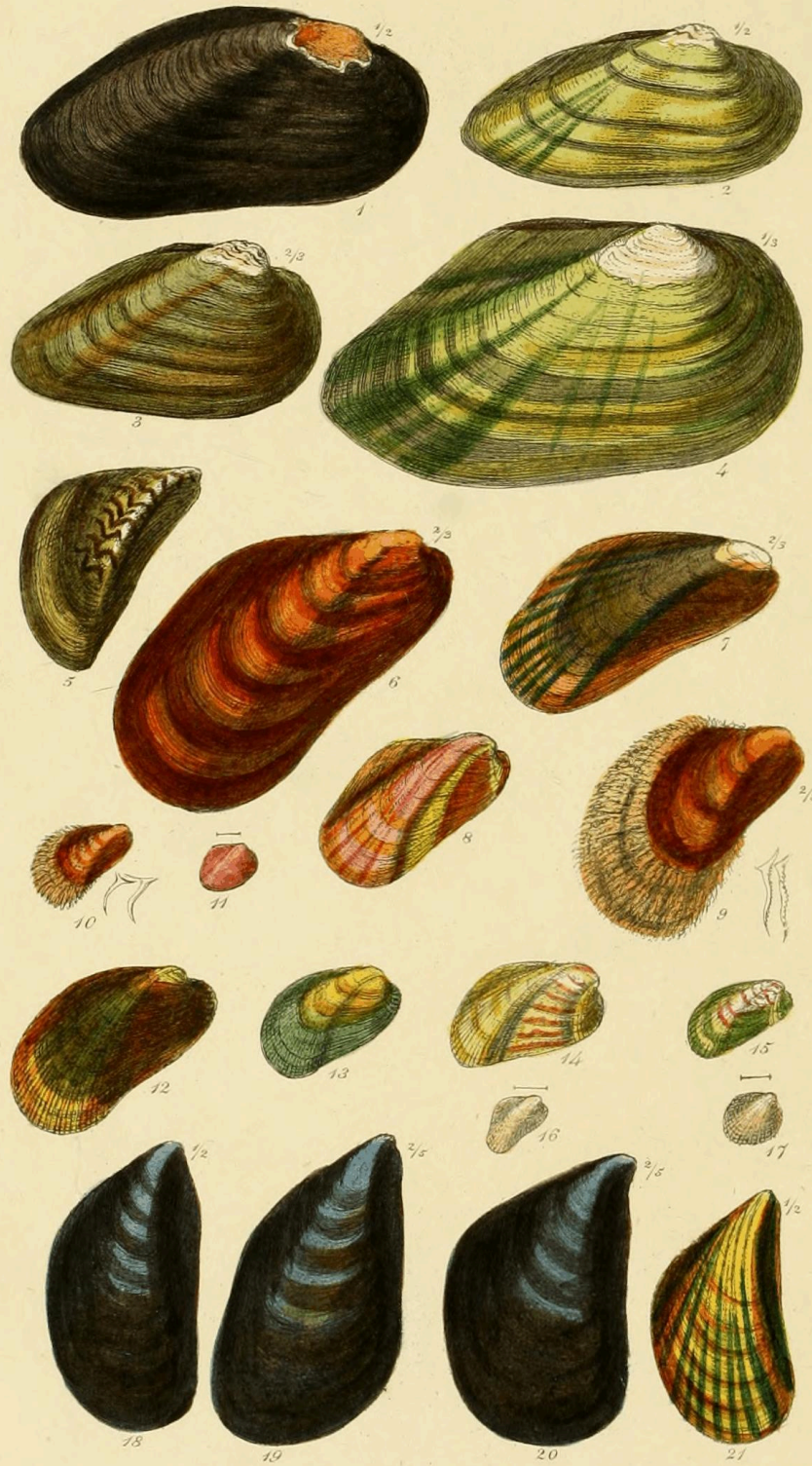
BRITISH SHELLS PL. VI.



G. B. Sowerby.

[back](#)

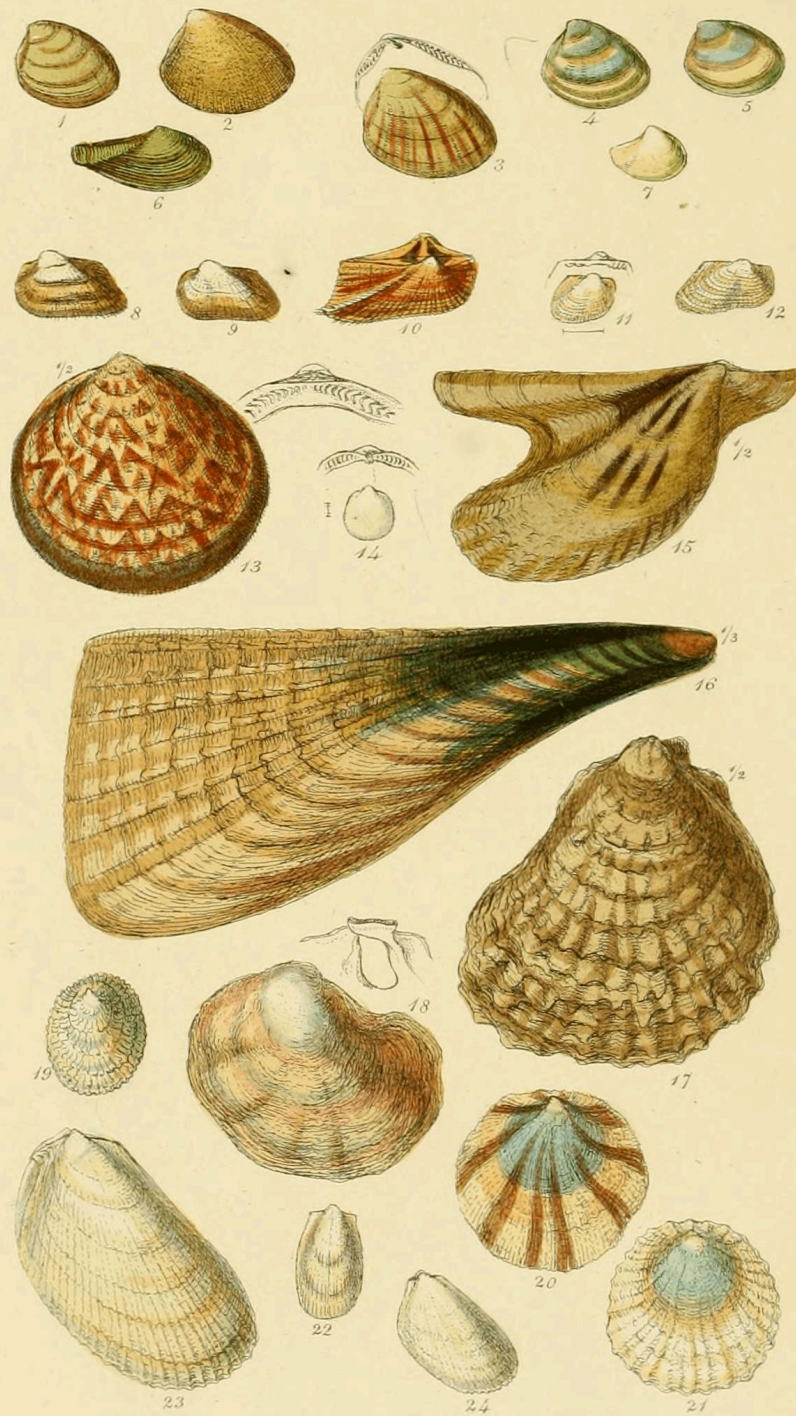
BRITISH SHELLS PL. VII.



G.B. Sowerby.

[back](#)

BRITISH SHELLS PL. VIII.



G. B. Sowerby

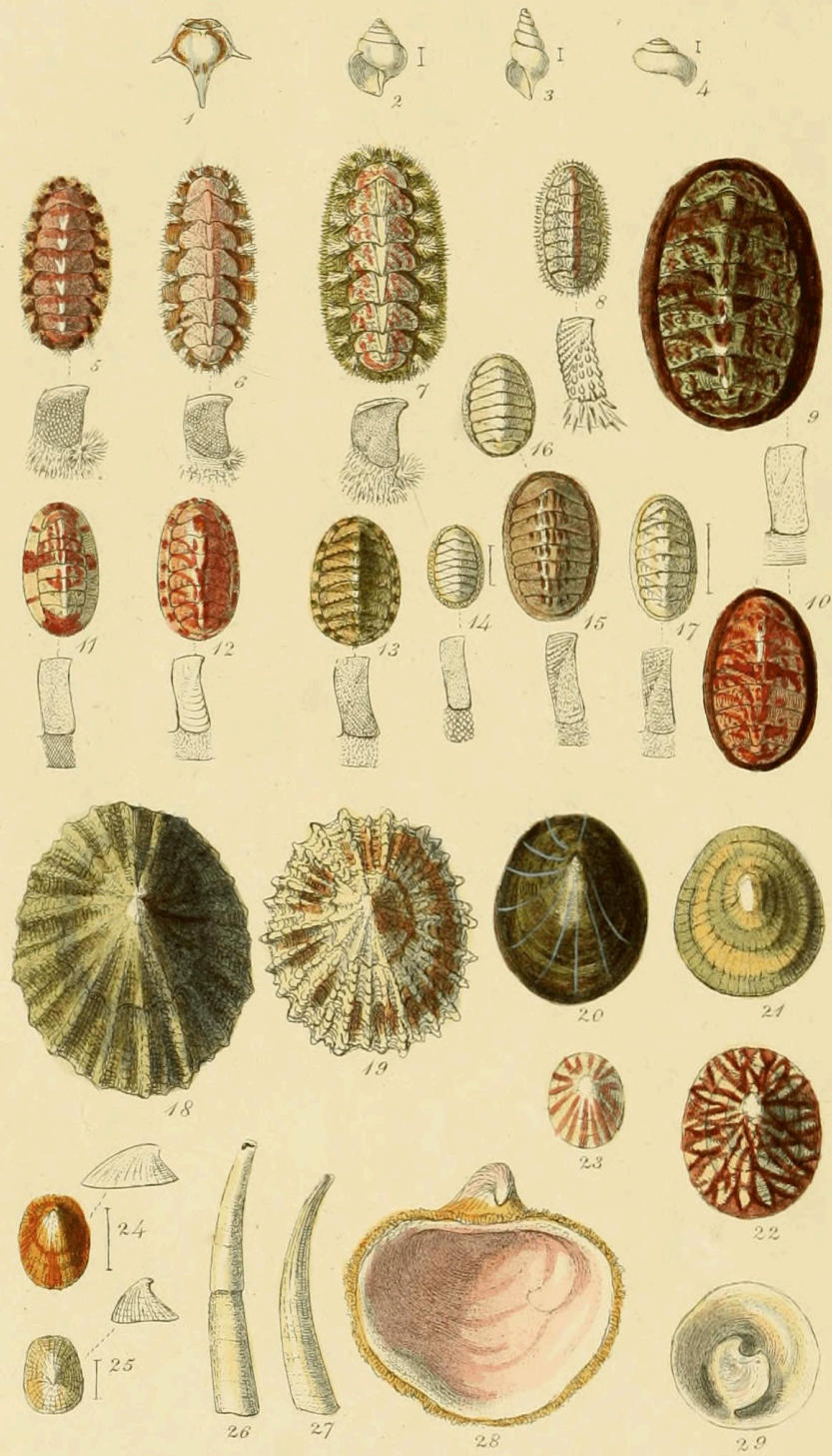
[back](#)

BRITISH SHELLS PL. IX.



[back](#)

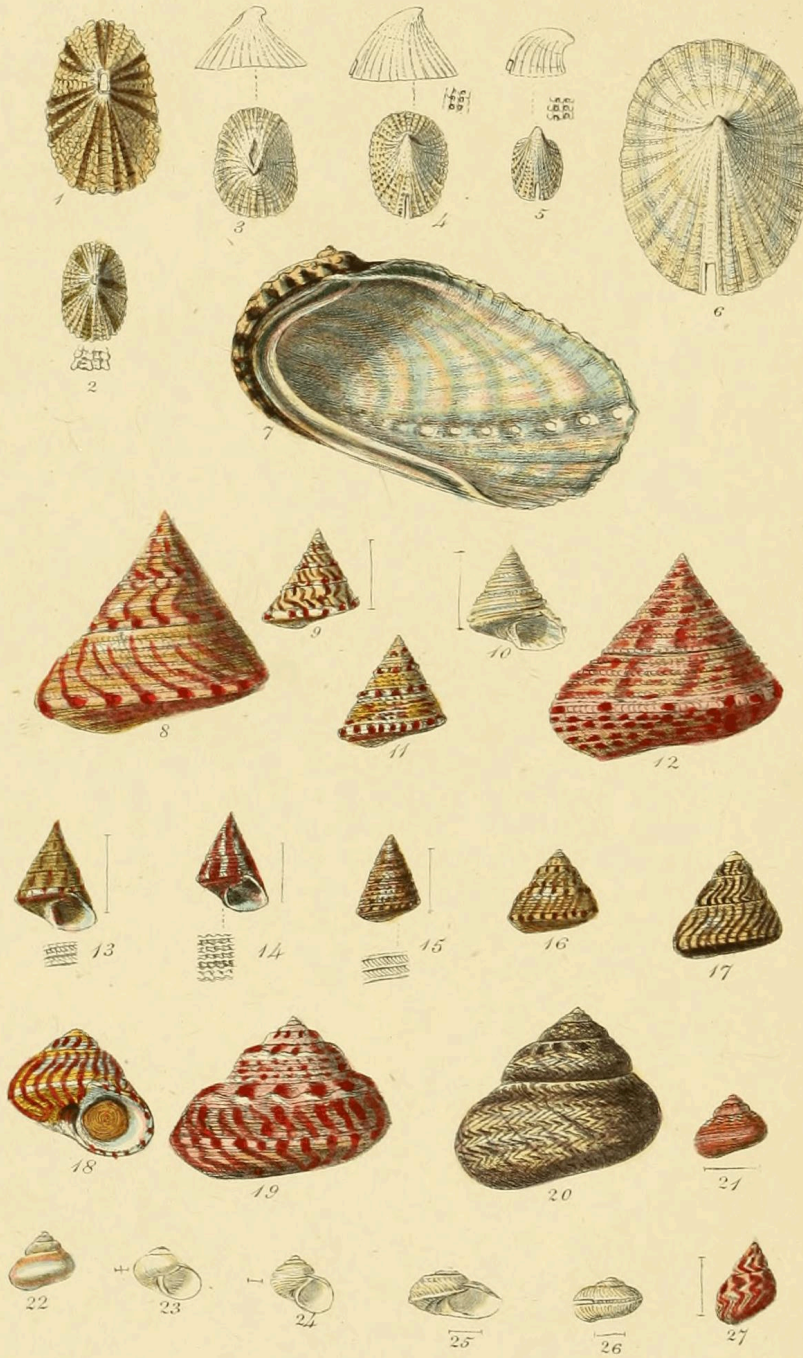
BRITISH SHELLS. PL.X.



G. B. Sowerby

[back](#)

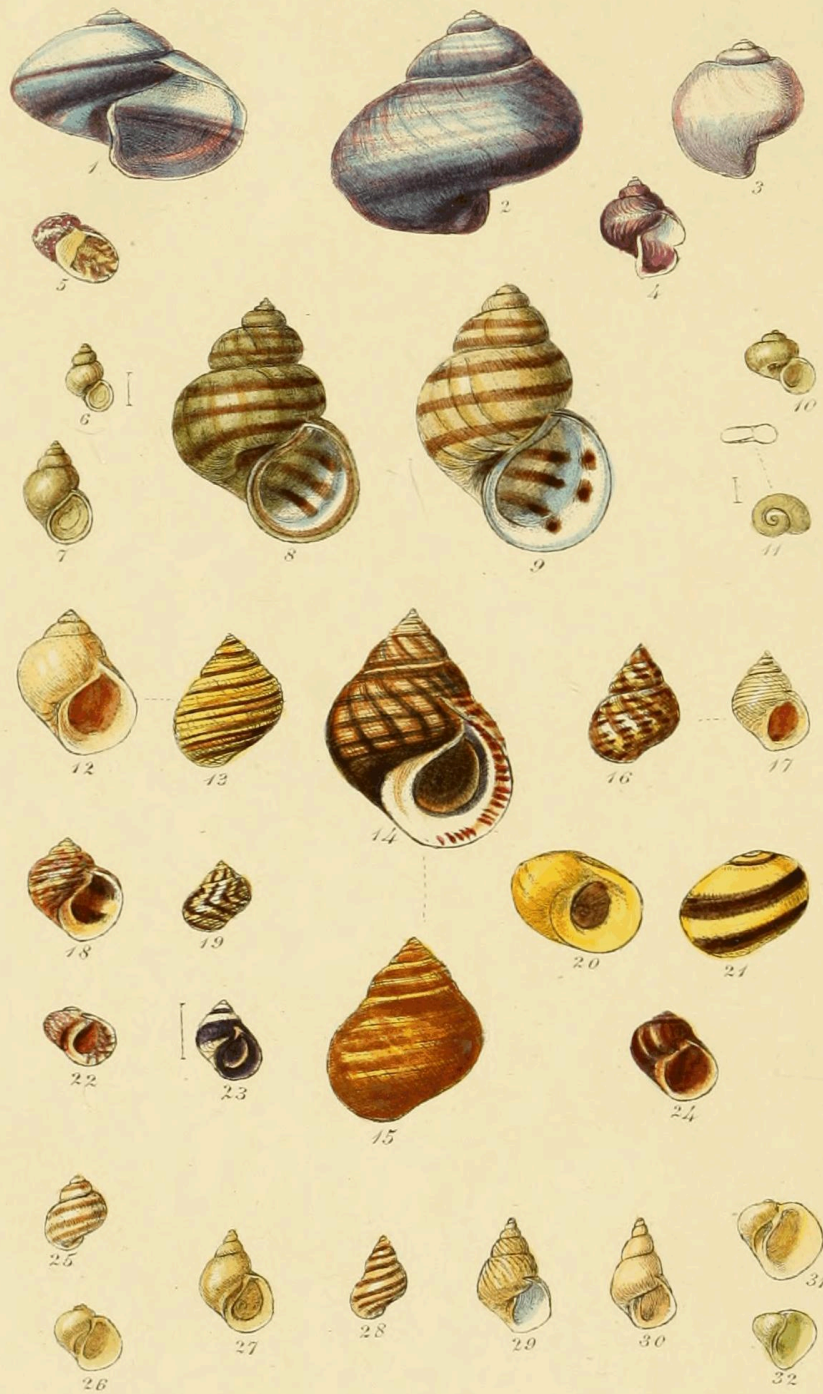
BRITISH SHELLS, PL. XI.



G. B. Sowerby

[back](#)

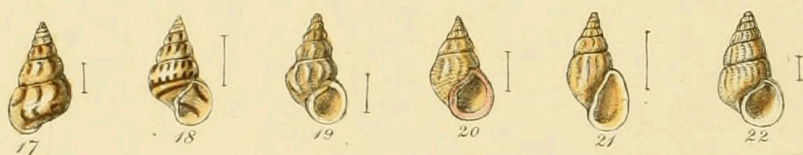
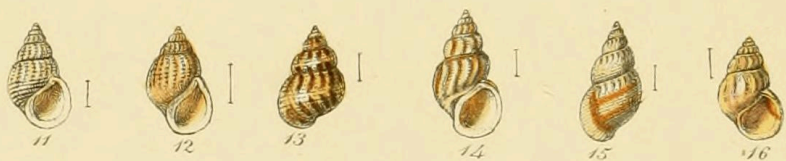
BRITISH SHELLS, PL. XII.



G.B. Sowerby.

[back](#)

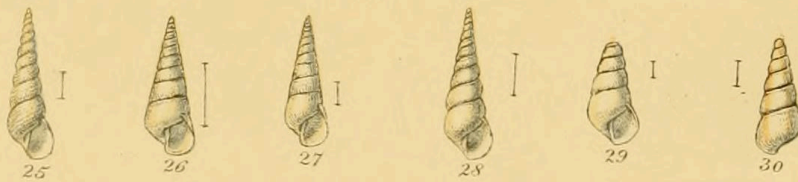
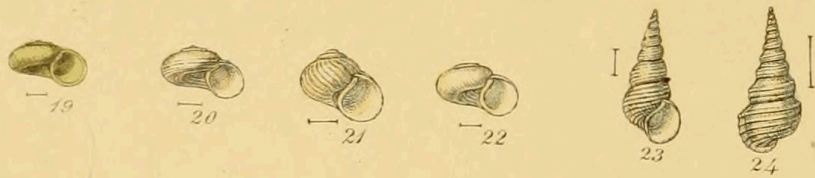
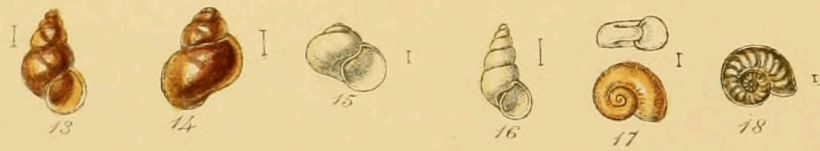
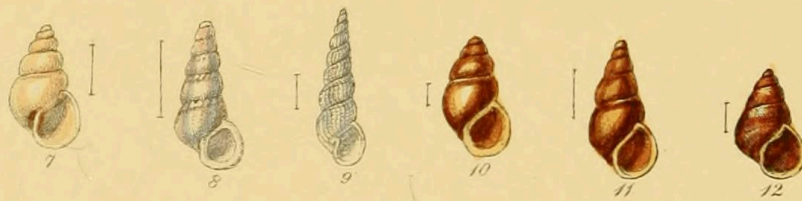
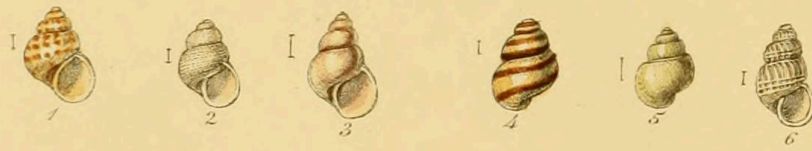
BRITISH SHELLS PL. XIII.



G. B. Sowerby.

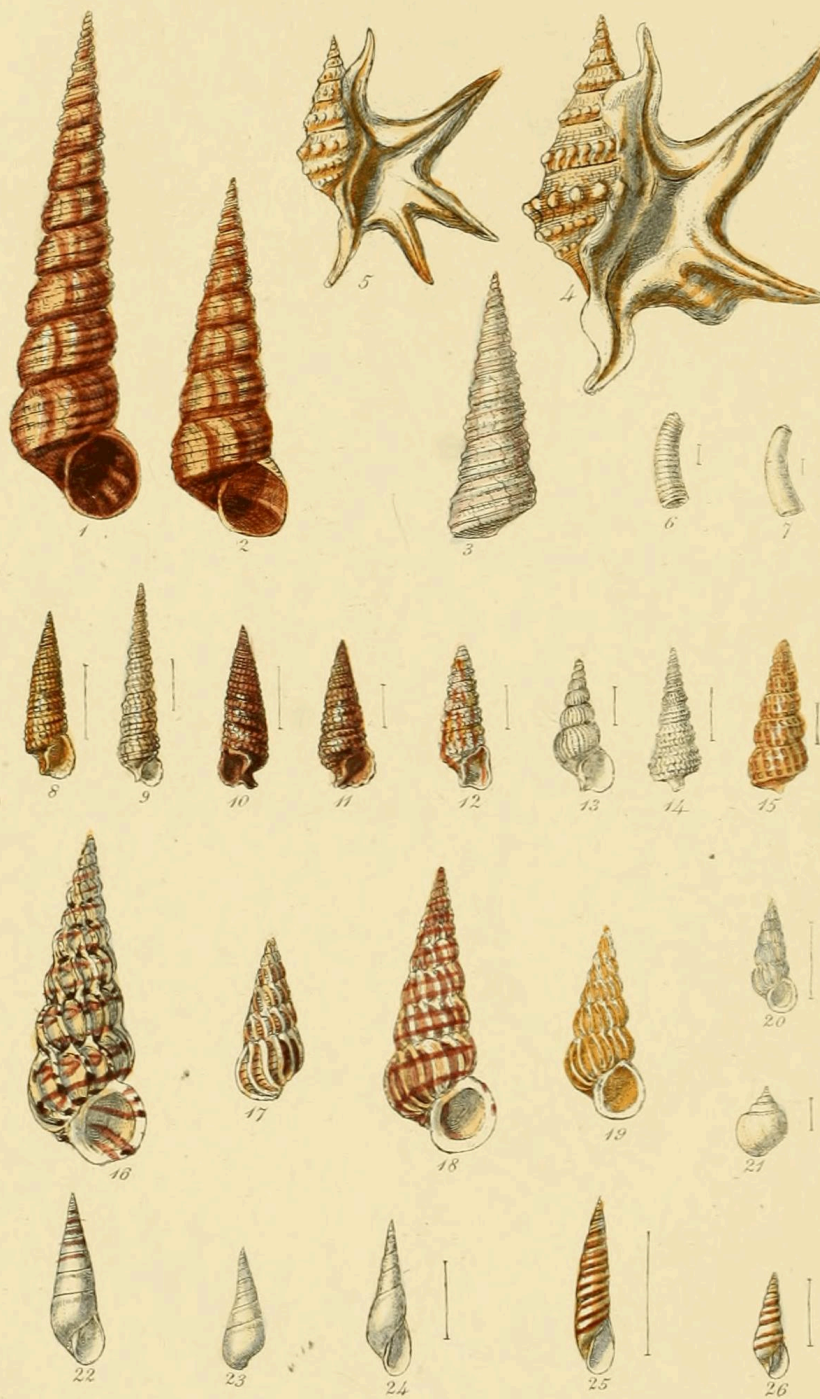
[back](#)

BRITISH SHELLS PL. XIV.



[back](#)

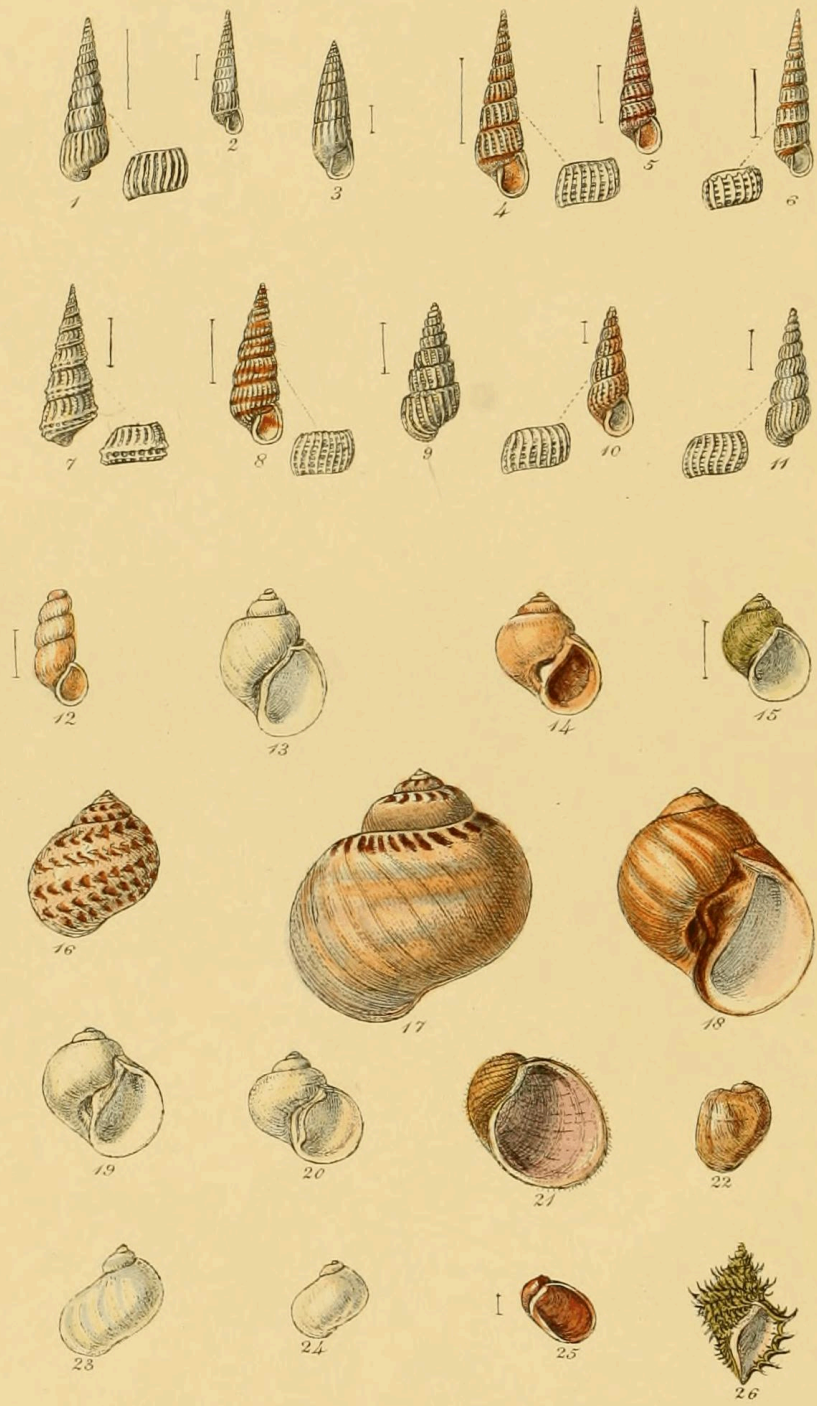
BRITISH SHELLS PL. XV.



G.B. Sowerby.

[back](#)

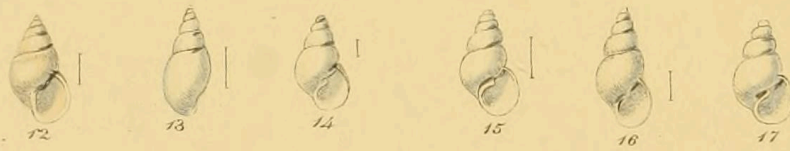
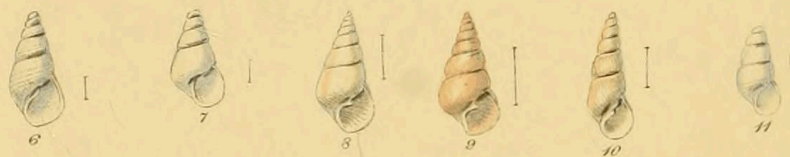
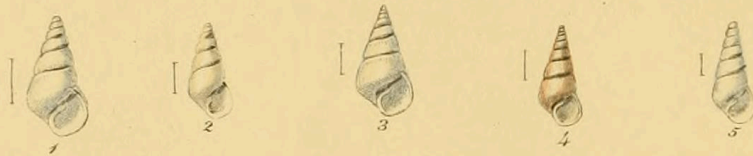
BRITISH SHELLS PL. XVI.



G. B. Sowerby.

[back](#)

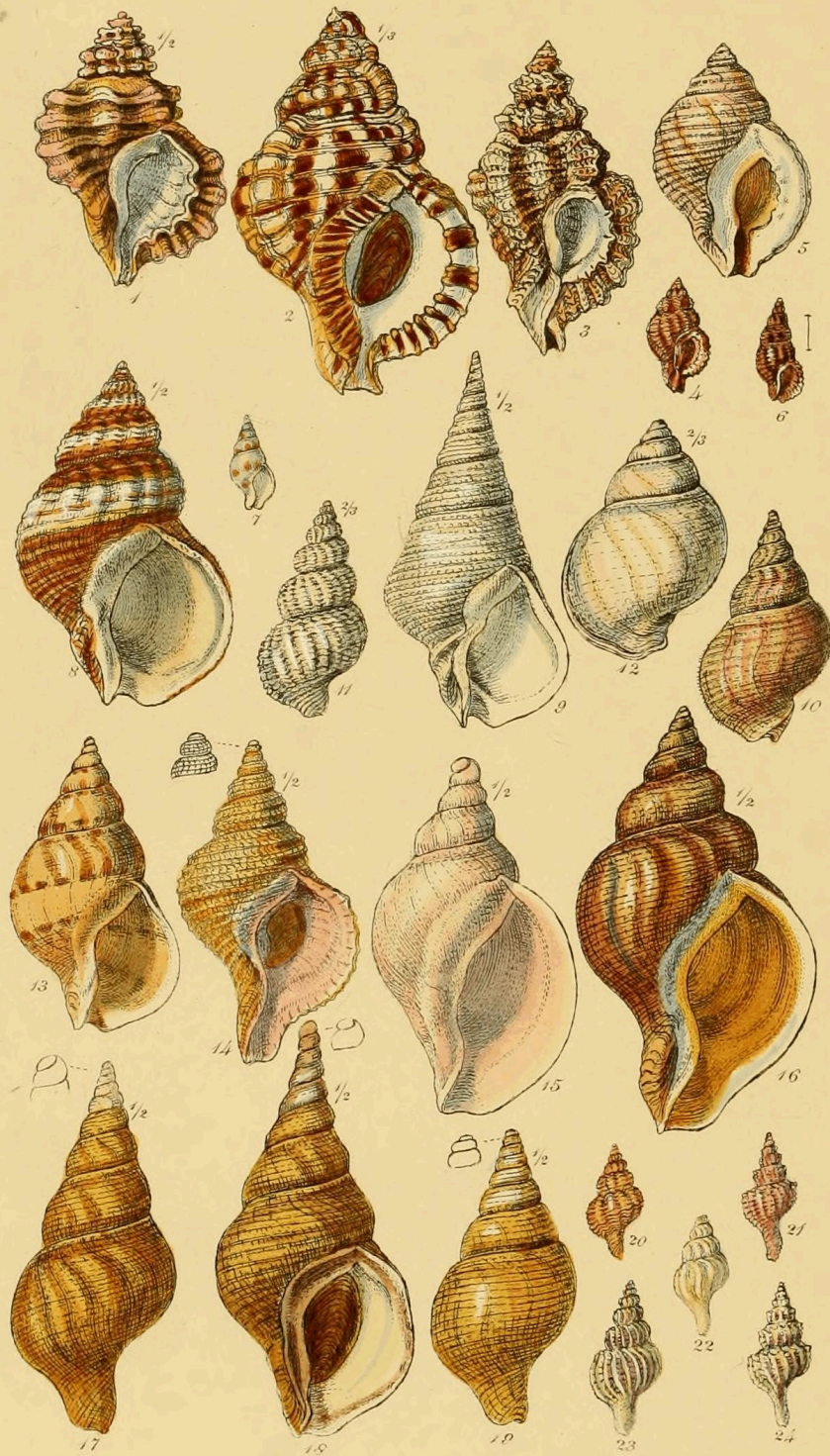
BRITISH SHELLS PL. XVII.



G. B. Sowerby

[back](#)

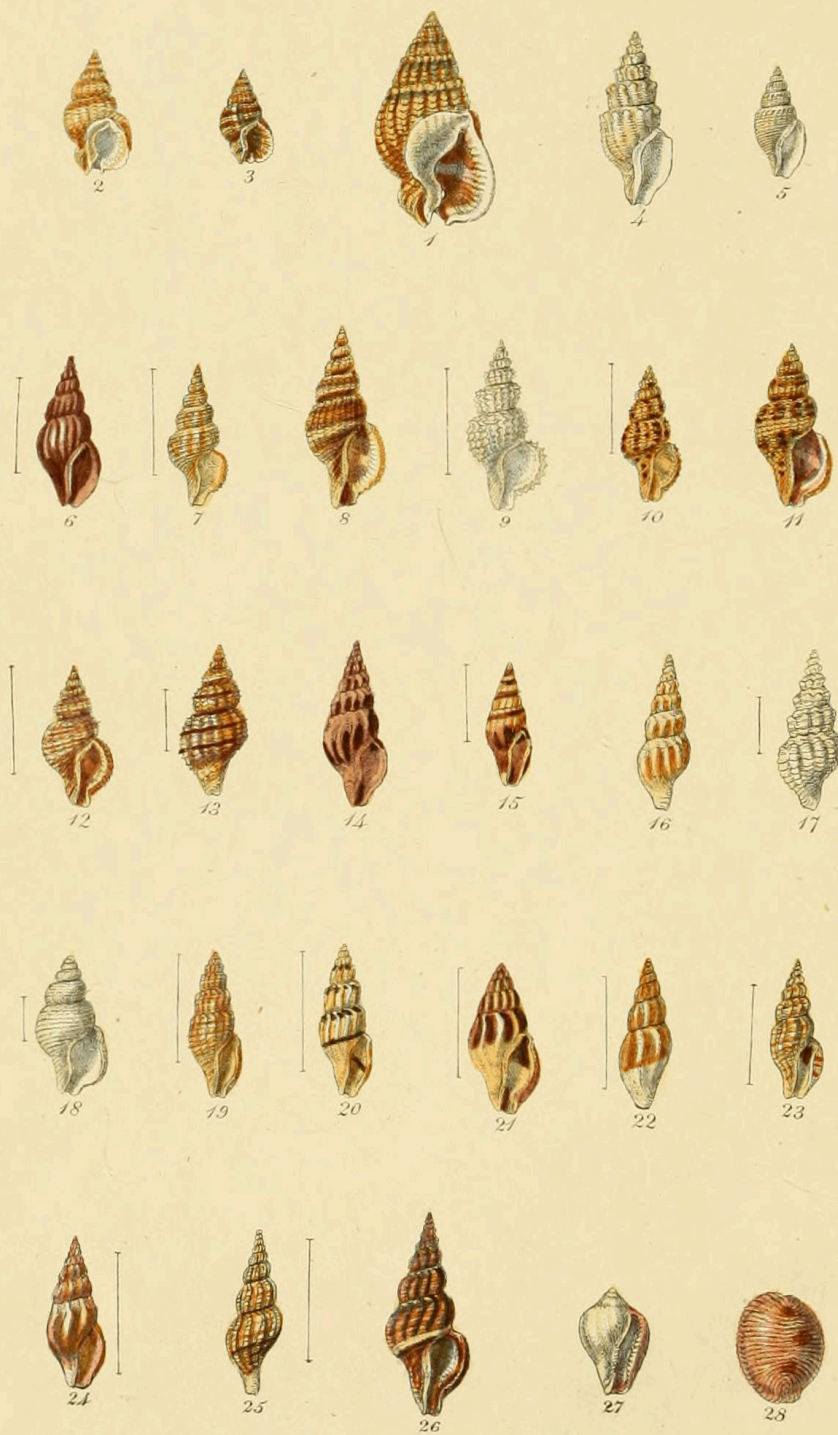
BRITISH SHELLS PL. XVIII.



G. B. Sowerby

[back](#)

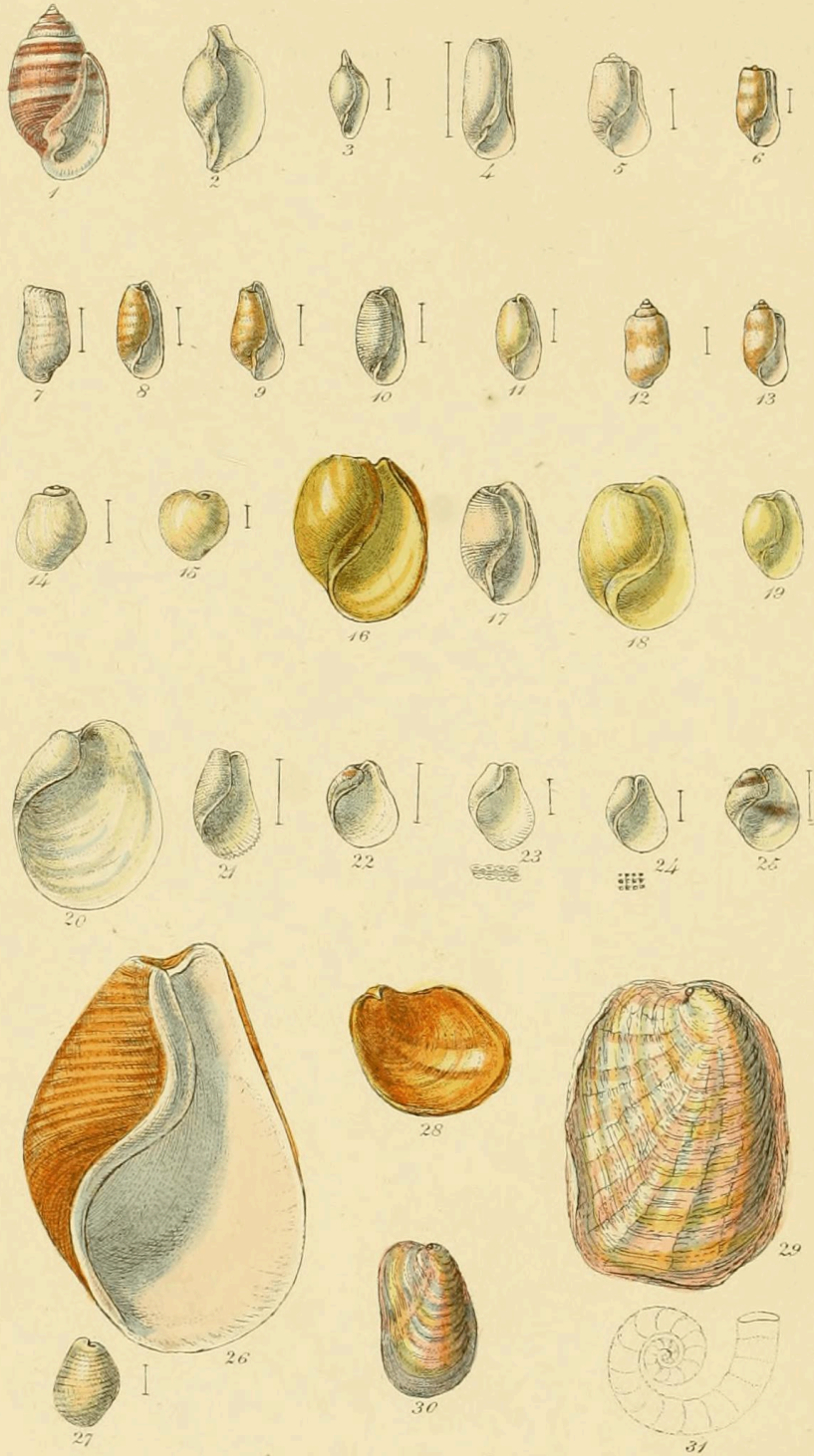
BRITISH SHELLS, PL. XIX.



G.B. Sowerby.

[back](#)

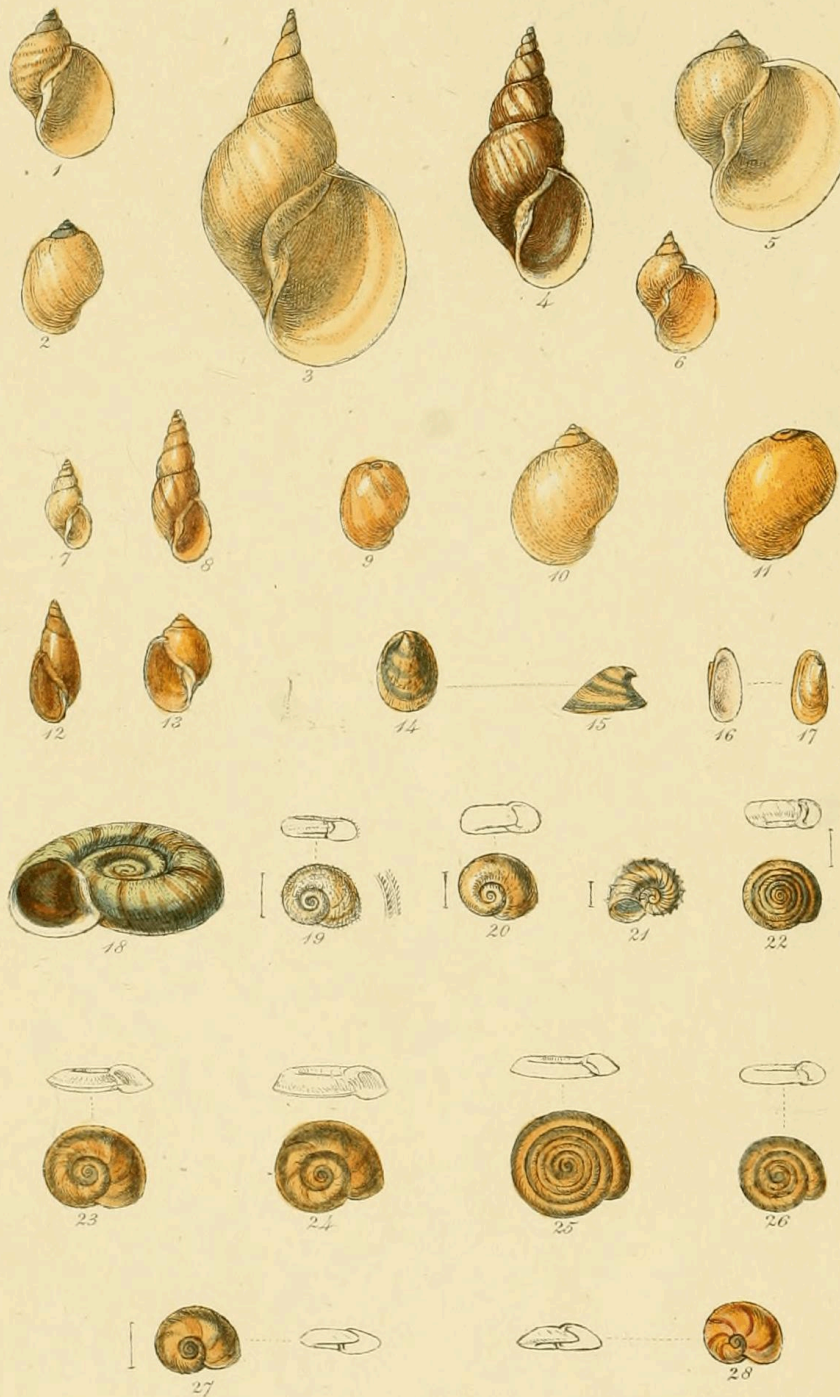
BRITISH SHELLS PL. XX.



G.B. Sowerby.

[back](#)

BRITISH SHELLS PL. XXI.



G. B. Sowerby

[back](#)

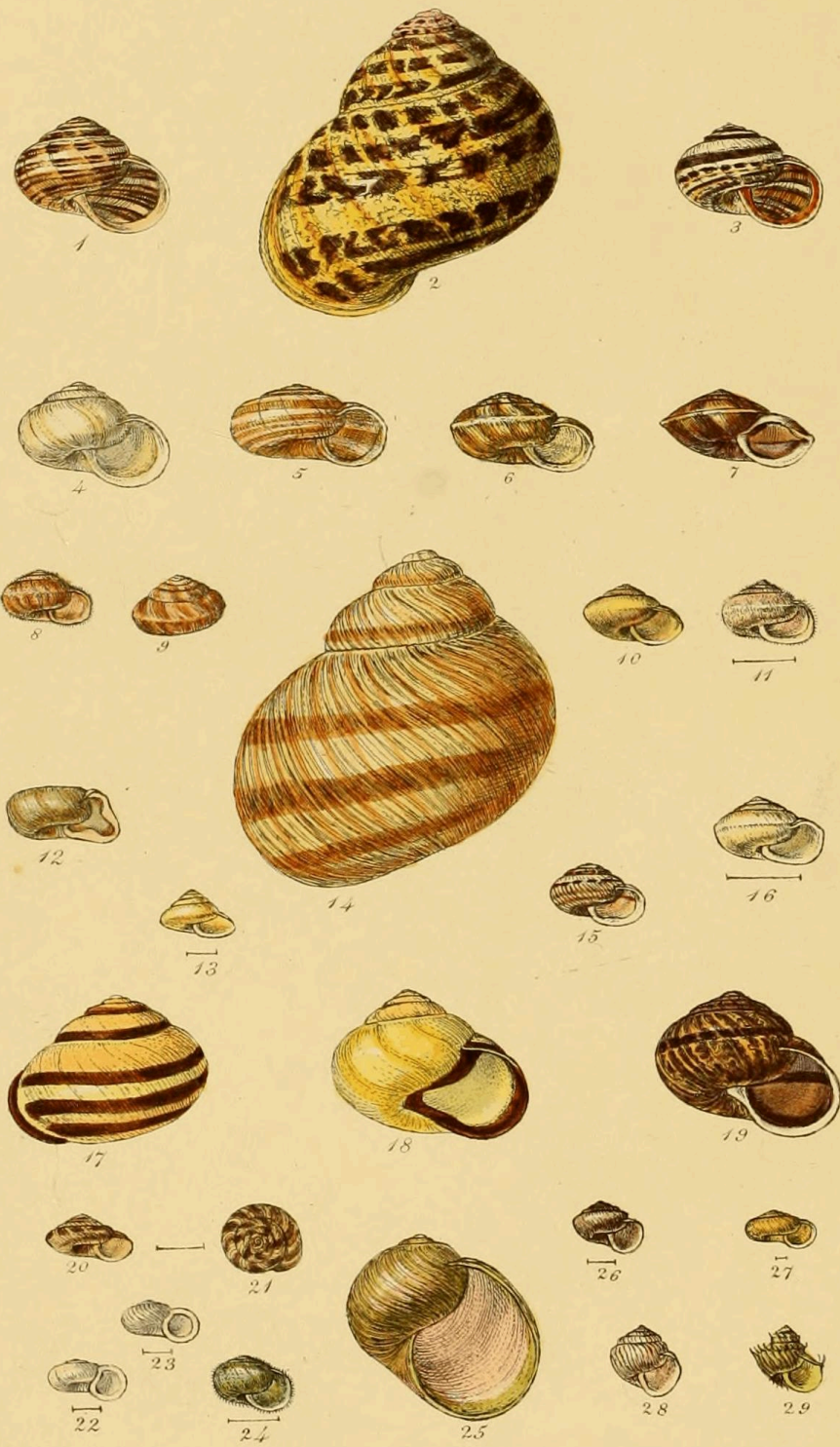
BRITISH SHELLS PL. XXII.



G. B. Sowerby

[back](#)

BRITISH SHELLS PL. XXIII.



G. B. Sowerby

[back](#)

BRITISH SHELLS. PL. XXIV.



G B Sowerby.

[back](#)